Final Analysis – RRRP3



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Head: André Franch

3R at a Glance

3R Petroleum is a **Brazilian Jr Oil** company which has been developing itself since their **IPO**. They operate in the **upstream** segment with **OnShore** and **OffShore** production, counting with **9 assets** of extraction.

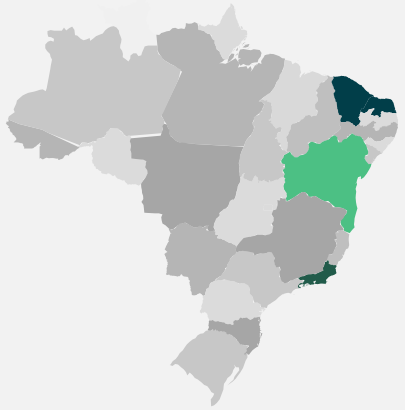
Business Model

Efficient, solid and diverse

- **Upstream** segment, **OnShore** and **OffShore** production
- **Revitalization** of mature assets
- Selling of the whole production to **Petrobras**

Location

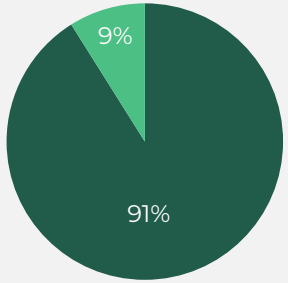
Spread and **diverse** asset portfolio



- Potiguar Cluster
- Recôncavo Cluster
- OffShore Cluster

Revenue Composition

Although they produce gas, their revenue is **mainly from oil**



Timeline

- **2010** — Ouro Preto Óleo e Gás SA foundation
- **2013** — Acquisition of BAR-M-387 and 35% of Pescada
- **2020** — IPO; Pescada and Macau acquisitions
- **2021** — Rio Ventura and Areia Branca acquisitions
- **2022** — Recôncavo, Peroá and Fazenda Belém acquisitions
- **2023** — Papa-Terra acquisition

Source: RI 3R

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Efficient, solid and diverse

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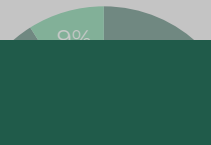
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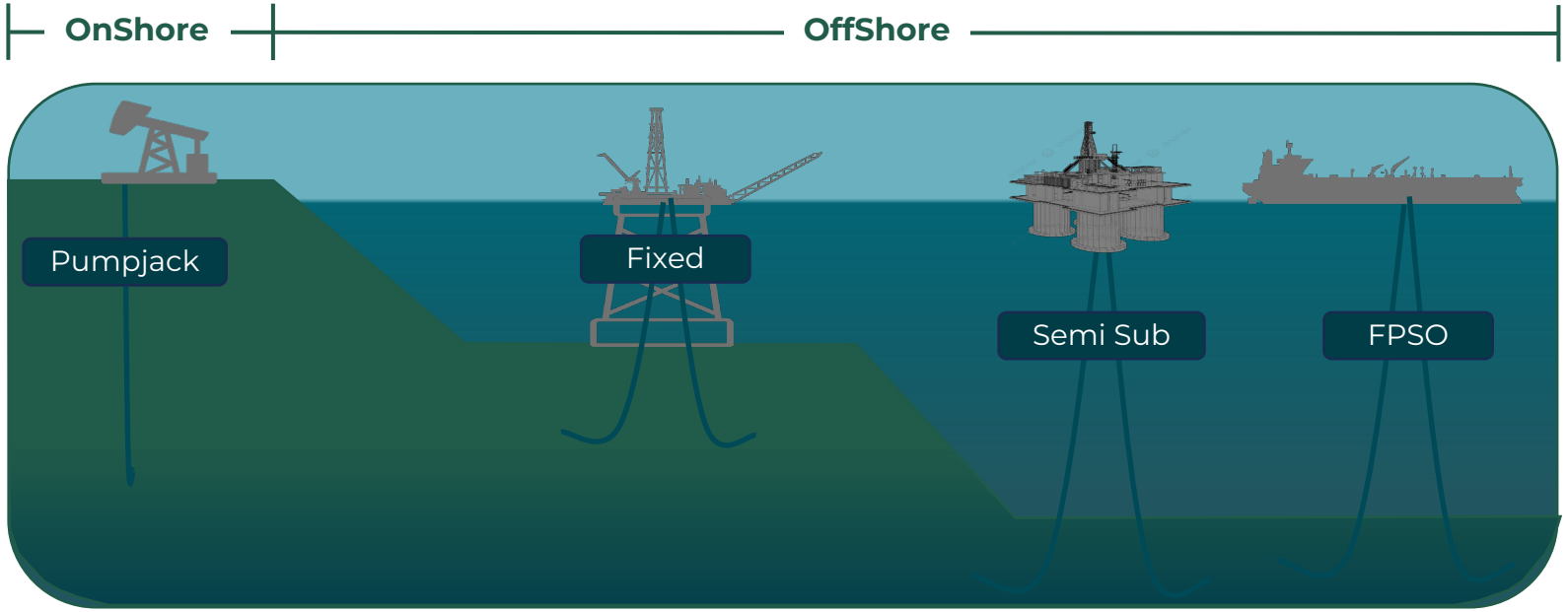
What is a Junior Oil company?



Source: RI 3R

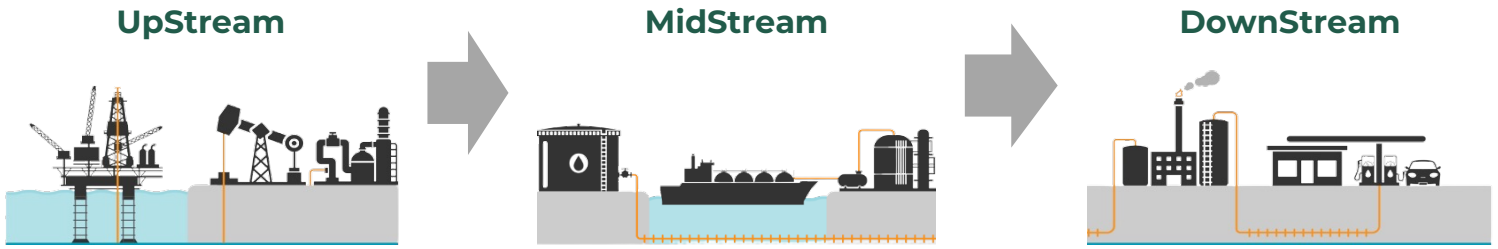
Investigating the Operations

Before diving in to Oil&Gas, its necessary to know how to classify an asset, by **location**, by **type of platform**, by operating **segment** and by the **quality** of their reserves.



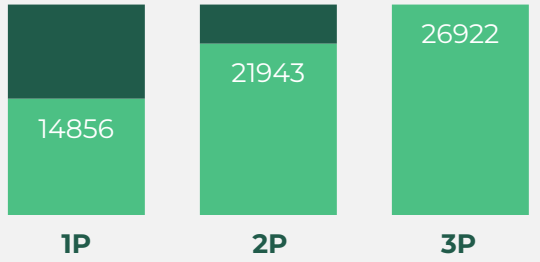
Oil&Gas Segments

Although the **energy sector** is composed by **three segments**, junior oils mainly work in the **upstream**



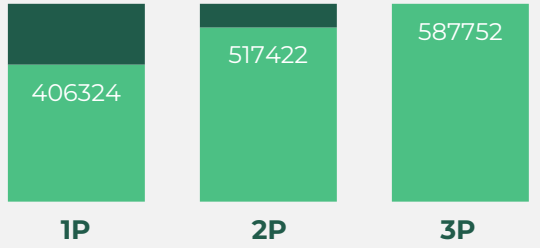
Brazil's Oil Reserves

In MMboe



Brazil's Gas Reserves

In MMm3



1P: **90% certainty** of commercial extraction
2P: **50% certainty** of commercial extraction (1P+2P)
3P: **10% certainty** of commercial extraction (1P+2P+3P)

Source: 3R RI, ANP

Tiny Giants

The O&G sector in Brazil can be seen by **two perspectives**: with and without **Petrobrás**, once it is by far the largest. Junior Oils are born by acquiring Petrobras's **mature wells** and **revitalizing** them.

O&G Companies MarketShare by Daily Oil Production

In Brazilian **Oil&Gas** sector, **Petrobras** is by far the **largest company**

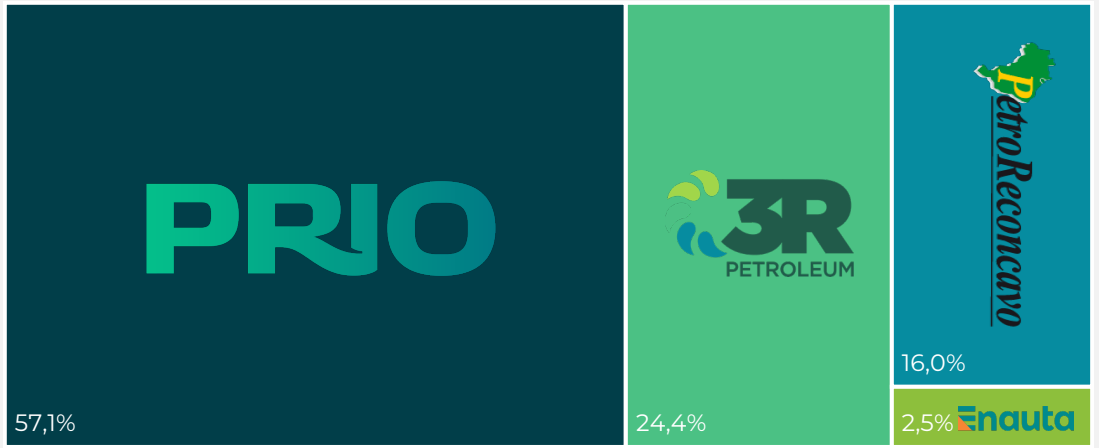
In % of oil boe/d



Junior Oils MarketShare by Daily Oil Production

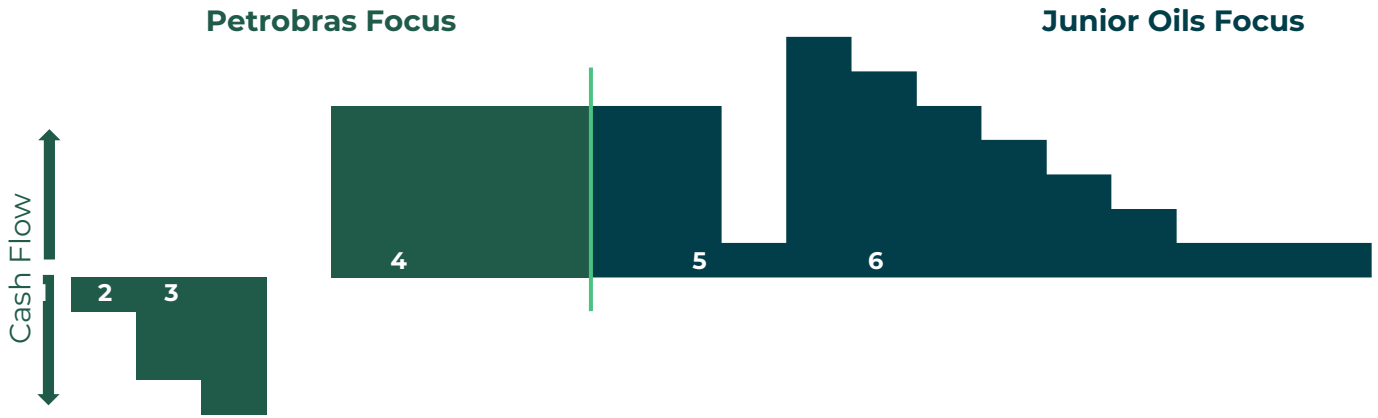
In the Brazilian **Junior Oil** sector, **PRIO** is the **largest one**, with 57,1% of MarketShare

In % of oil boe/d



Trajectory of an O&G Well

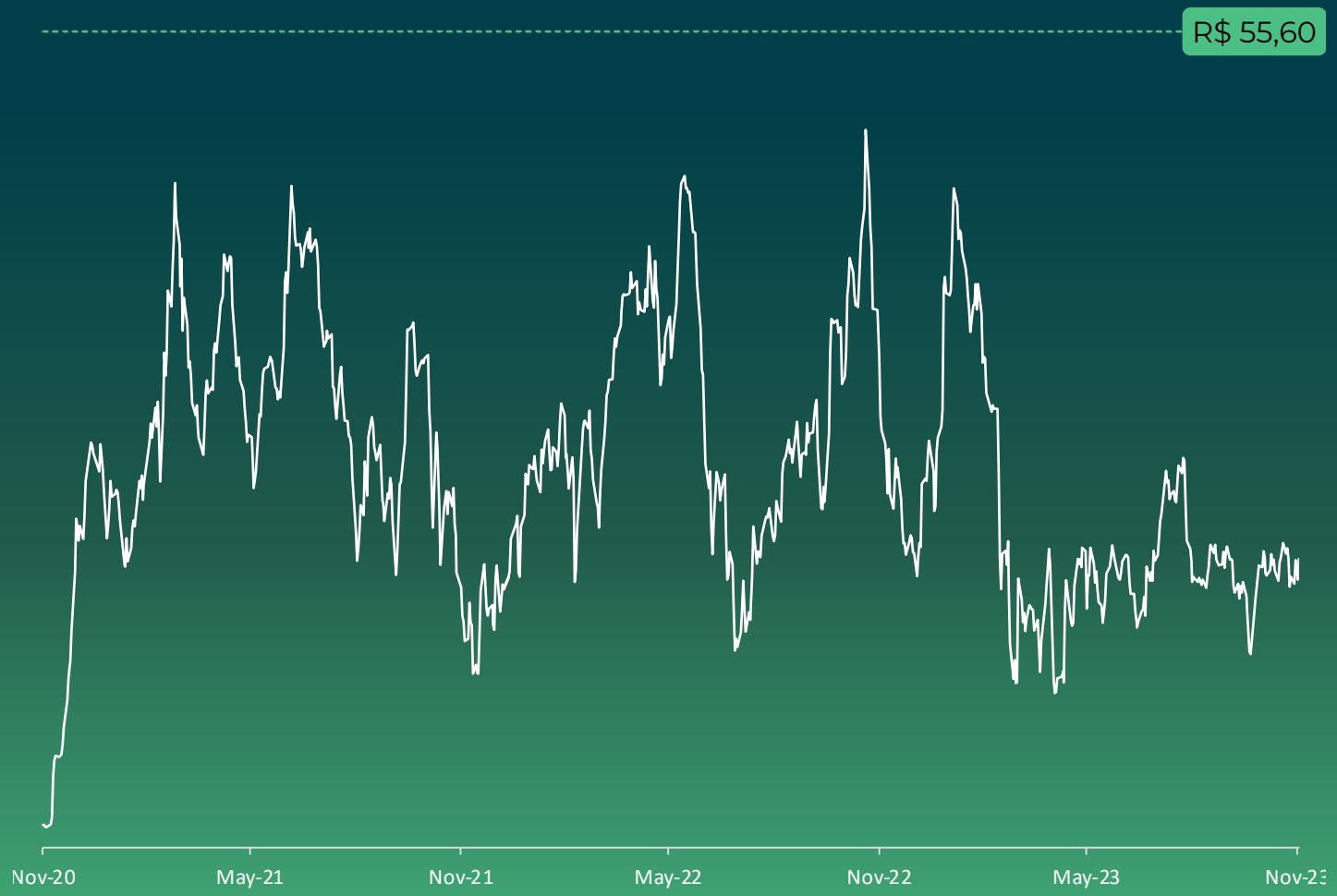
The **first part** of a well's E&P is **Petrobras responsibility**. After some time, **Junior Oils** buy this **mature assets** and **revitalize** them, extracting was having been left



1 Exploration
2 Appraisal
3 Development
4 Production
5 Redevelopment
6 Production Recovery

Source: 3R RI, XP

RRRP3 - BUY



BUY

32,95
Actual Price

55,60
Target Price

68,7% UPSIDE

35,6% IRR

RRRP3 Investment Thesis

Stepping on a solid field
Oil&Gas sector is worldwidely consolidated and the demand for it is guaranteed.

Increase in Production
With the new management combined to a consolidated asset portfolio, 3R Petroleum is already showing growth in its production.

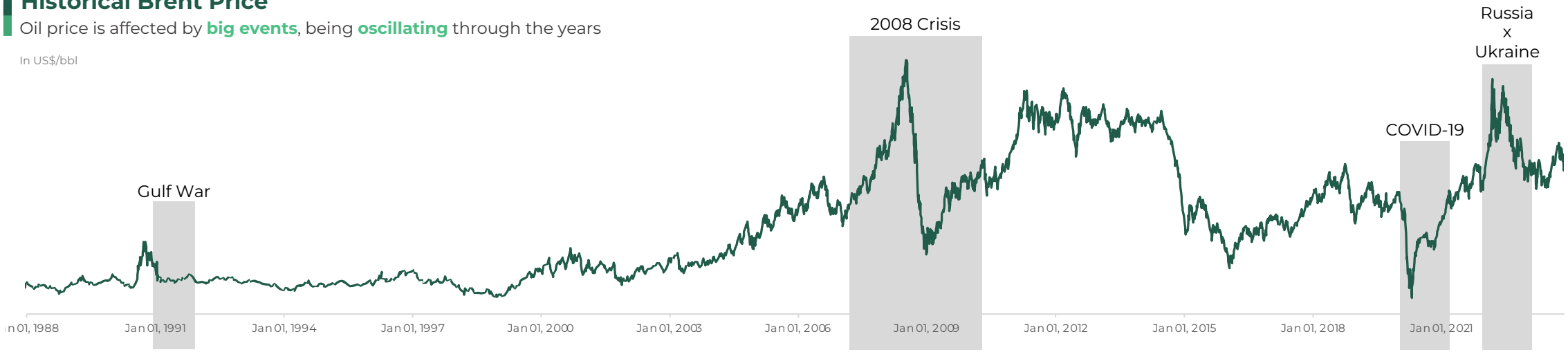
Significant upside, conservative premises
Even using extremely conservative premises, and without new Clusters, there is a significant Upside.

Looking from the Bigger Picture

O&G sector is directly **dependent** on the **macroeconomic scenario**. Brent price varies due to **uncontrollable big events**, which dictates junior oil's prices.

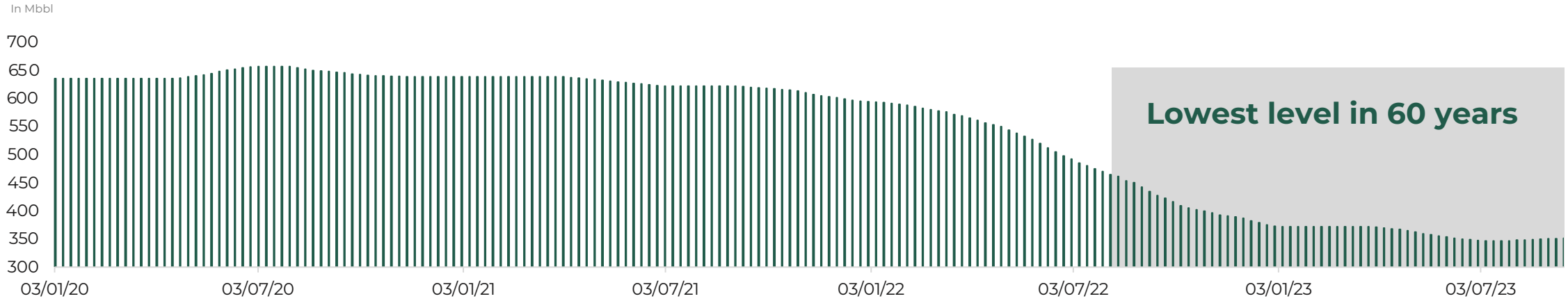
Historical Brent Price

Oil price is affected by **big events**, being **oscillating** through the years



U.S. Ending Stocks of Crude Oil in SPR

The U.S. strategic reserve is at its **lowest level in the last 60 years**, hovering at a concerning number. Its likely that the U.S. government will need to repurchase a portion of the reserves.



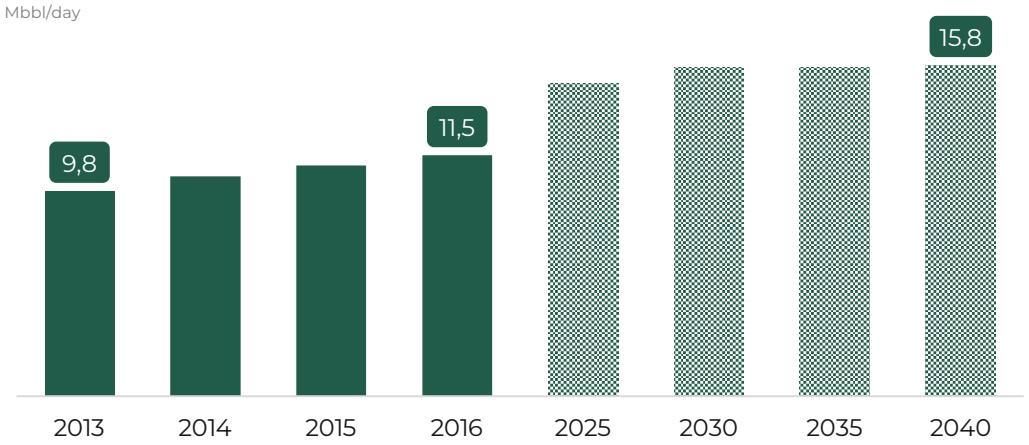
Source: US EIA

The World Urges for Oil!

Even with **new renewable ways of energy** coming along, the demand for oil&gas remains **solid** and **inelastic** for the next years. Powerful countries, like China, and emerging countries, like India, will **keep urging** for more oil!

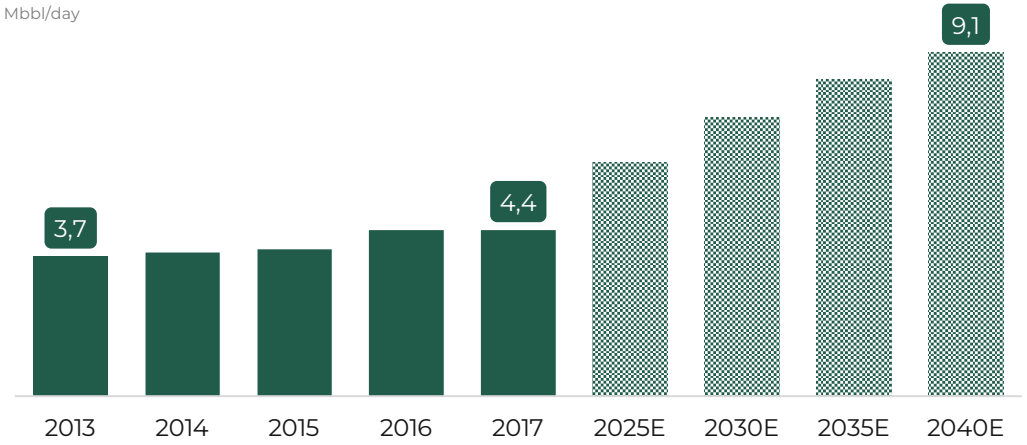
Projected Oil Demand in China

For the next years, one of the most **powerful countries** will **increase demand** for oil



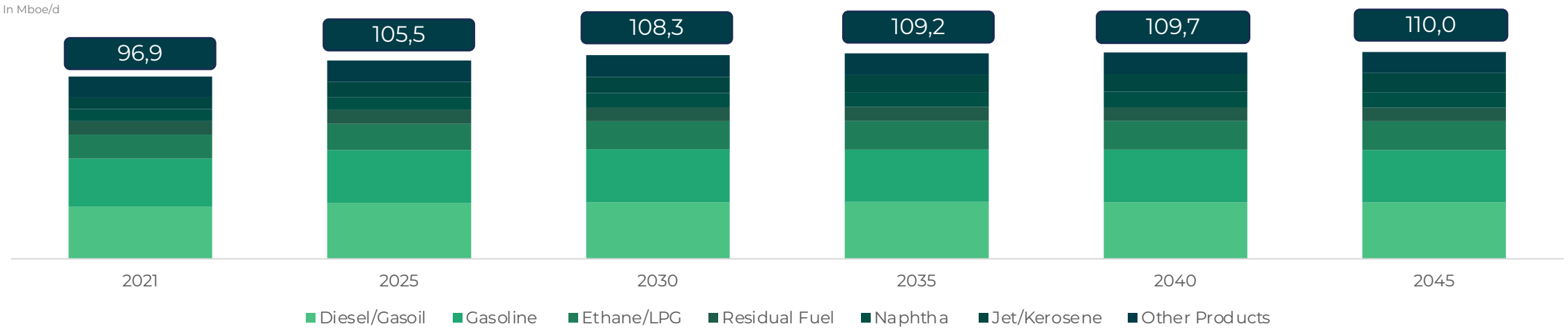
Projected Oil Demand in India

The **emerging countries**, like India, will **need oil** like never before!



Global Demand for Oil per Product

Global demand tends to **stabilize** for the next years, but **never to decrease**. The world will never be petroleum independent, at least not for now



Source: Statista

The World Urges for Oil!

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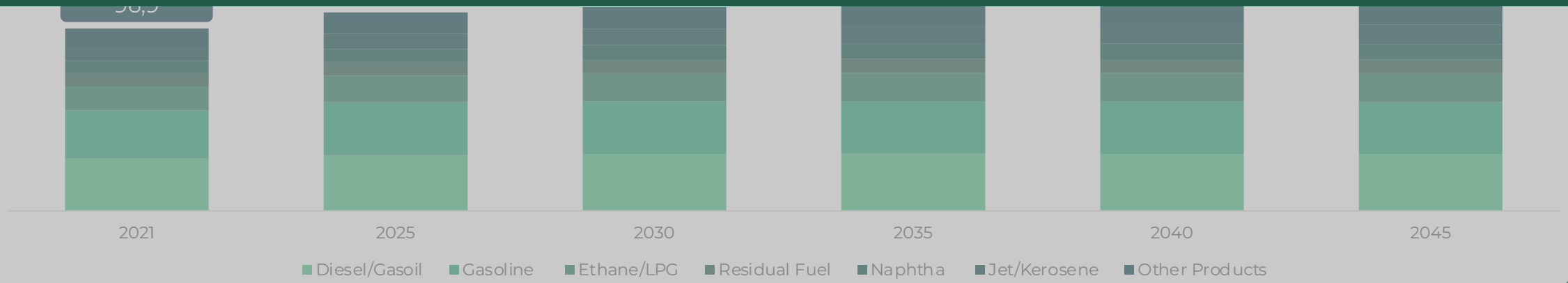


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What About the Energetic Transition?



Source: Statista

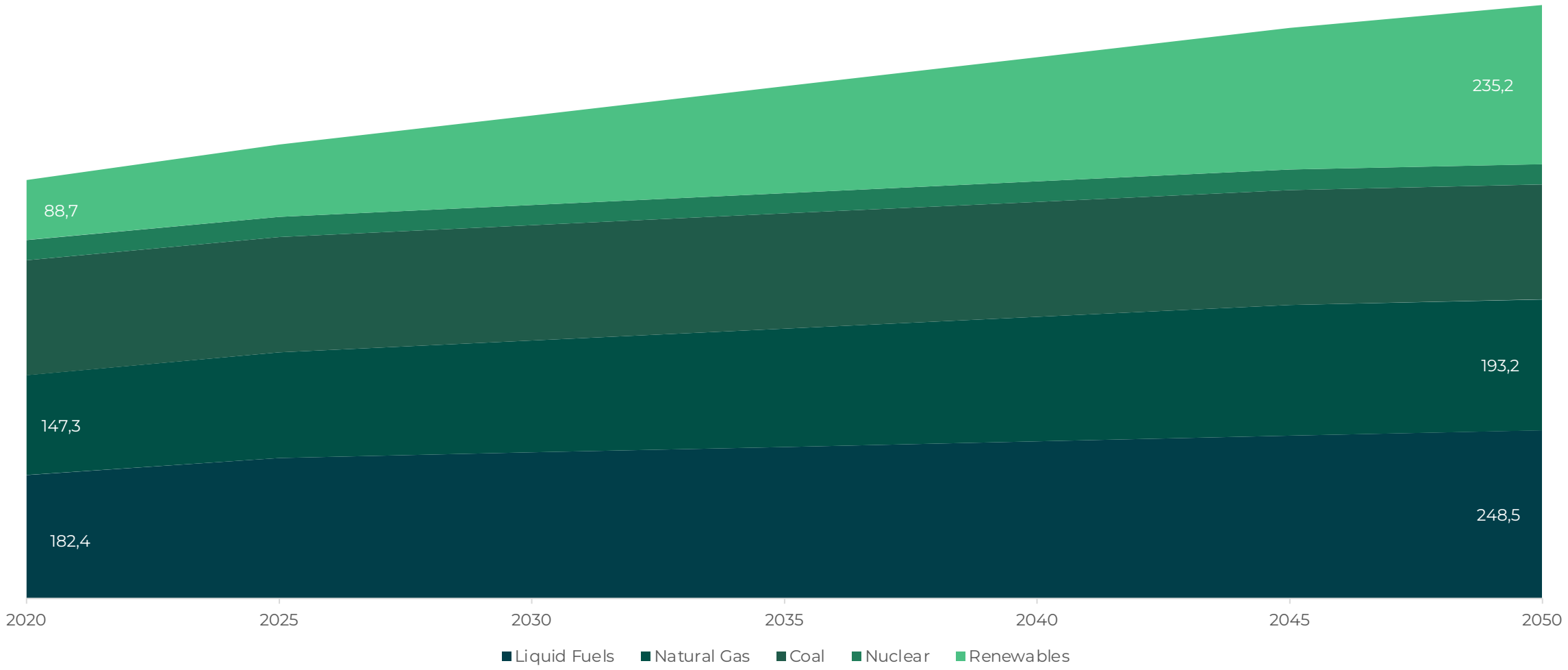
Oil Stands the Test of Time

As time goes by, **energy demand grows**. Instead of widely increasing oil demand, companies start to supply their **new necessities with renewable energy**. Oil keeps being used, but not with new matters so it keeps **constant** through the years.

Types of Energy Consumption for the Next Years

Renewable energy supplies the new demand, but the oil and gas's demand stay **loyal and solid**

In quadrillion British thermal units



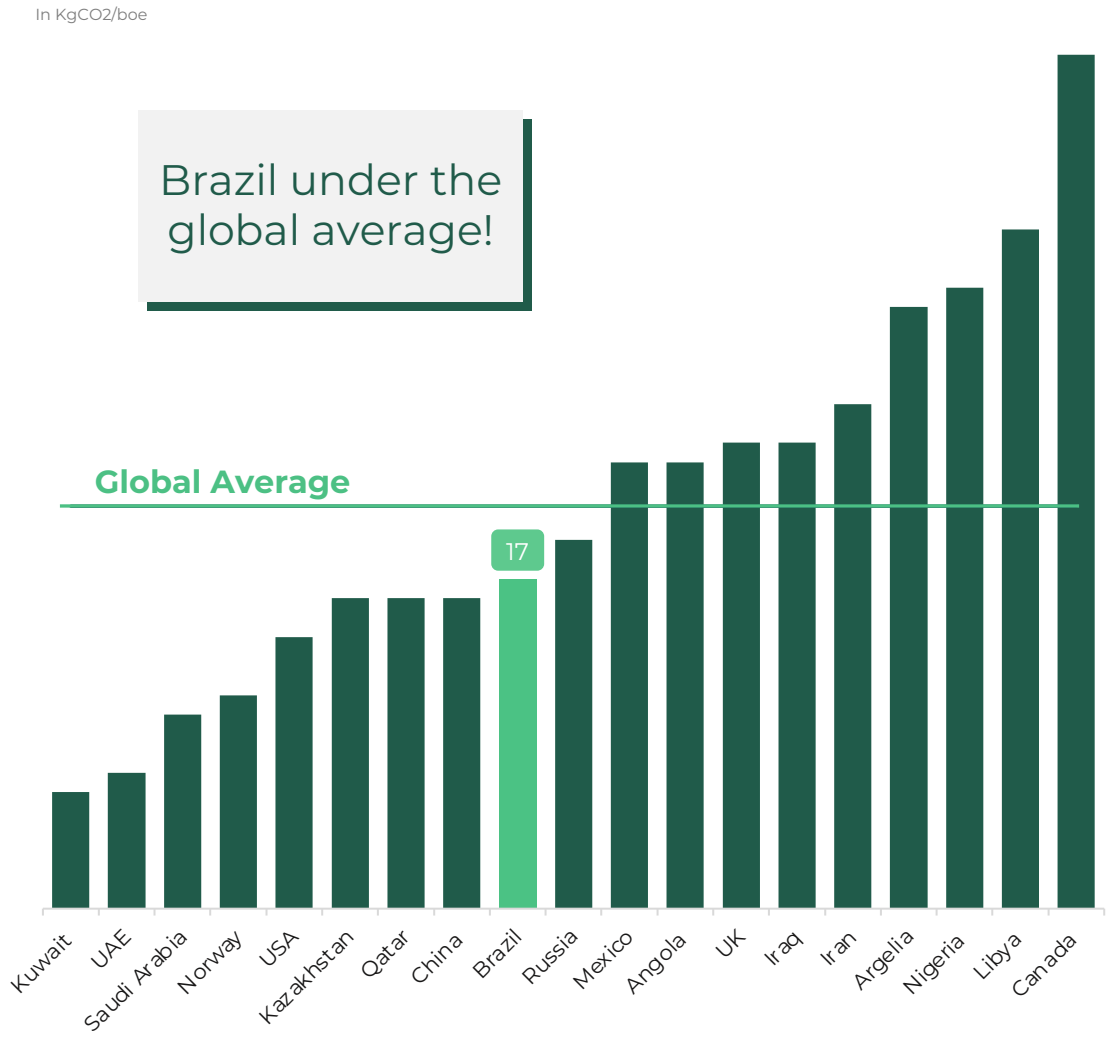
Source: IEA

Brazil as an ESG Reference!

As environmental matters get stronger, the forecast is that countries with **less carbon emissions** will have some kind of **benefit**, therefore Brazil is in a **great position** for this changes!

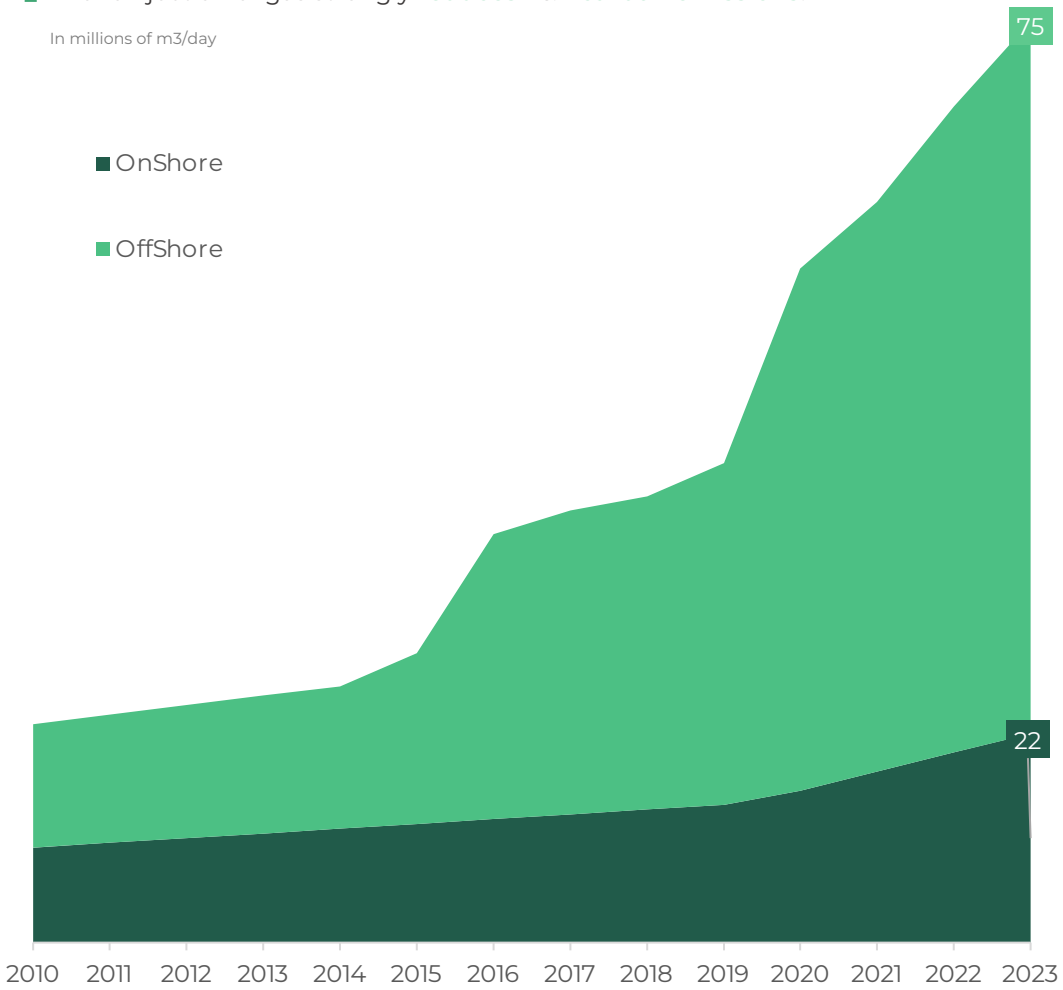
Intensity of Carbon Emissions in Oil Production in 2019

Brazil's rate is under the global average!



Reinjection of gas in OnShore and OffShore Platforms

The reinjection of gas strongly **reduces** E&P **carbon emissions**!



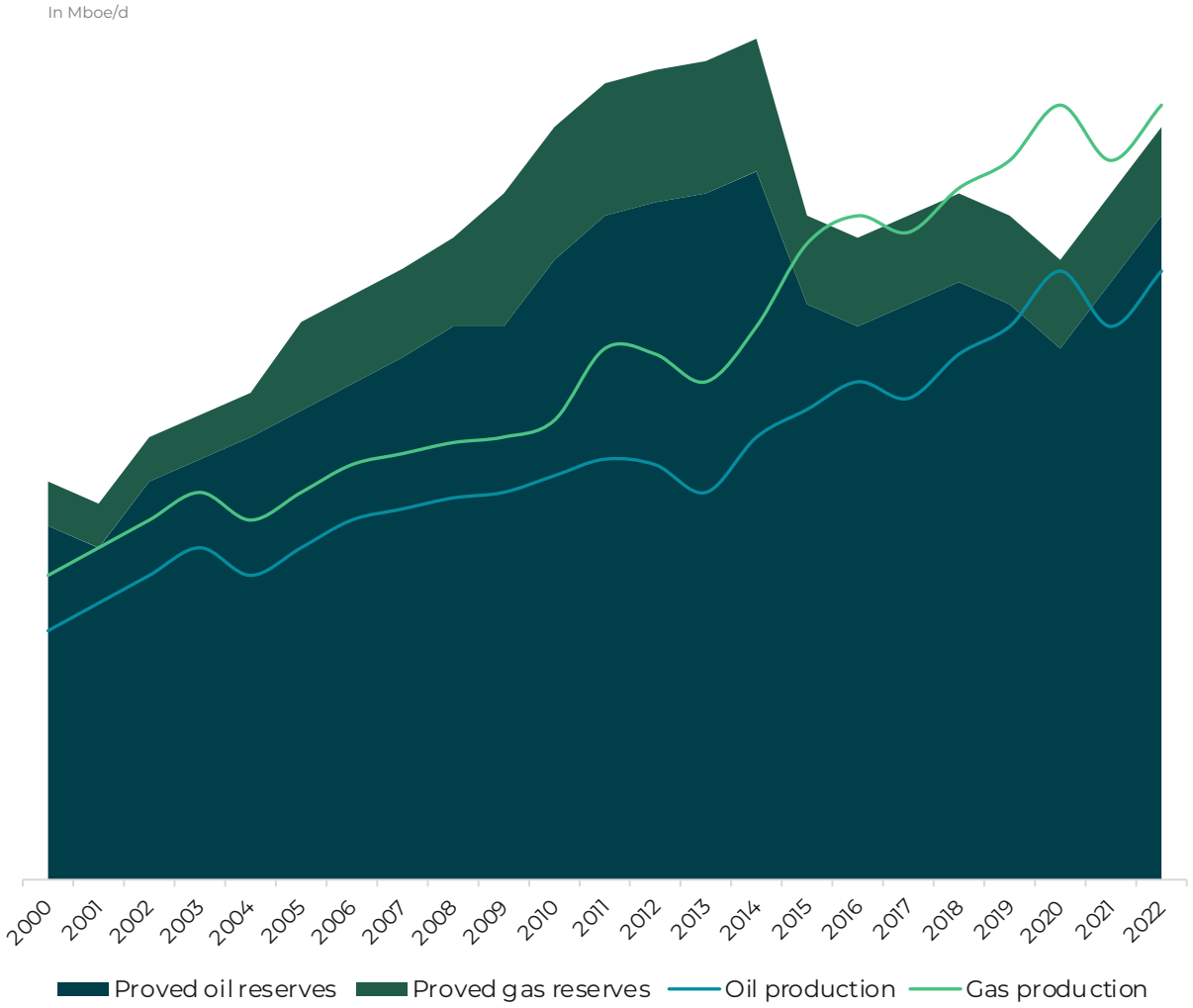
Source: IBP, Statista

Brazilian Black Gold

With an expressive participation in Brazilian economy, the O&G sector has been **constantly growing** in the country. Brazil is one of the **world's big oil producers**, with **3% of MarketShare**!

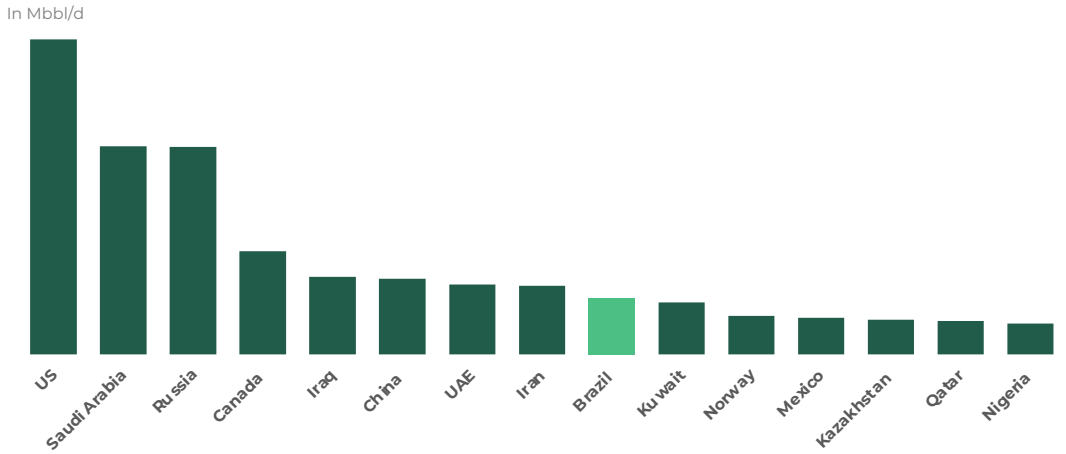
Nacional Evolution of 1P and 2P Reserves and of O&G Production

O&G production **keeps intensely rising** in Brazil!



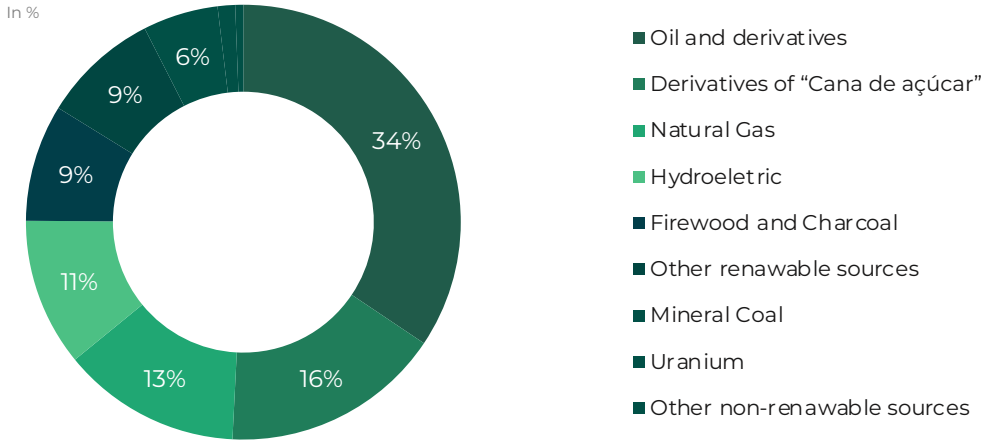
Top 15 Global Oil Producers

Brazil is the **9th largest** oil producer of the world!



Brazil's energy matrix

O&G are **essential for existence**, being responsible for **50%** of national energy



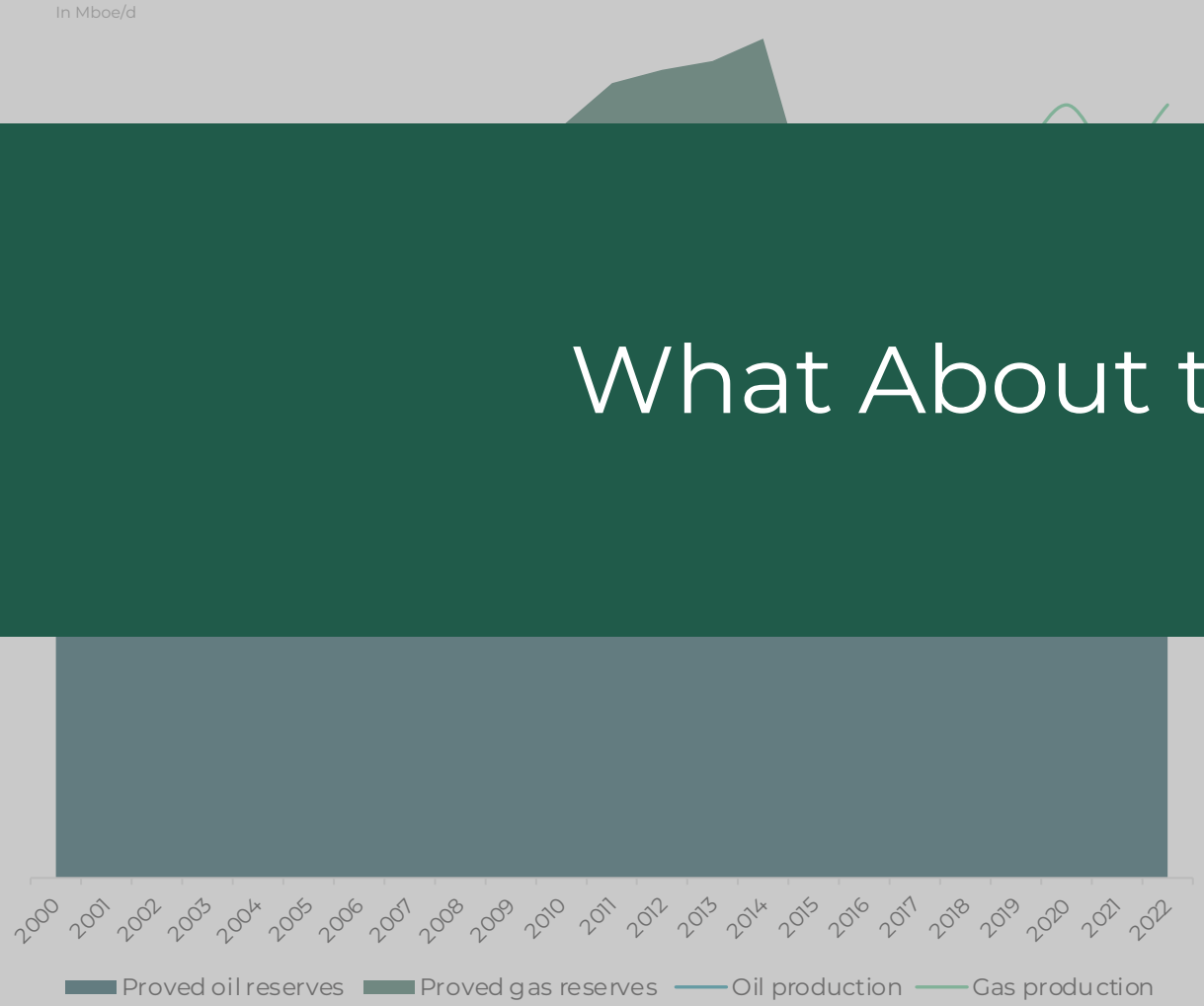
Source: IEPE, RI 3R, Statista

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What About the Peers?

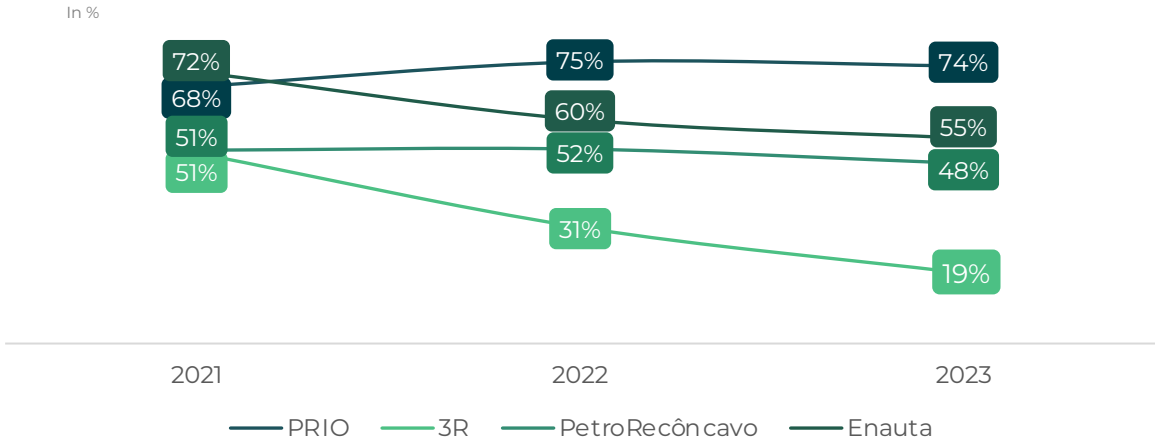
Source: IEPE, RI 3R, Statista

Unattractive Appearance

At first sight, 3R is definitively not the best company in the field. This happens because her **peers** are **older** and, therefore, **more consolidated**, and **3R is new**, its just **starting its extractions**. They are in **different stages**!

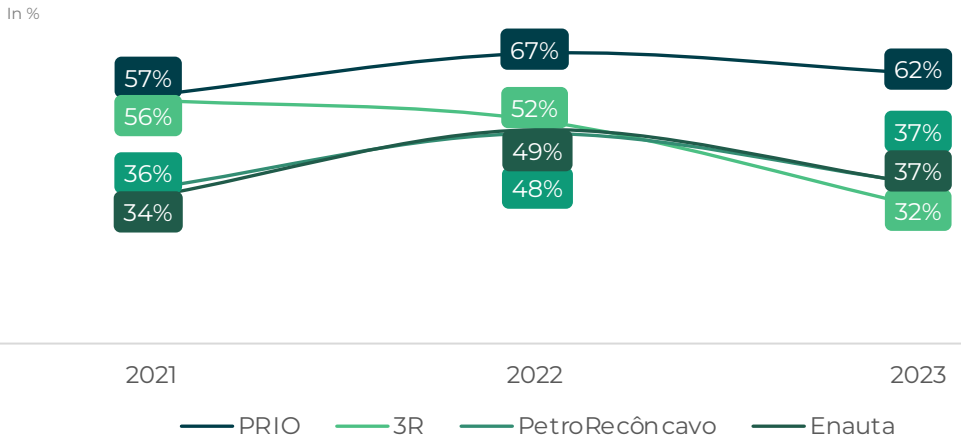
Ebitda Margin

When compared to peers, 3R has a **low** EBITDA Margin



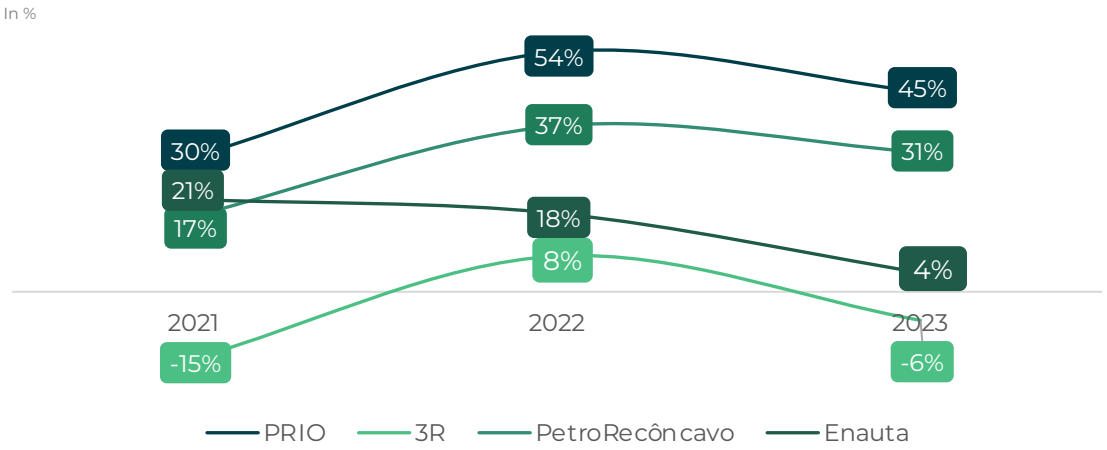
Gross Margin

When compared to peers, 3R has an **average** Gross Margin



Net Margin

When compared to peers, 3R has a **low** Net margin



Lifting Cost

When compared to peers, 3R has a **high** lifting cost



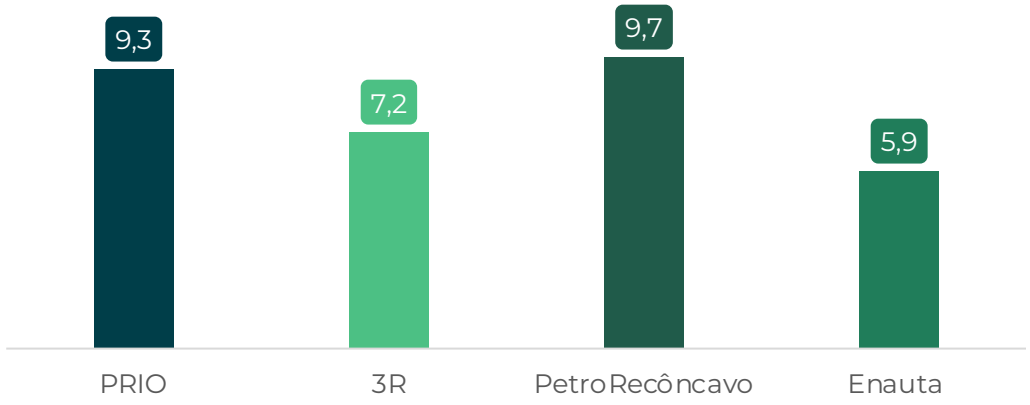
Source: Bloomberg, 3R RI

Promising Future

3R may not be the biggest peer of the sector, but it certainly is the **most growing one**. The low margins are due to the fact that 3R is in the **begging of their operations**, and the results are just about to come!

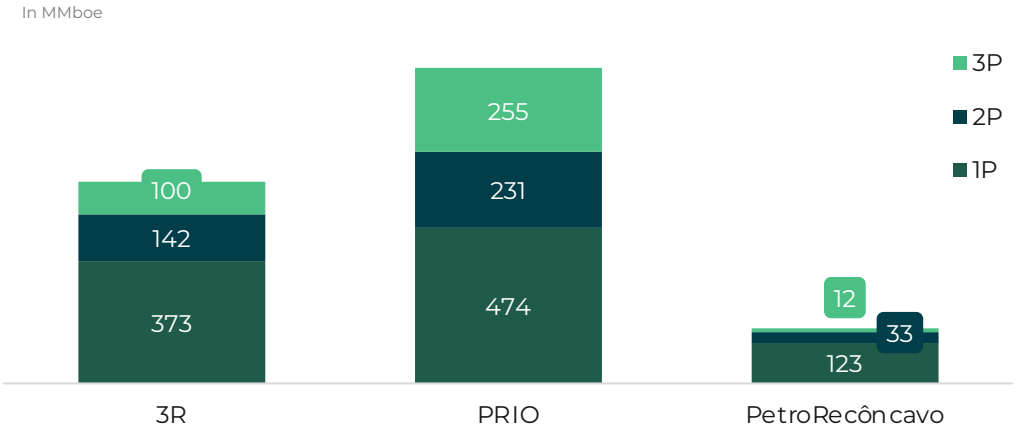
EV/2P of Junior Oils

With **more 2P** reserves than PetroRecôncavo, 3R faces a **growth opportunity**



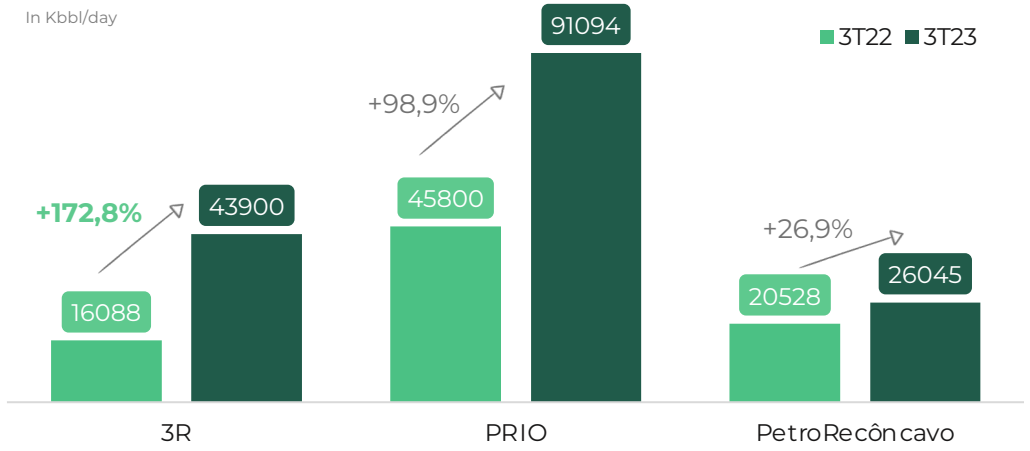
Certified 2P Reserves of each Junior Oil

Although PRIO has more proved reserves, 3R has a **higher proportion of 1P**.



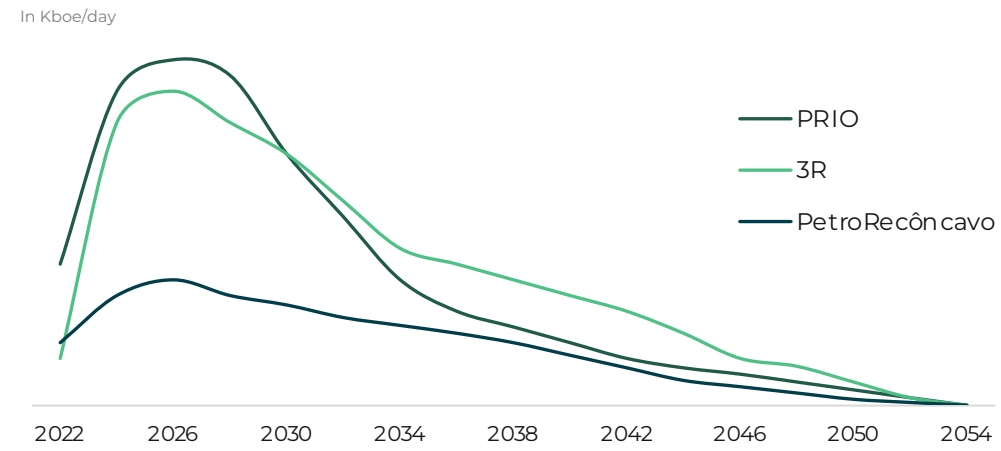
Evolution of the Peers Daily Production

Although PRIO produces more, 3R had the **biggest production growth**



Expected Daily Production Curve (2P)

It's likely that with this current assets, **3R's production surpasses PRIO's** by 2030



Source: 3R RI, PRIO RI, RECV RI, ITAU BBA

RRRP3 Investment Thesis

Stepping on a solid field
Oil&Gas sector is worldwidely consolidated and the demand for it is guaranteed.

Increase in Production
With the new management combined to a consolidated asset portfolio, 3R Petroleum is already showing growth in its production.

Significant upside, conservative premises
Even using extremely conservative premises, and without new Clusters, there is a significant Upside.

A Heavyweight Team

After all acquisitions, 3R's **executives were all changed** in 2023, with the objective of focusing on the **organic growth**. 3R's board is also really strong and efficient, considering that **83% of their salaries are based on shares**.

New executives, new decisions

All executives were **changed,** being now more **capable** and **efficient**



CEO: Matheus Dias

promon engenharia | Starboard ASSET | Hidrovias do Brasil



CFO: Rodrigo Pizarro

promon engenharia | Starboard ASSET



E&P: Maurício Diniz

BR

ShareHolding Composition

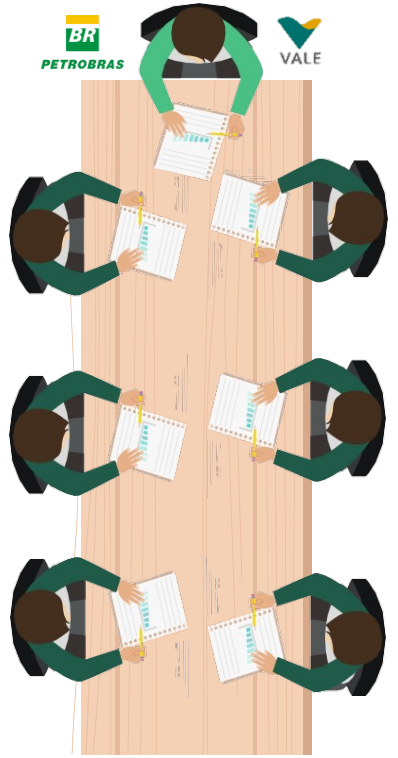
Besides of other shareholders, **Gerval** and **BTG** are the most representative



An Even More Experienced Board

3R counts with a **diverse** and **efficient** board, with **perspectives** from all possible areas

Roberto Castello Branco
Board President



Richard Gerdau
GERDAU

Guilherme Ferreira
teorema CAPITAL

Paula Kovarsky
raízen

Harley Scardoelli
GERDAU

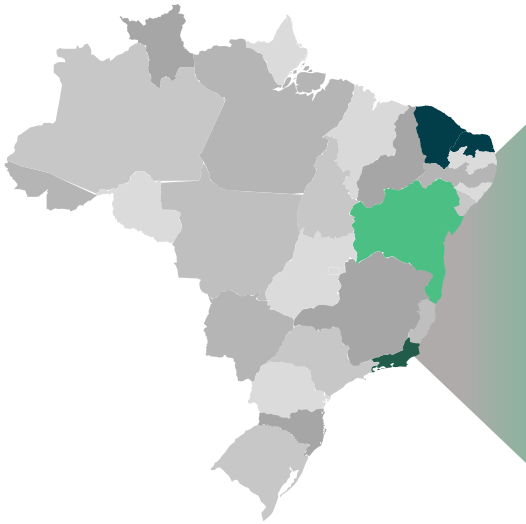
André Bartelle
Grendene
VULCABRAS

Carlos Alberto de Oliveira
BR PETROBRAS

Source: 3R RI

The M&A Powerhouse

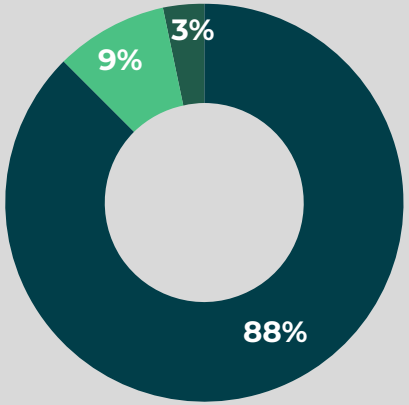
Known as the **M&A machine**, the Oil&Gas sector, mainly between the **juniors,** grew due to an enormous quantity of mergers and acquisitions. During **Petrobras disinvestment policy**, the juniors took advantage of it and grew.



3R Asset Portfolio

- **Potiguar Cluster**
- **Recôncavo Cluster**
- **OffShore Cluster**

Consolidated Reserves



Total:
142M Oil BBl

- Campos
- Potiguar
- Reconcavo

2010 — Ouro Preto Óleo e Gás SA foundation

2013 — Acquisition of **BAR-M-387** and **35% of Pescada**

2020 — IPO
Pescada and **Macau** acquisitions

2021 — **Rio Ventura** and **Areia Branca** acquisitions

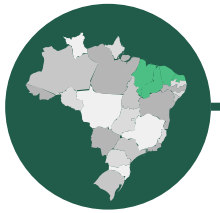
2022 — **Recôncavo**, **Peroá** and **Fazenda Belém** acquisitions

2023 — **Papa-Terra** and **Potiguar** acquisition

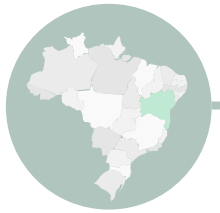
Source: 3R RI

Taking a Closer Look

The market has its doubts about the company's execution capacity, but with the **change in management** and **the excellent quality of its assets**, this perception **will change**.



Potiguar Cluster



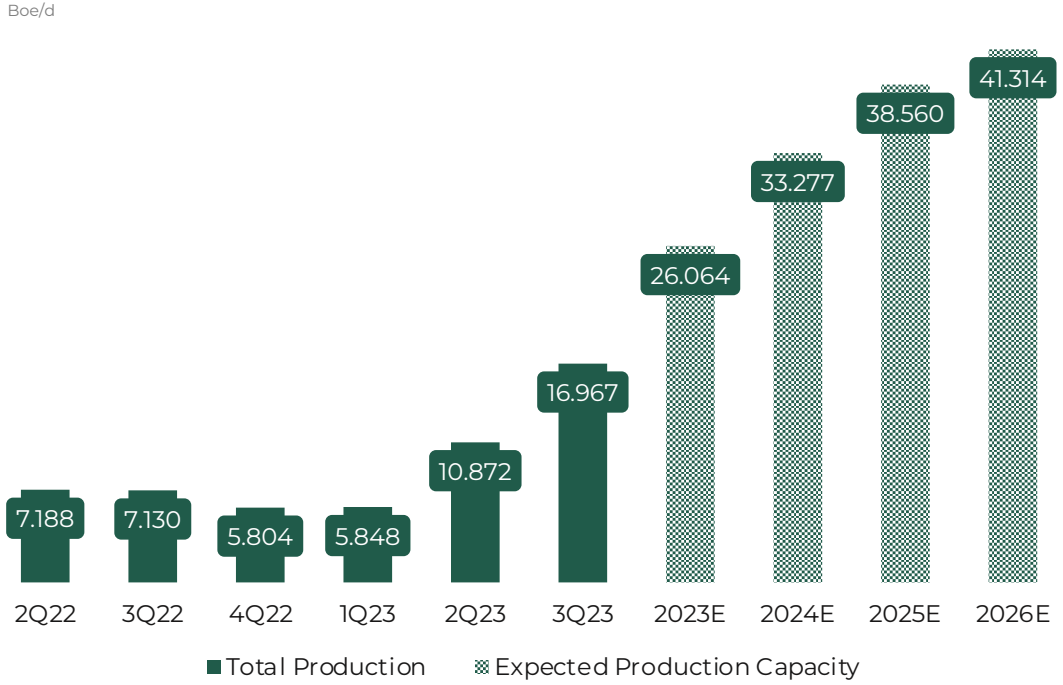
Recôncavo Cluster



3R Offshore Cluster

Total Production x Expected Production

The **promiscuous Potiguar Field** will keep showing its results!

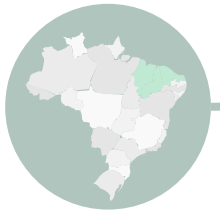


- **5 Fields**: Macau, Pescada, Areia Branca, Fazenda Belém and Potiguar
- 08/06/23 – **Incorporation** of the Potiguar Field.
- Two **new rigs** started campaigns in the Areia Branca Field.
- **Expansion** of the production infrastructure of the Fazenda Belém Field.

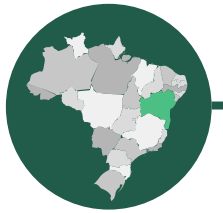
Source: 3R RI, ANP

Taking a Closer Look

The market has its doubts about the company's execution capacity, but with the **change in management** and **the excellent quality of its assets**, this perception **will change**.



Potiguar Cluster



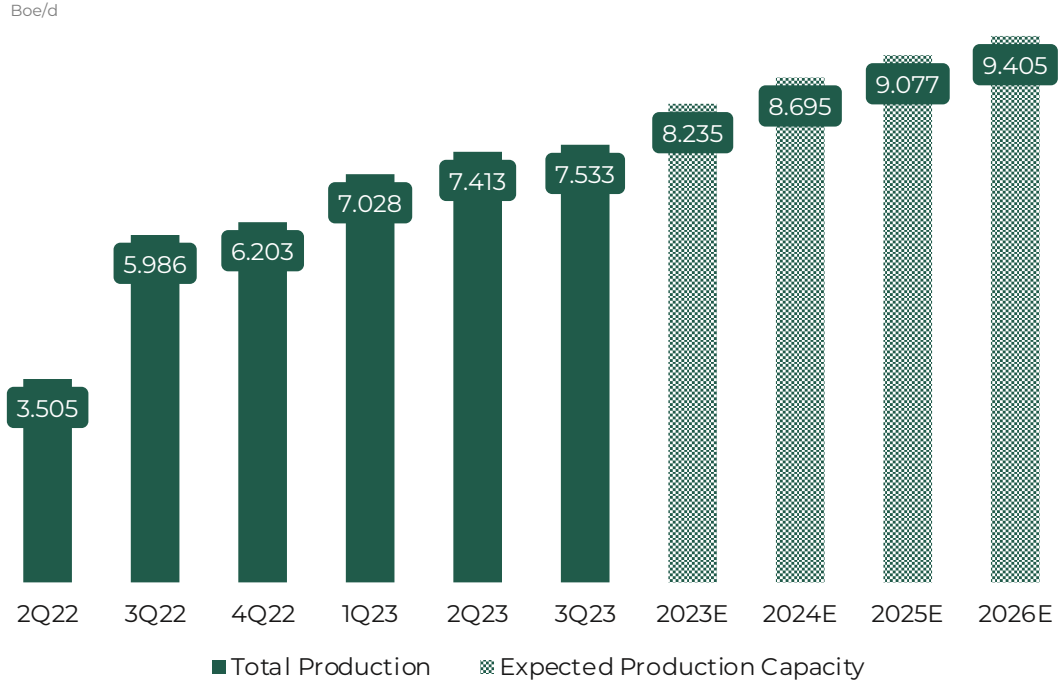
Recôncavo Cluster



3R Offshore Cluster

Total Production x Expected Production

The Recôncavo Cluster has been **showing growth in its production.**

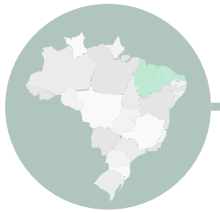


- **2 Fields**: Recôncavo and Rio Ventura
- **Interventions** with rigs, especially well reactivation, workover, and pulling performed on the assets. 64% of produced **gas is being reinjected**!
- **Connection** of reactivated and drilled wells to the production network.
- New **perforations campaign** starting in 4T23

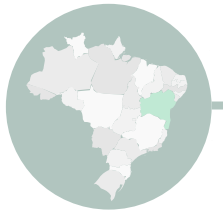
Source: 3R RI, ANP

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Potiguar Cluster



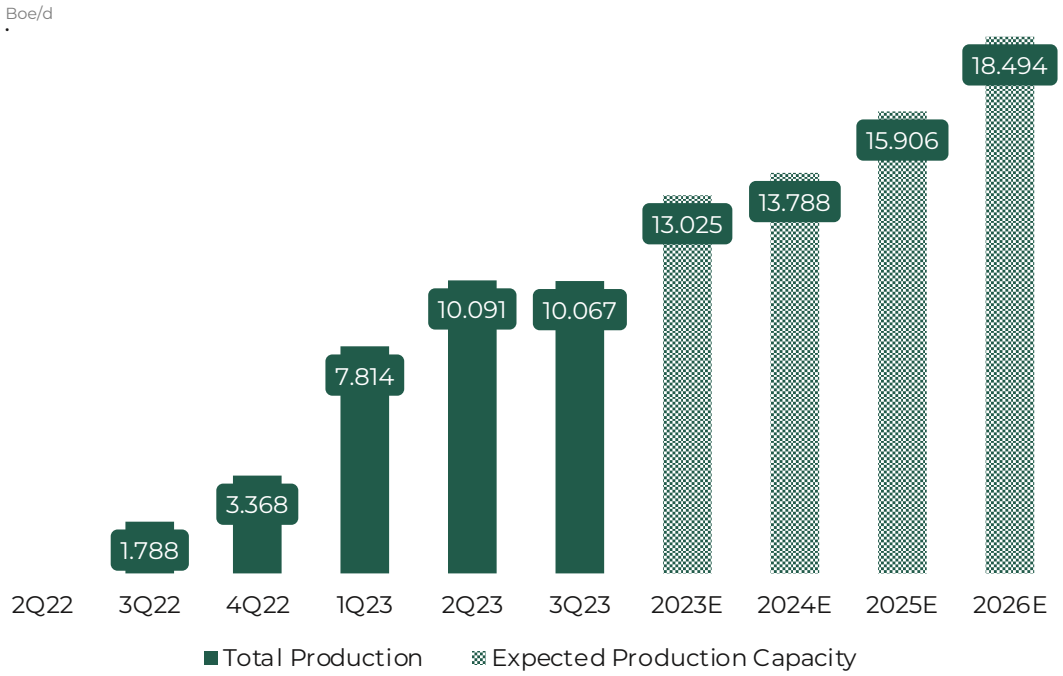
Recôncavo Cluster



3R Offshore Cluster

Total Production x Expected Production

The 3R Offshore Cluster has been **showing growth in its production**



2 Fields: Peroá and Papa-Terra

Corrective maintenance on the power generation, offloading, gas compression, and boiler systems, aiming to **increase operational efficiency**.

Connection of reactivated and drilled wells to the production network.

Higher **operational efficiency** recorded at the Papa Terra Field.

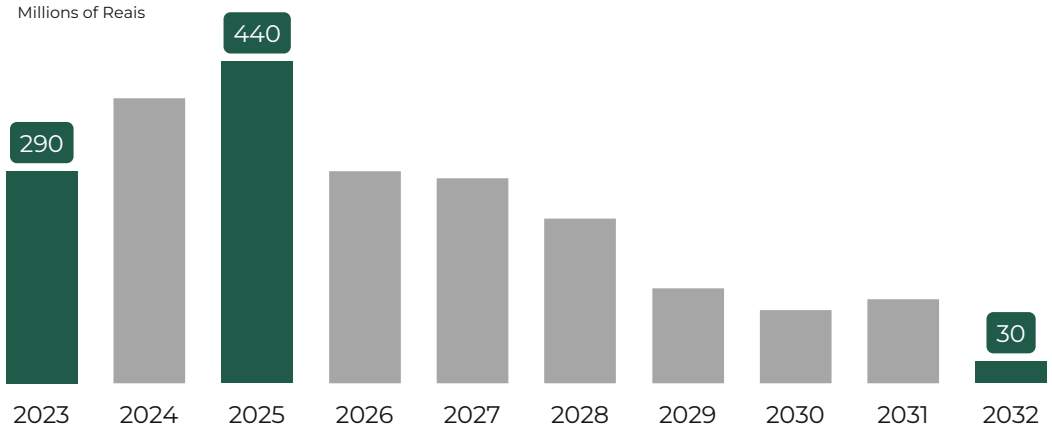
Source: 3R RI, ANP

Expensive, But Not For Long

Operational costs are expected to **peak in the short term**, with a forecasted maximum in 2025 in order to **invest** as much as possible in assets to **extract more and more oil.**

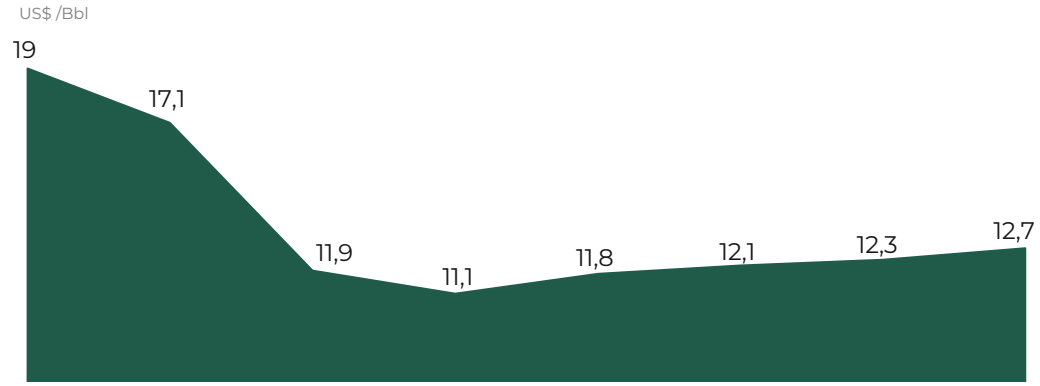
Projected Capex

The expectation is for **sharp growth** in the short term, to **finance the increase** in production.



Projected Opex

Operational cost will **decrease** considerably, having surpassed the phase of greatest investment.





Rodrigo Pizarro
CFO and Director of RI at 3R Petroleum

"Company's preparation this quarter for the start of a **more intensive activity and investment in the second half** [...] we intend to get to an order of magnitude of **a total volume of $280 million** to $290 million of CapEx [...] For **2024, business plan is about $400 million"**

Source: 3R RI

Keeping an Eye on the Debt

To finance the **M&As** and **increase in its CapEx & operating costs**, seeking to increase production, the company had to incur a **lot of debt** from various sources to consolidate an **extremely large cash position**.

Financing History

4Q20
R$ 690 Mn – IPO
Macau Cluster

2Q21
R$ 822 Mn - Follow On
Recôncavo Cluster

4Q21
R$ 2,1 Bn – Follow On
Potiguar Cluster

3Q22
R$ 900 Mn – Debenture
Potiguar Cluster

2Q23
R$ 5,1Bn - Debenture
R$ 900Mn - Follow On
R$ 2,5Bn – Loan

All Acquisitions + Ops.

Debt Overview

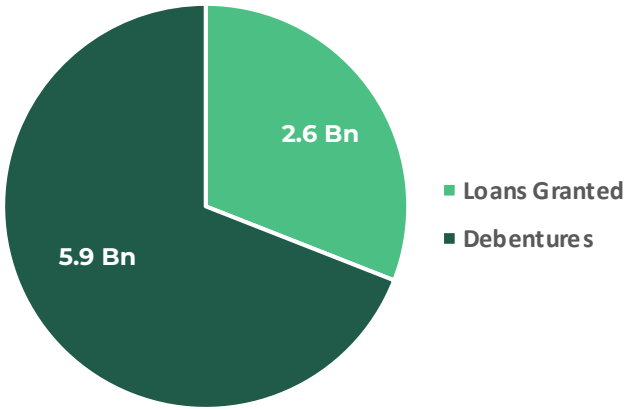
The company has a high volume of debt, mainly in the form of debentures, representing approximately 70%.

São Paulo, 15 de Outubro de 2023

New Debênture:

+ R$ 1Bn

IPCA + 8,41%



Debt Cost

Even with a high volume of debt, the interest coupons are relatively attractive.

Debênture R$ 900Mn: **CDI + 3%**

Debênture R$ 5,1 Bn: **10,16%**

Loan R$ 2,5 Bn: **SOFR + 6,25%**

Source: Morgan Stanley, 3R RI, Bloomberg

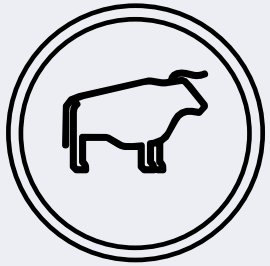
RRRP3 Investment Thesis

Stepping on a solid field
Oil&Gas sector is worldwidely consolidated and the demand for it is guaranteed.

Increase in Production
With the new management combined to a consolidated asset portfolio, 3R Petroleum is already showing growth in its production.

Significant upside, conservative premises
Even using extremely conservative premises, and without new Clusters, there is a significant Upside.

Scenarios



Bull Scenario

- **Lower** discount in total production
- **Higher brent** price premise
- **Higher** USD/BRL rate



Base Scenario

- **Standard** discount in total production
- **Standart brent** price premise
- **Standard** USD/BRL rate



Bear Scenario

- **Higher** discount in total production
- **Lower brent** price premise
- **Lower** USD/BRL rate

Scenarios

27 Scenarios

Production Discount (%)	Brent Discount (%)	Forex Discount (%)
Bear: **20%**	Bear: **20%**	Bear: **5%**
Base: **10%**	Base: -	Base: -
Bull: **5%**	Bull: **-10%**	Bull: **-4%**



Bull Scenario

- **Lower** discount in tota production
- **Higher brent** price premise
- **Higher** USD/BRL rate

- **Standard** USD/BRL rate



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- **Lower** USD/BRL rate

Chasing the Production

Before building up the revenue, some **premises** were taken to calculate the **projected production**, so that the valuation results turned out in the most **realistic** scenario.

Finally, Production.

After selecting the 2P reserve curve, apply a discount on the guidance and a depletion discount, then you have the forecast production.

Lastly, the Depletion

It involves a discount due to the pressure loss in the wells.

Discount on the Guidance

A standard market discount of about **10%** is applied to the estimate.

The 2P extraction curve

Chosen at the basis for production as it faithfully represents the extraction caoacity of each well



Source: RI 3R, Morgan Stanley

Chasing the Production

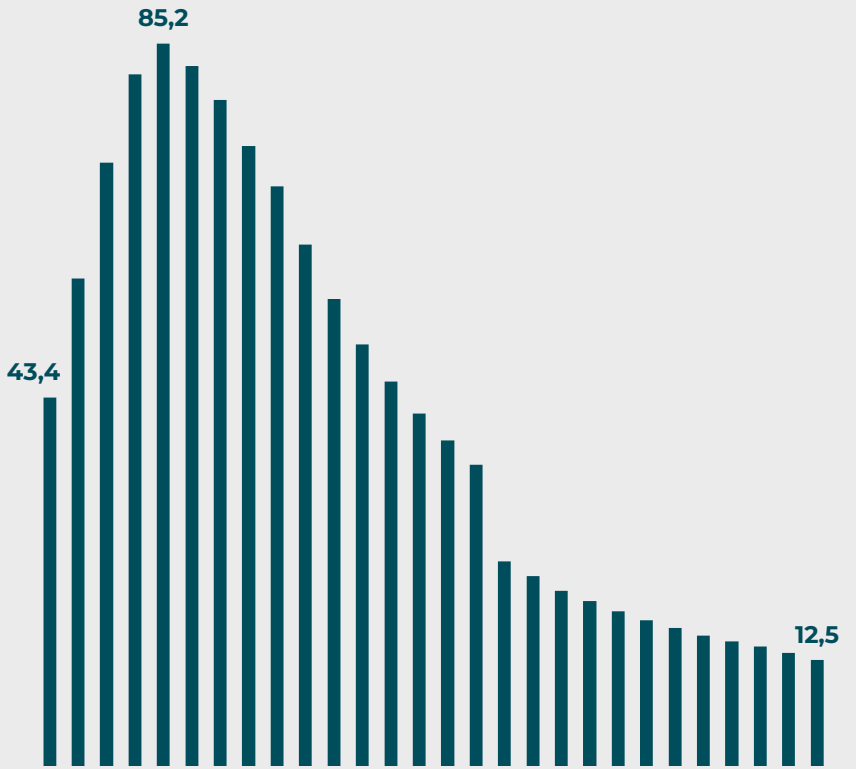
Before building up the revenue, some **premises** were taken to calculate the **projected production**, so that the valuation results turned out in the most **realistic** scenario.

The 2P extraction curve

2P Production Curve

Production forecasted by the 2P reserves certification

In Kbbl/d



Comes from the reserve certification report produced by an independent, unbiased company.

Market imposes a defined discount of 10% based on historical corrections on the guidance from the report.

A technical adjustment that reflects the pressure loss in the wells as extraction occurs.

Arriving at the Oil Production, which provides a conservative projection, also, leading to the gas projection.

Source: RI 3R, Morgan Stanley

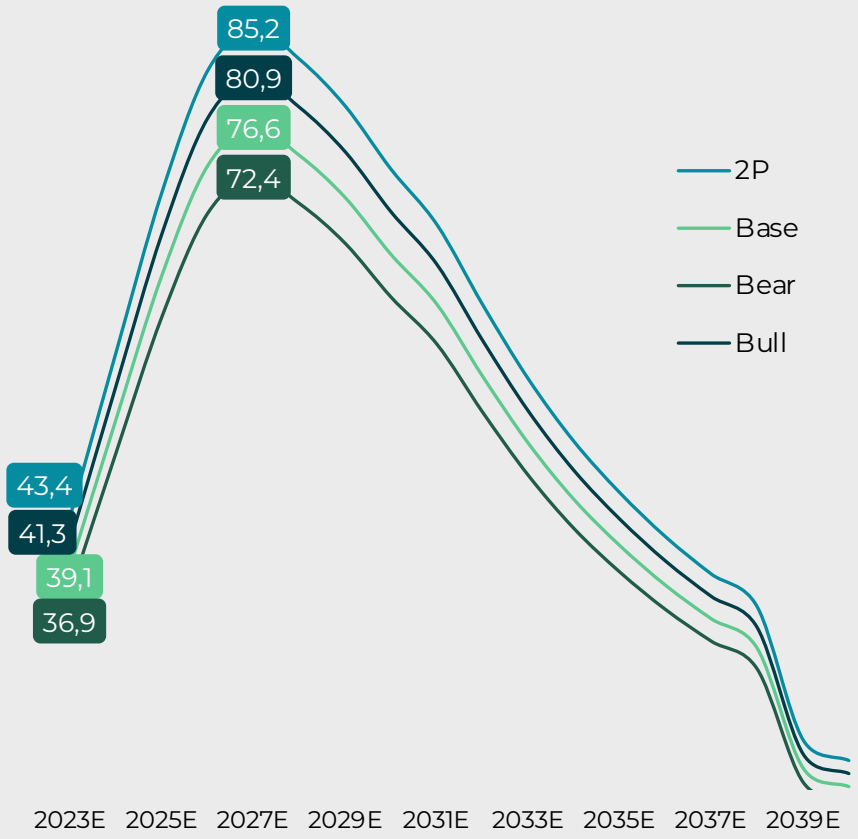
Chasing the Production

Before building up the revenue, some **premises** were taken to calculate the **projected production**, so that the valuation results turned out in the most **realistic** scenario.

Discount on the guidance

2P Curve with scenarios discount

For a realistic projection, a discount of 5%, 10% and 15% is applied.



Arriving at the Oil Production, which provides a conservative projection, also, leading to the gas projection.

A technical adjustment that reflects the pressure loss in the wells as extraction occurs.

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Source: RI 3R, Morgan Stanley

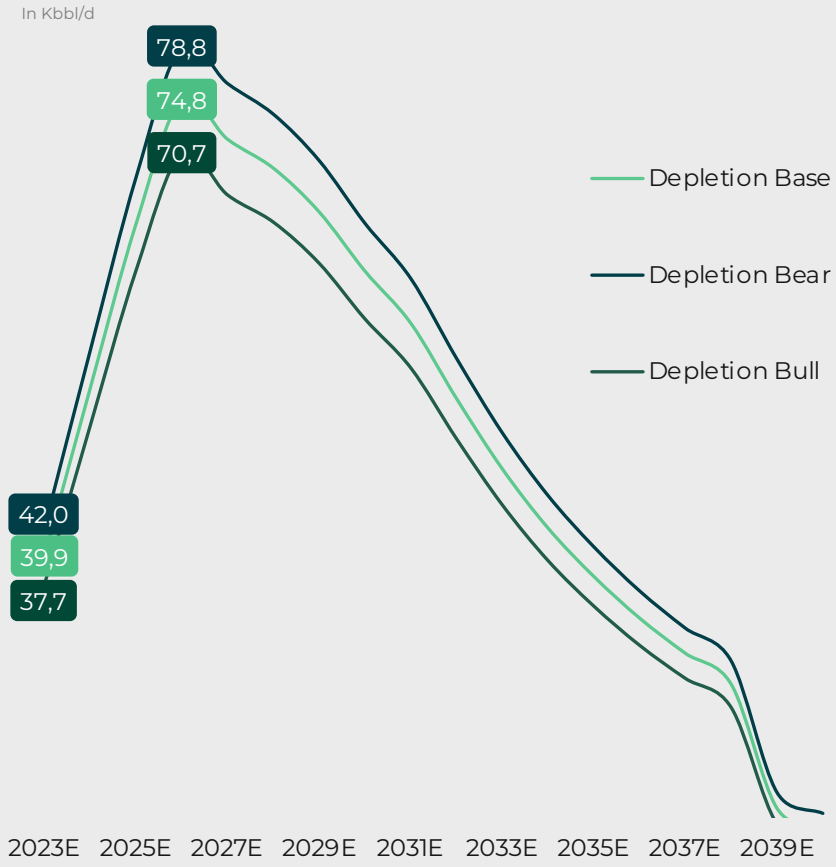
Chasing the Production

Before building up the revenue, some **premises** were taken to calculate the **projected production**, so that the valuation results turned out in the most **realistic** scenario.

Lastly, the Depletion

2P Curve with Depletion

Due to pressure loss, all clusters have another discount



Arriving at the Oil Production, which provides a conservative projection, also, leading to the gas projection.

A technical adjustment that reflects the pressure loss in the wells as extraction occurs.

Market imposes a defined discount of 10% based on historical corrections on the guidance from the report.

Comes from the reserve certification report produced by an independent, unbiased company.

Source: RI 3R, Morgan Stanley

Chasing the Production

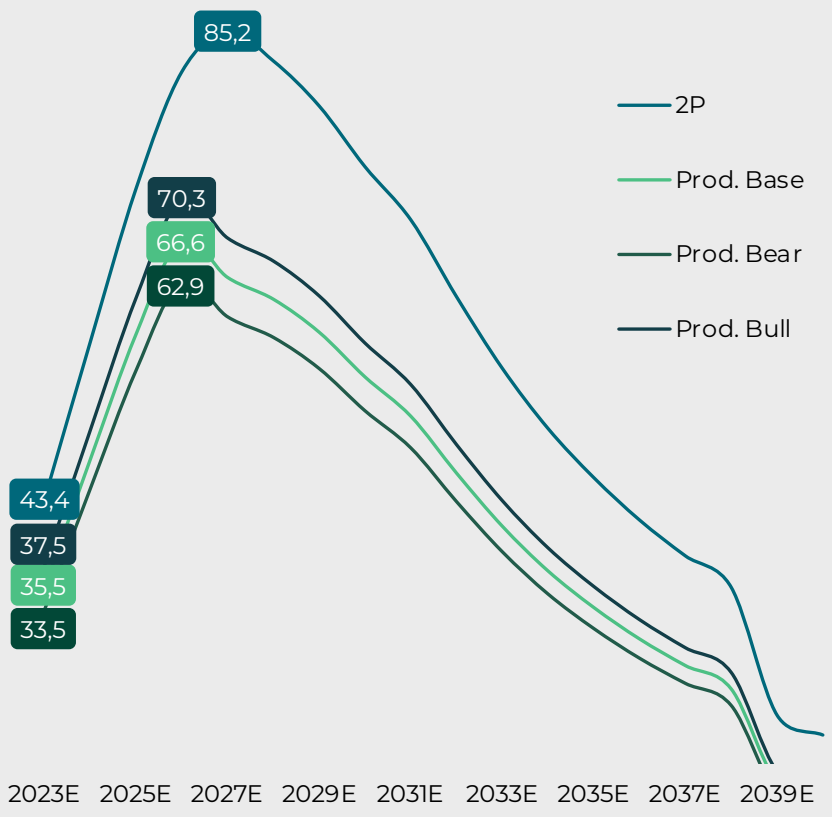
Before building up the revenue, some **premises** were taken to calculate the **projected production**, so that the valuation results turned out in the most **realistic** scenario.

Finally, Production.

Final Production Curves

After guidance and depletion discounts

In Kbbl/d



Arriving at the Oil Production, which provides a conservative projection, also, leading to the gas projection.

A technical adjustment that reflects the pressure loss in the wells as extraction occurs.

Market imposes a defined discount of 10% based on historical corrections on the guidance from the report.

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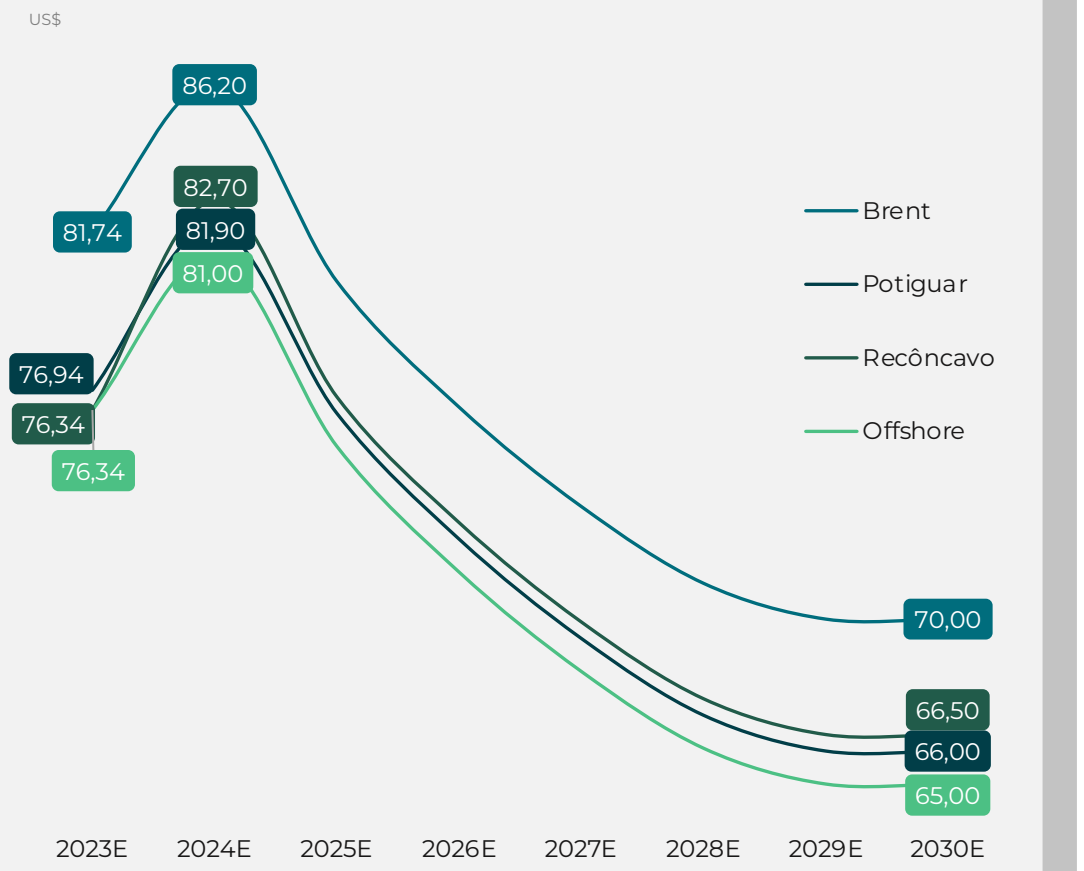
Source: RI 3R, Morgan Stanley

An Elaborately Simple Recipe

In principle, the revenue is nothing more than the **projected production of each cluster**, applied to its discounts, and **multiplied by the price of oil exercised in each field**.

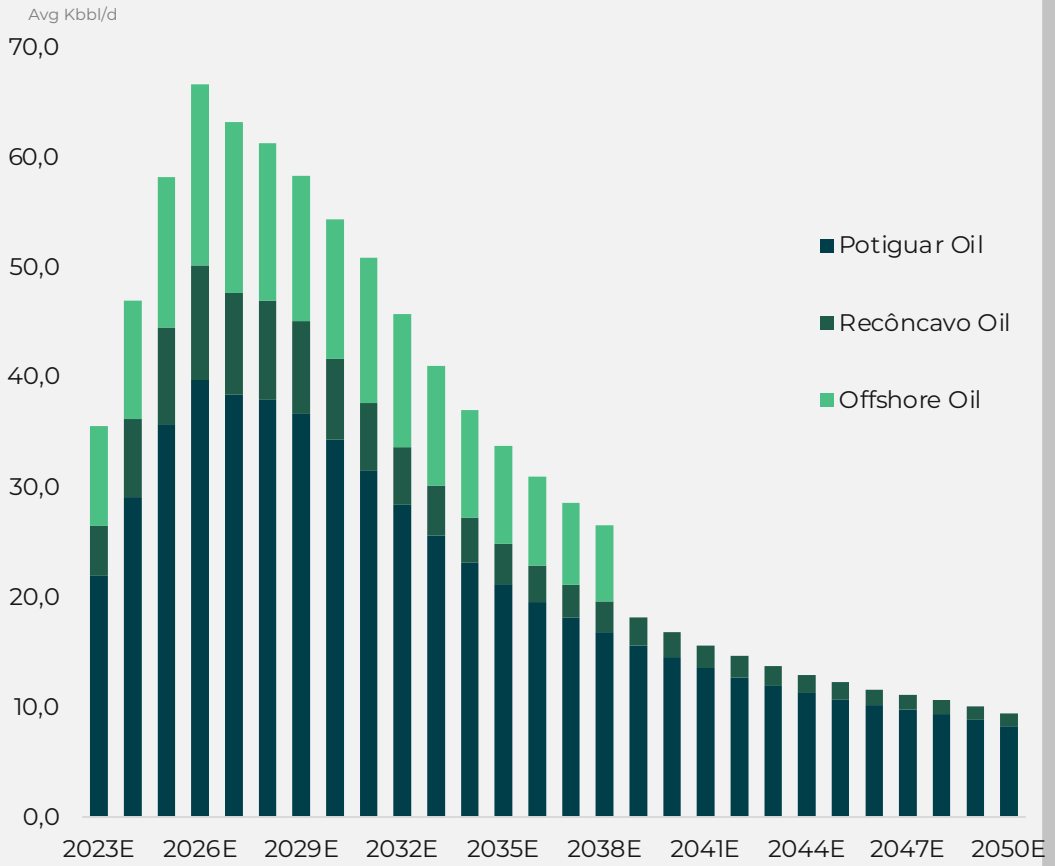
Clusters's Oil Price

Each complex sells their oil at a **different price** according to its **quality**.



Forecasted Production per Cluster

Estimated **useful life** of the clusters



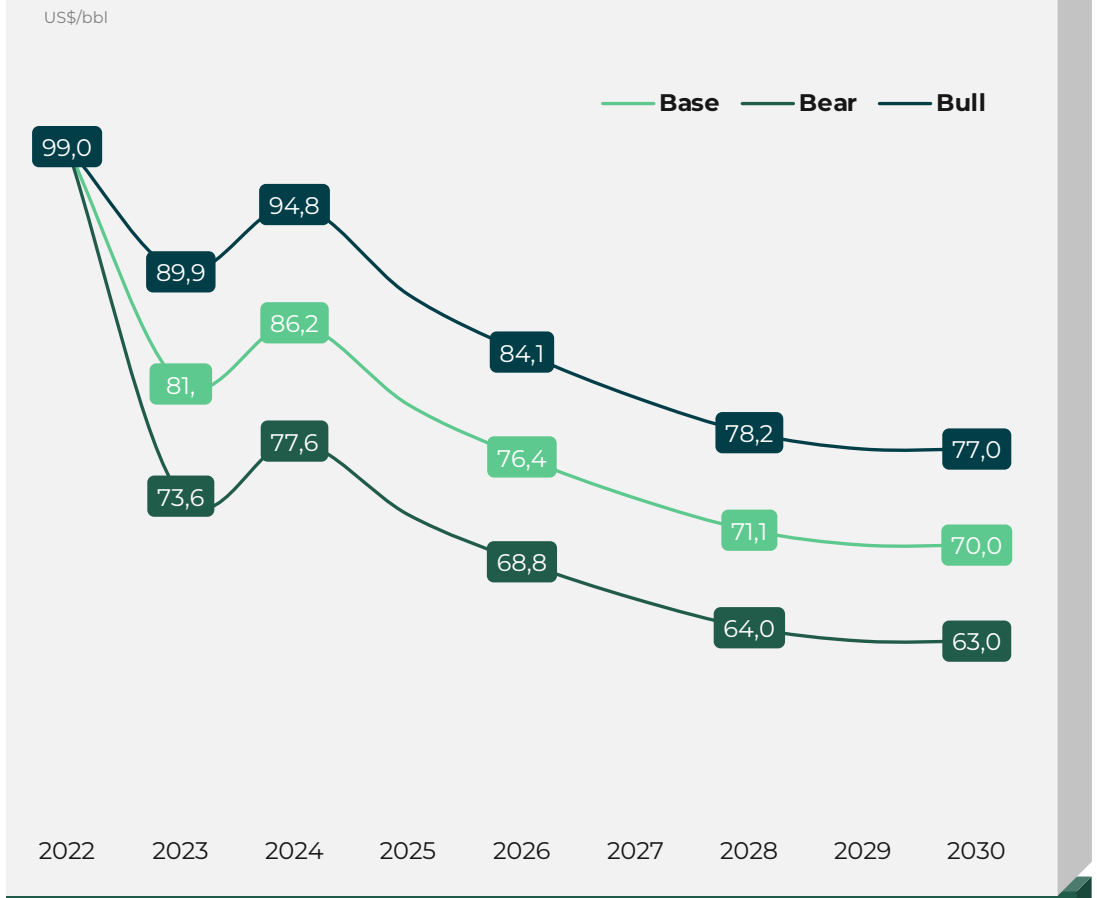
Source: 3R Valuation, Bloomberg

Macro Impacts on Revenue

Macroeconomics have a huge impact on 3R's revenue. It depends not only on the **oil and gas prices**, determined by global offer and demand, but also on the **exchange rate**, once their products are **sold in dollars**.

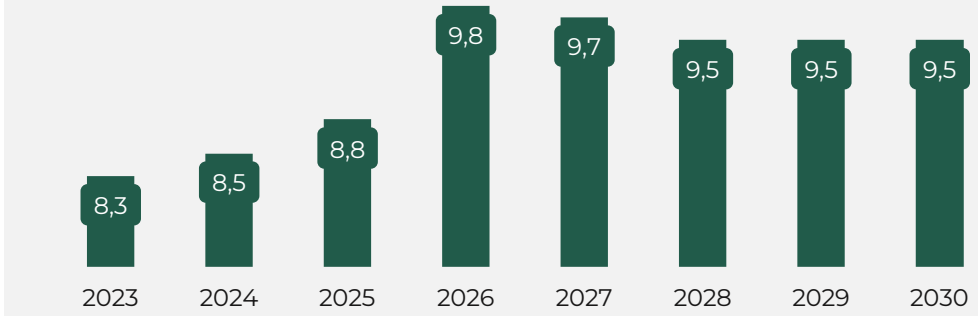
Brent price high impact

Variation of the **Brent projection** in the three scenarios.



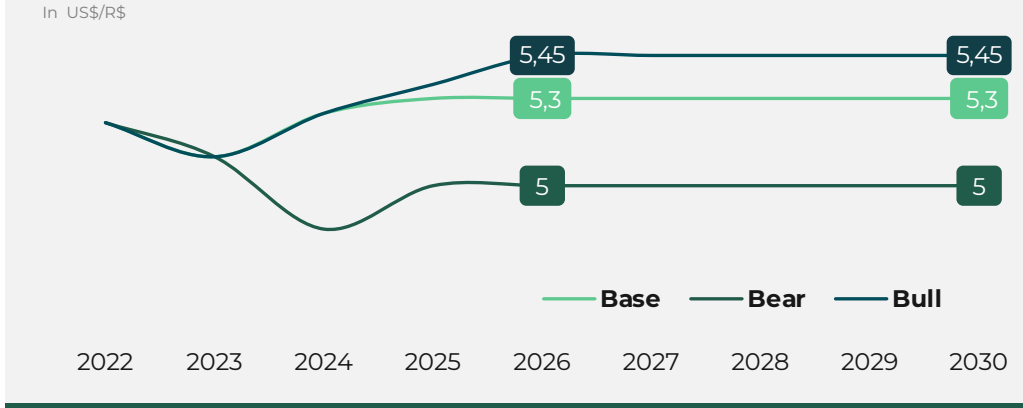
Gas price forecast

Our premise is that the price for Gas Boe price will **stabilize at around US$9,5**



Exchange rate for the following years

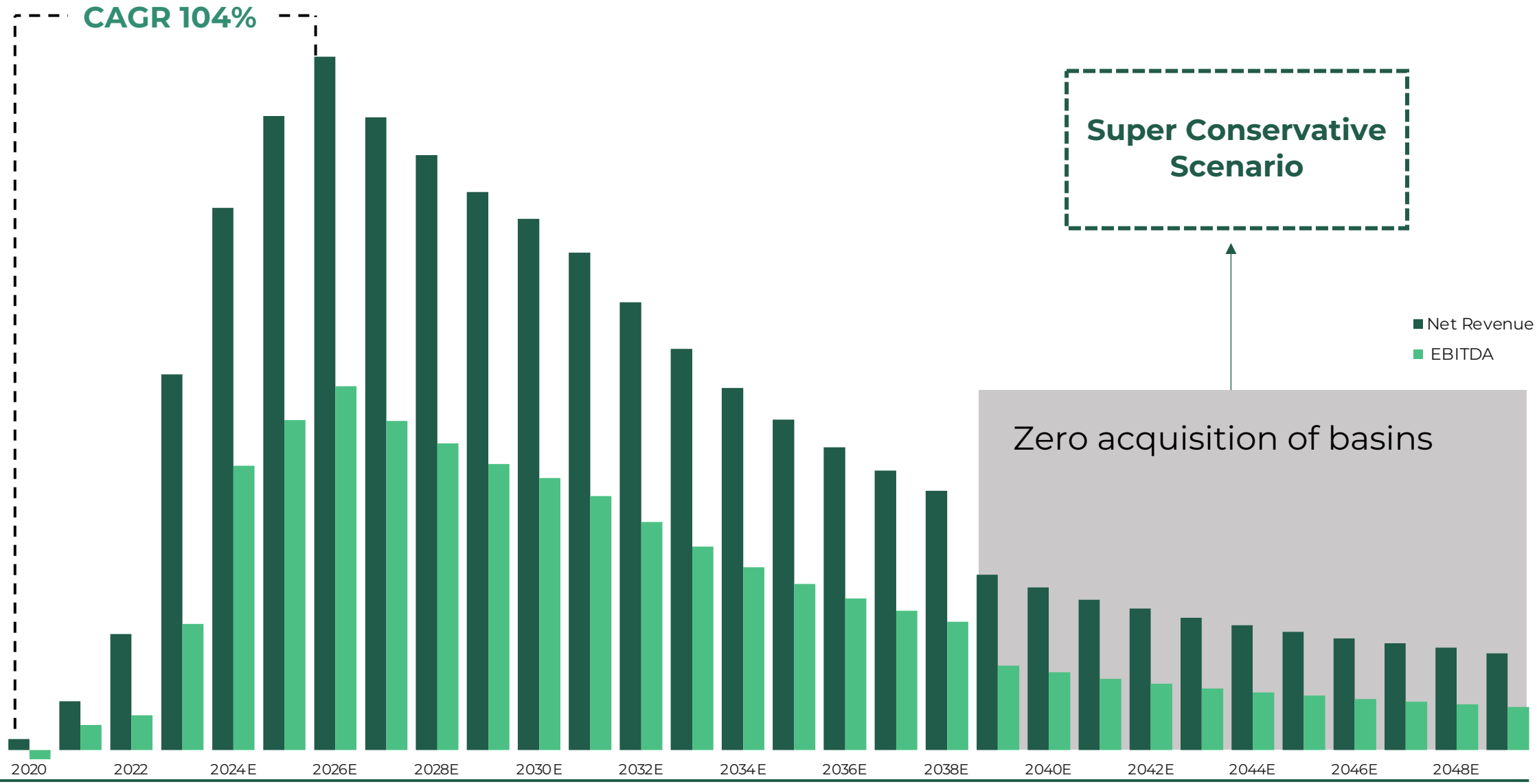
Our assumption is that the exchange rate will **stabilize at 5.3**



Source: Bloomberg, Morgan Stanley

Good Results Coming Along!

Based on 3R's **2P reserves**, the projection is that as their **volume of extraction grows**, which is already happening, their revenue also **increases**, and numbers don't lie!



Source: Valuation

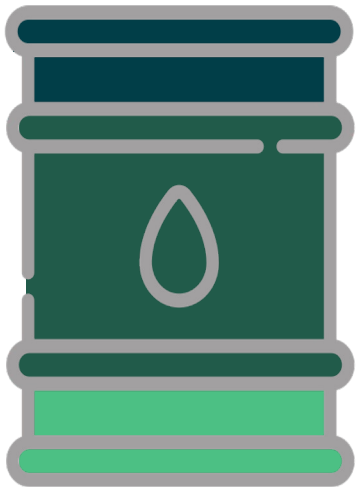
Analyzing Costs

In the end of the day, everything follows the **productions curve**. All margins and expenses are **higher in the extraction peak** and **decrease after 2030**.

"Costs and Capex are much **higher** in the **beginning of the operations**. After the wells are established, the expenses are **only for maintenance**."



Raphael Oliveira
RI 3R Petrolem



2026E

- **R$383,10** of Net Revenue per barrel sold
- 40% of COGS = R$153,20
- 27% SG&A = R$103,50
- 33% Net Income = **R$126,40** per barrel

Analyzing Margins

Gross Margin, EBIT Margin and EBITDA Margin



Source: 3R RI, Valuation

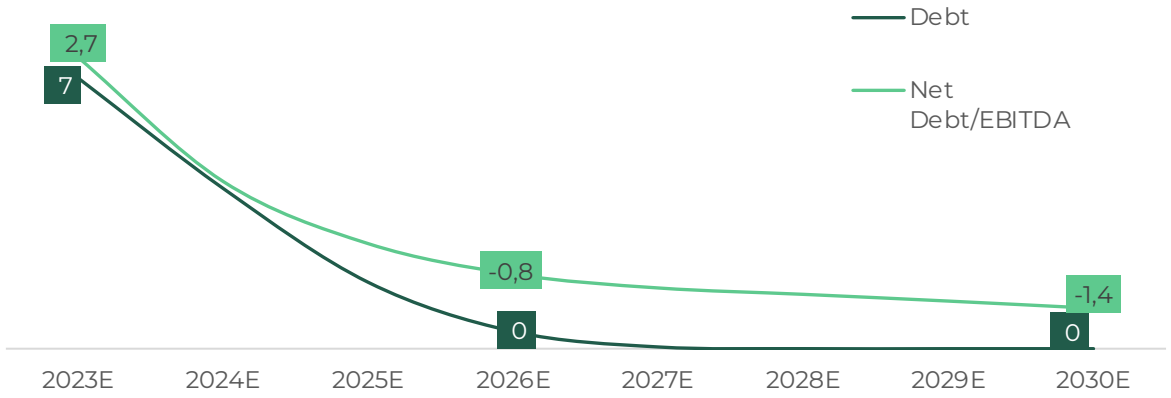
A Closer look at the Debt

Considered as one of the main attention points of the company, upon closer inspection, even though the **amount is high**, the **amortization schedule should settle the debts by 2028**. Also, if **cash and investments are excluded**, the amount is **much less concerning.**

Total Debt x Net Debt/EBITDA

The company's total debt should **de-leverage** as it follows the **amortization schedule** and overcomes the phase of greater capital requirement.

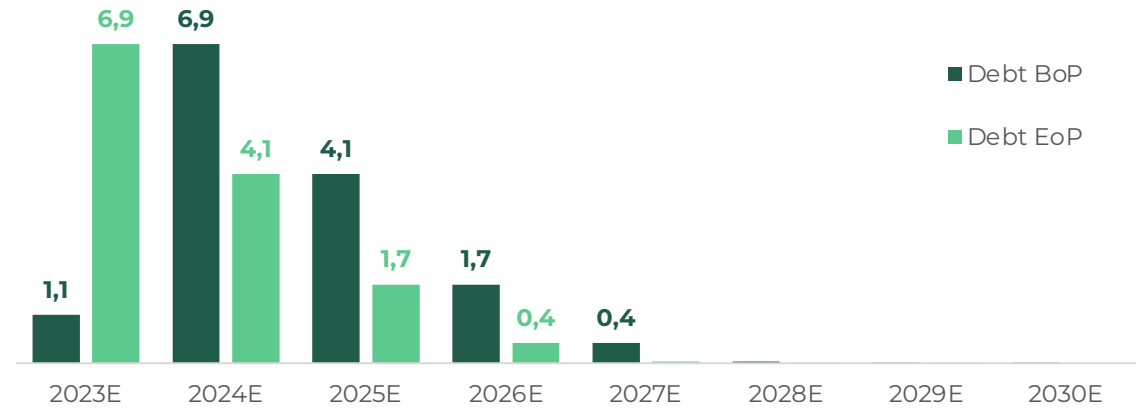
In Billions of R$



Debt Amortization

Debt Amortization Schedule id planned to **end in 2028**

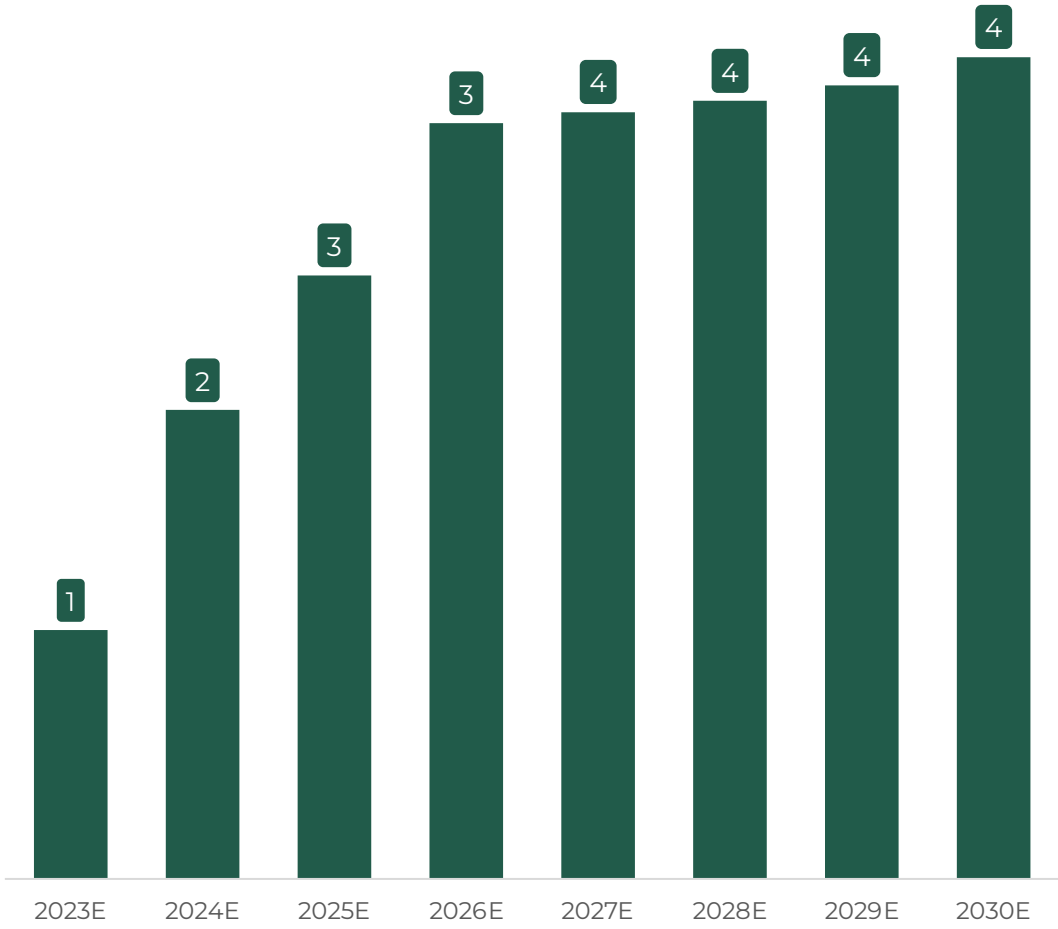
In Billions of R$



Cash Flow generated from Operations

The company should start to **consistently generate cash** in the coming years and **meet its debts**.

In Billions of R$



Source: Valuation

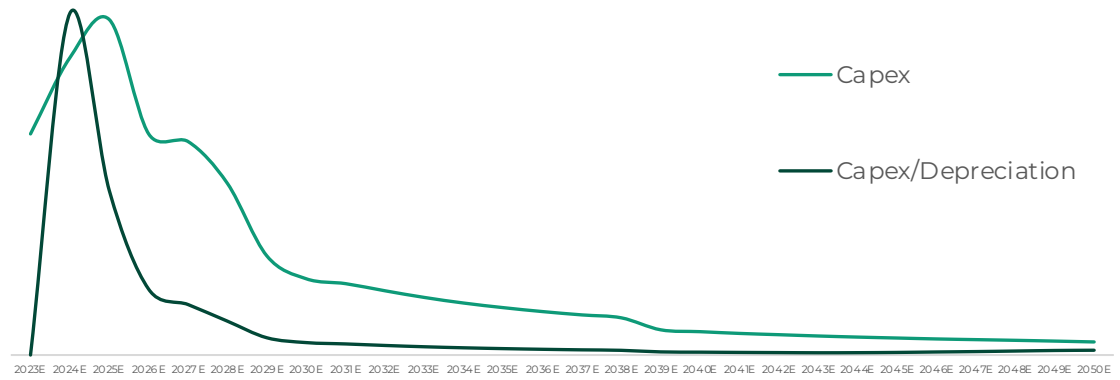
Roots to FCFF

Before diving in to the FCFF, its important to understand its roots. With this **variables established**, the projected FCFF has a **118% raise between 2023 and 2026**!

Capex/Depreciation x Depreciation

Evolution of capex and depreciation through time

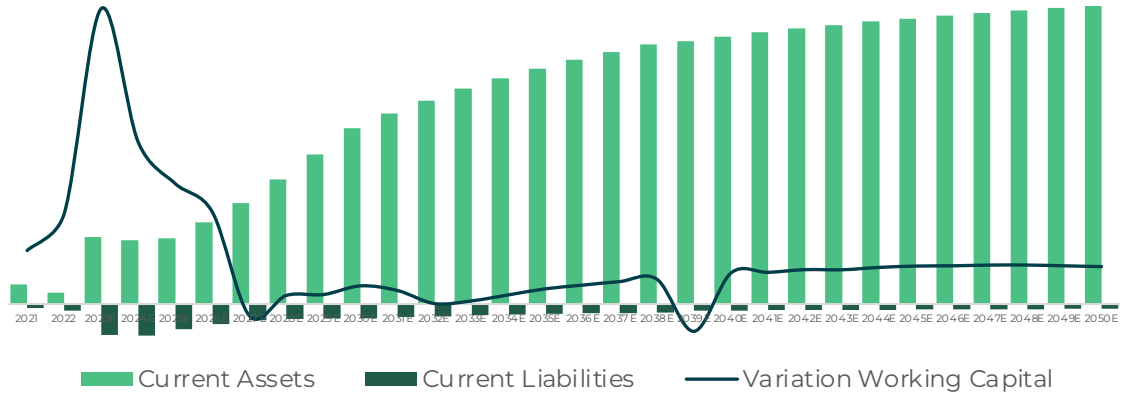
In Thousands of R$



Variation on Working Capital

Evolution of Working Capital through time

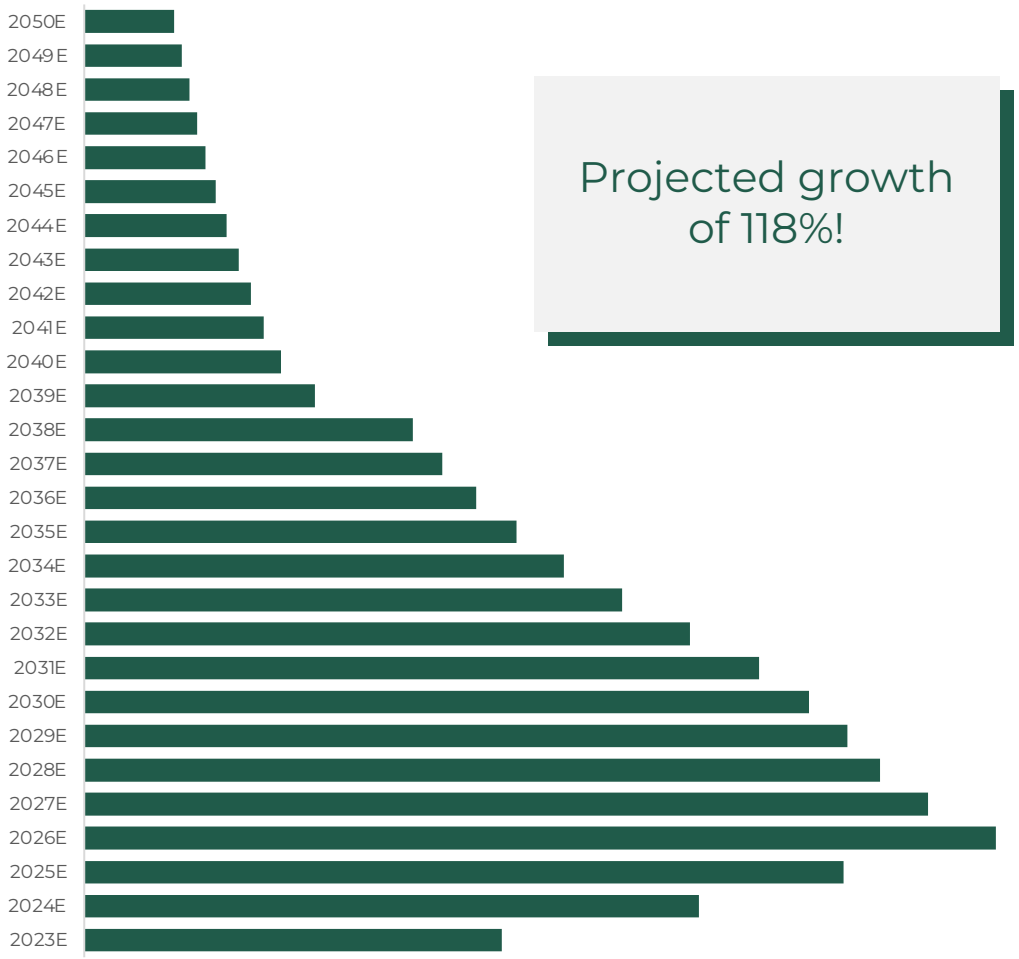
In Thousands of R$



Projected FCFF

FCFF forecasted with **118% of growth**

In Thousands of R$



Projected growth of 118%!

Source: Valuation

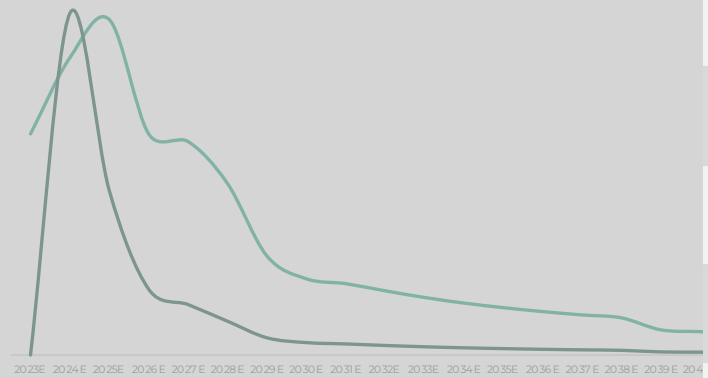
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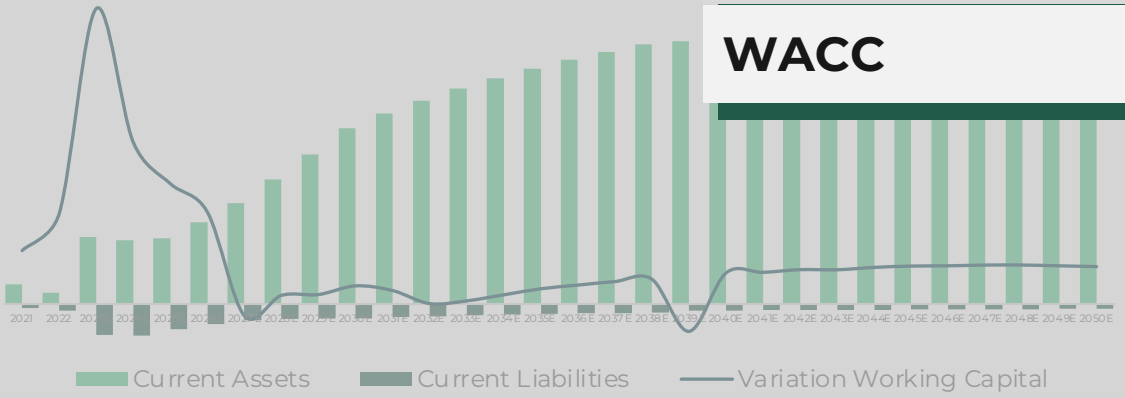
In Thousands of R$



Variation on Working Capital

Evolution of Working Capital through time

In Thousands of R$

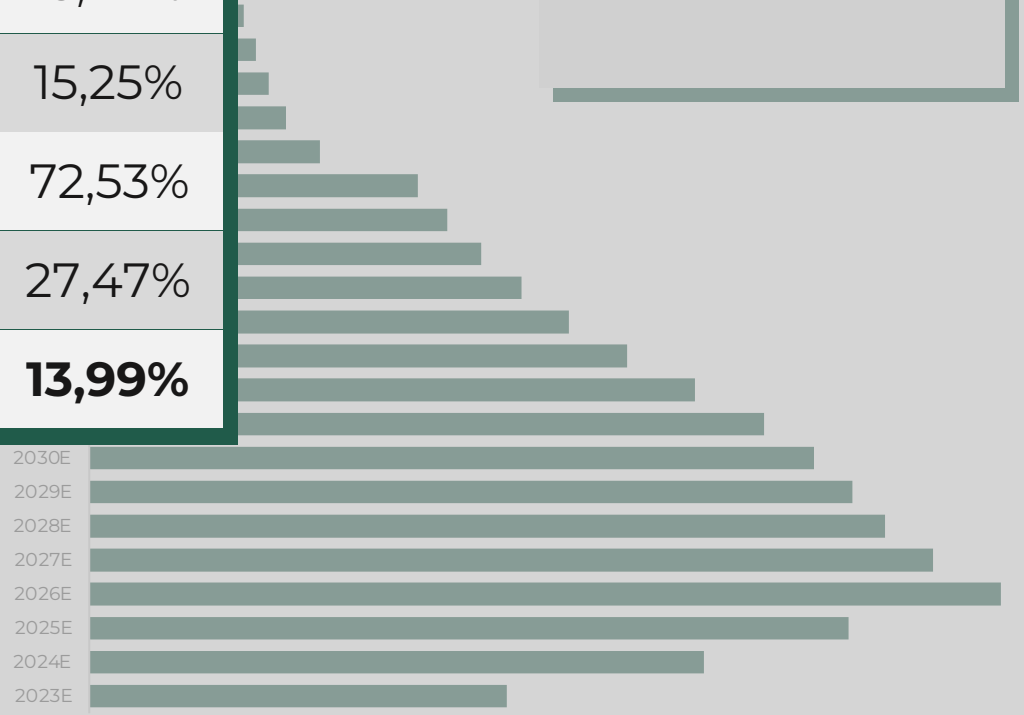


Ke US	15,04%
Ke BR	16,62%
Kd	13,24%
Tax	15,25%
E/(D+E)	72,53%
D/(D+E)	27,47%
WACC	**13,99%**

Projected FCFF

FCFF forecasted with **118% of growth**

In Thousands of R$

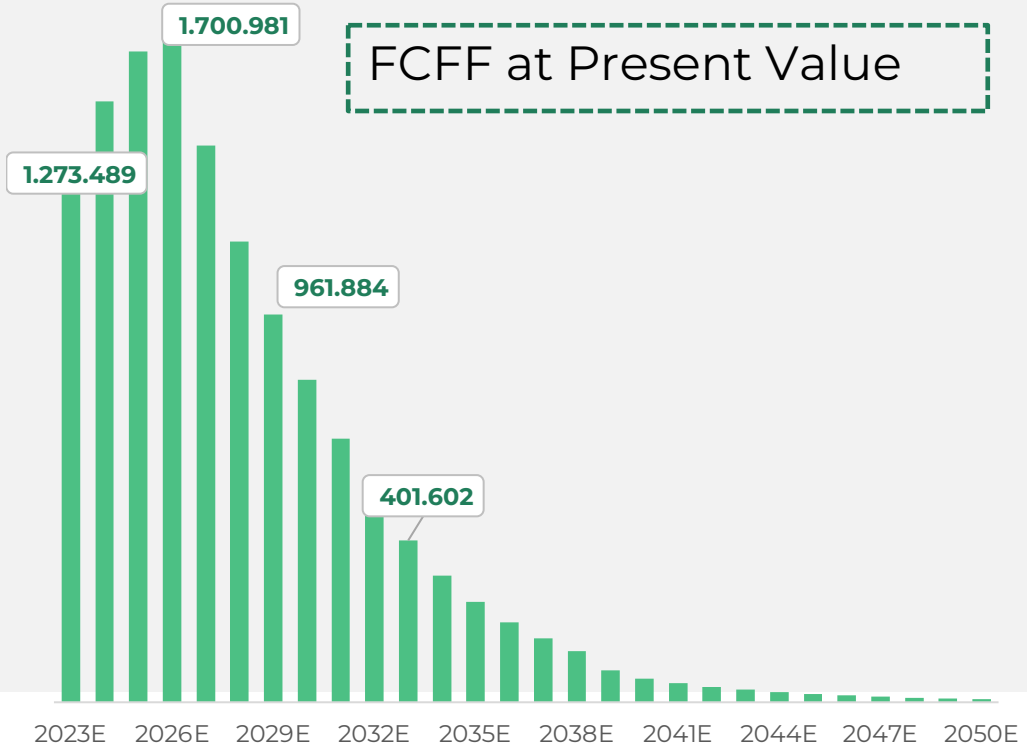


Projected growth of 118%!

Source: Valuation

Can’t Escape the Curve

With all the **revenue based on the production curve**, the **present value FCFF also follows this trend**. With the main values concentrating before 2040.



DCF Valuation

Enterprise Value

TEV: R$ 13.3 Bi

(-) Net Debt: R$ 251,000

Equity Value: R$ 13 Bi

Target Price: R$ 55,6

Upside: + 68,7%

Source: Valuation

Can’t Escape the Curve

With all the **revenue based on the production curve**, the **present value FCFF also follows this trend**. With the main values concentrating before 2040.

3R Scenario Distribution



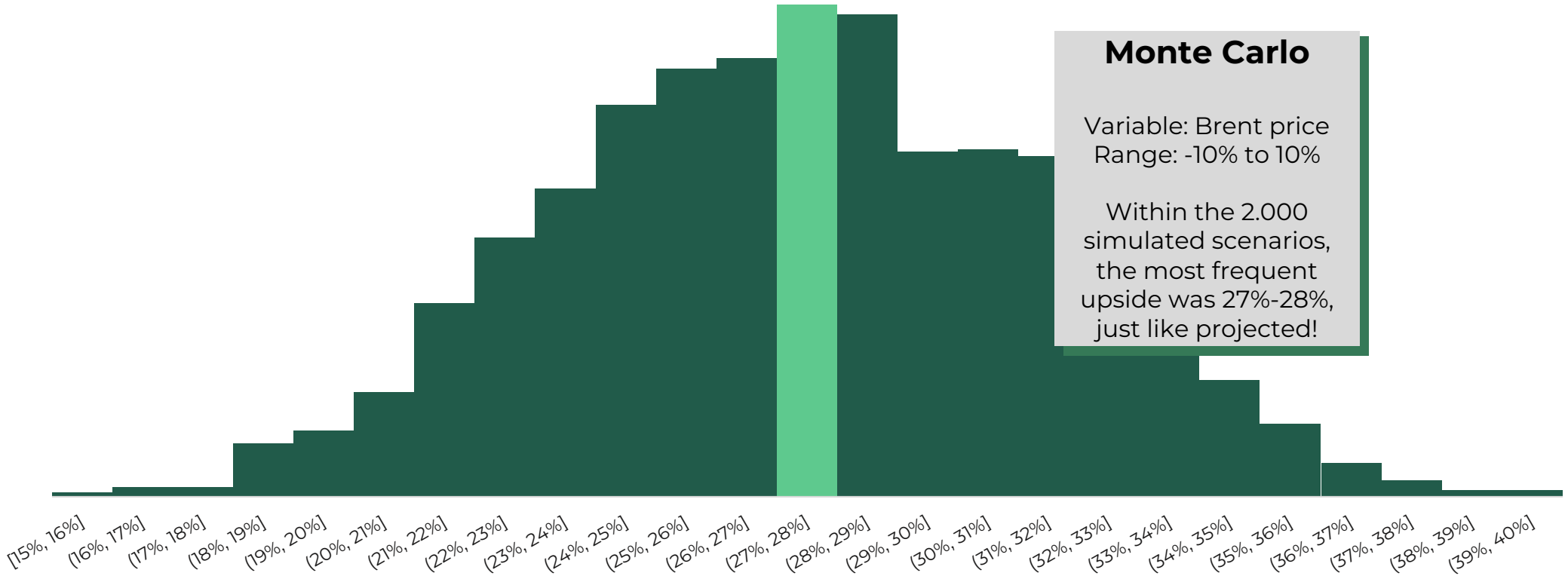
Production	Brent	Forex	Upside
Bull	Bull	Bull	111%
Bull	Bull	Base	107%
Base	Bull	Bull	96%
Bull	Bull	bear	93%
Base	Bull	Base	91%
Bull	Base	Bull	87%
Bull	Base	Base	83%
Base	Bull	bear	79%
Base	Base	Bull	73%
Bull	Base	bear	71%
Base	Base	Base	69%
bear	Bull	Bull	65%
bear	Bull	Base	61%
Base	Base	bear	57%
bear	Bull	bear	50%
bear	Base	Bull	44%
bear	Base	Base	41%
Bull	bear	Bull	39%
Bull	bear	Base	35%
bear	Base	bear	31%
Base	bear	Bull	27%
Bull	bear	bear	26%
Base	bear	Base	24%
Base	bear	bear	15%
bear	bear	Bull	4%
bear	bear	Base	1%
bear	bear	bear	-7%

Source: Valuation

Checking One More Time

The 27,66% upside was **proved once** again through the Monte Carlo valuation method. Using a **variation on the brent price** and analyzing the **upside variation**, this one was the **most frequent**.

Upside: 27% to 28%



Monte Carlo

Variable: Brent price
Range: -10% to 10%

Within the 2.000 simulated scenarios, the most frequent upside was 27%-28%, just like projected!

Exit Multiple Optics

Another metric to determine the value of a company is by **the comparison of the multiples** of the players of the sector, once again showing 3R's advantages in relation to its peers

Multiple Comparison

3R has the highest multiple, but it should **stabilize to the average of peers over the years**

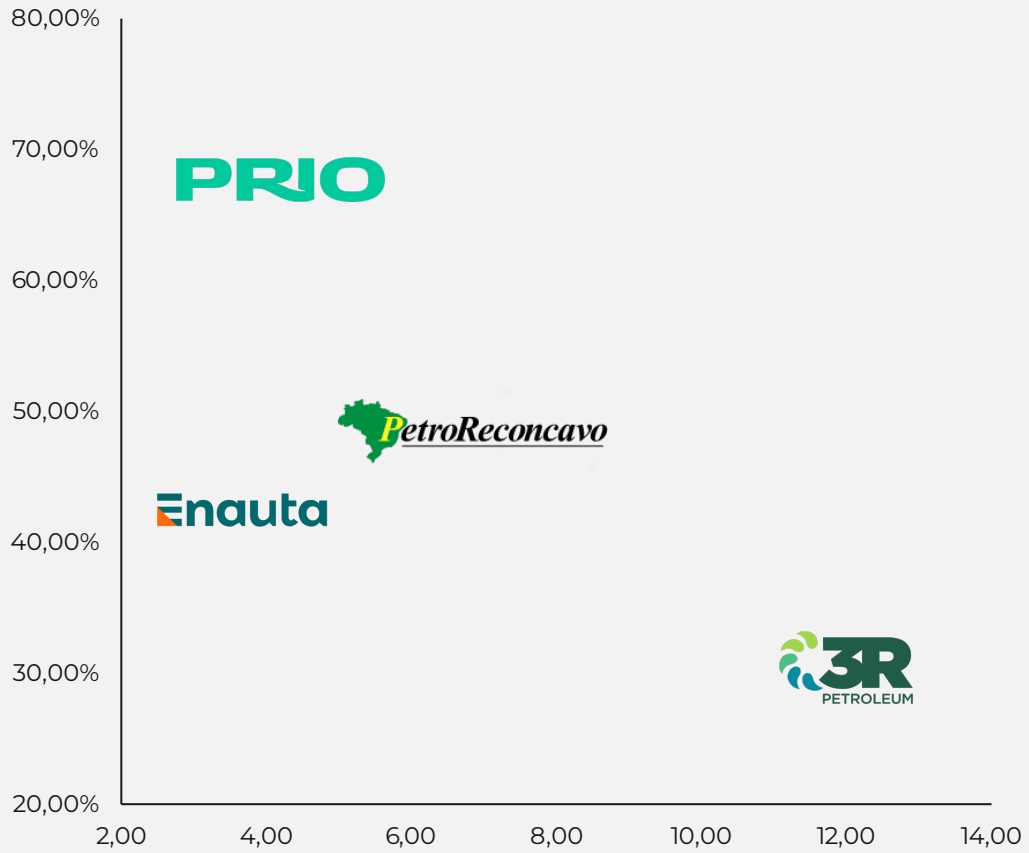
Peers	EBITDA Margin LTM	EV/EBITDA
3R PETROLEUM	30,45%	12,09x
Enauta	42,01%	3,67x
PetroReconcavo	48,22%	6,73x
PRIO	68,31%	4,11x

Mean EV/EBITDA of peers: **5,42** → **Exit Multiple**

IRR: 35,60%

EBITDA Margin x EV/EBITDA

The **margin of 3R is lower** than its competitors, hence the higher multiple.



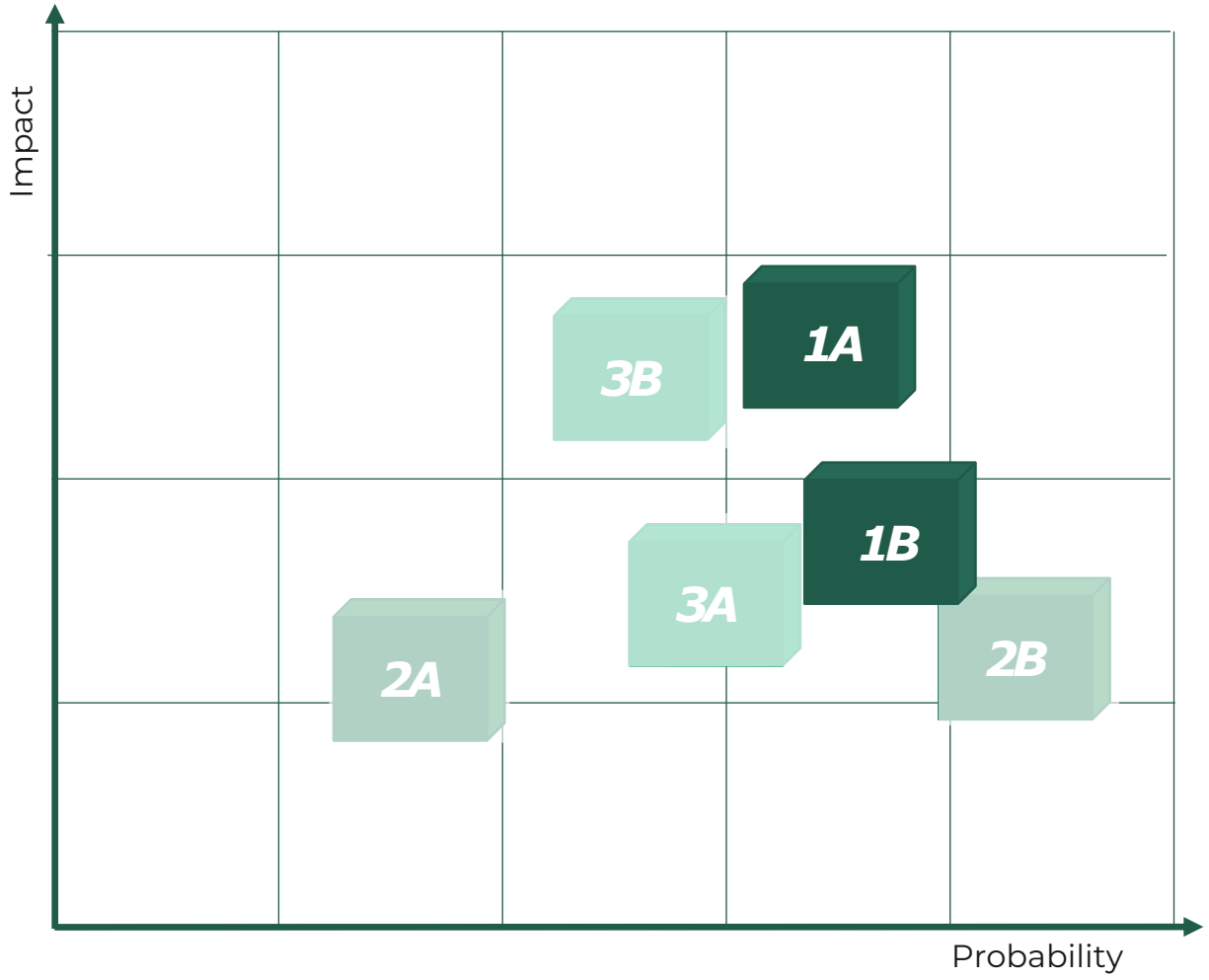
Source: Capital IQ

Risk Assesment

The matrix produces an intriguing analysis between the **probability** and the **impact** of a certain risk.

Risk Matrix

Highlighting the Brent price, which **is highly volatile** and has **a significant impact** on oil and gas companies



1. Macroeconomic Risks

A. Fluctuation in the Brent Price

B. Currency Fluctuation

2. Industry Risks

A. Increase in taxes and royalties

B. Energy Transition

3. Specific Risks

A. High debt

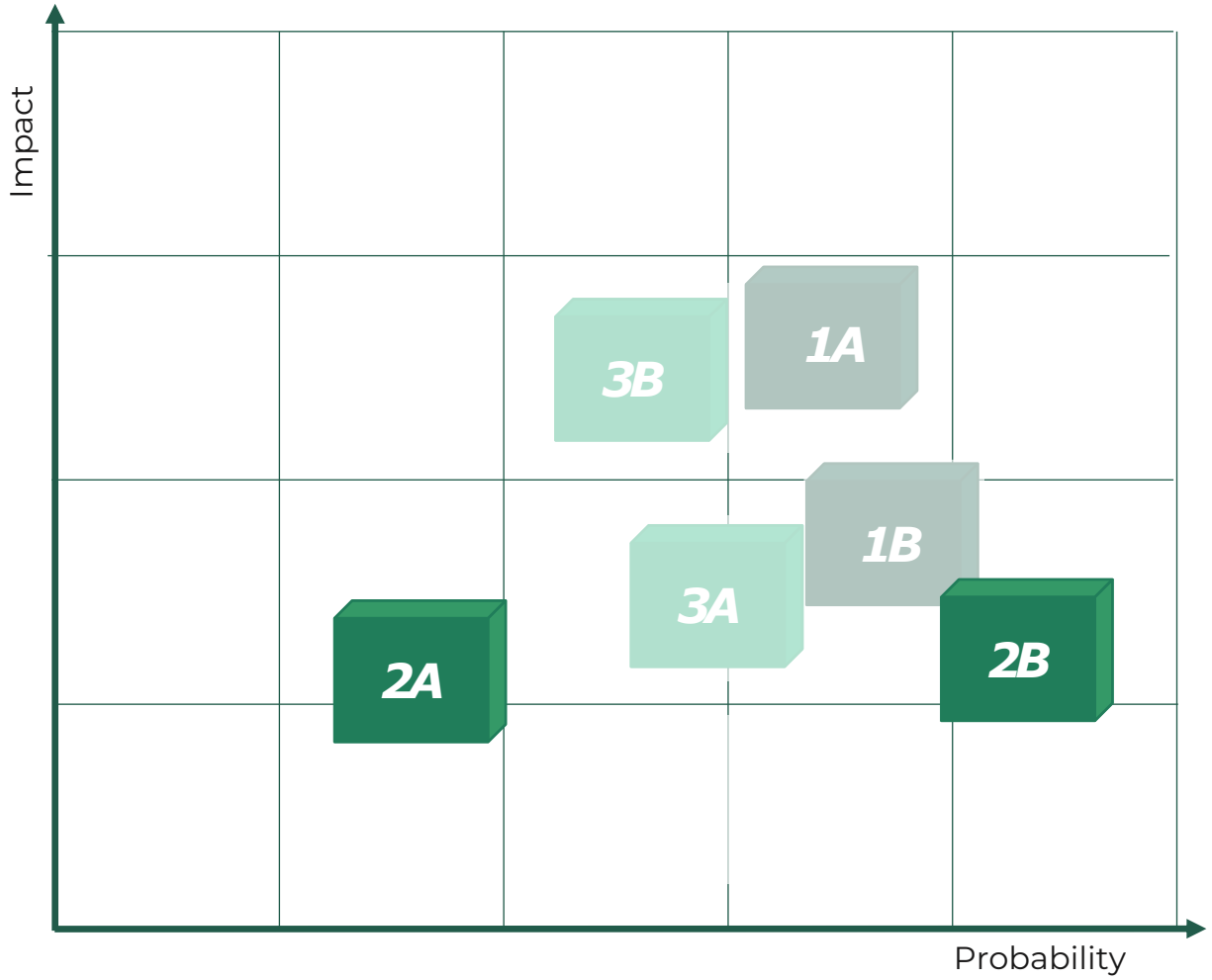
B. Damage to facilities and equipaments

Risk Assesment

The matrix produces an intriguing analysis between the **probability** and the **impact** of a certain risk.

Risk Matrix

Highlight for the **energy transition** that is already happening, but it **won't significantly affect the sector**.



1. Macroeconomic Risks

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A. High debt

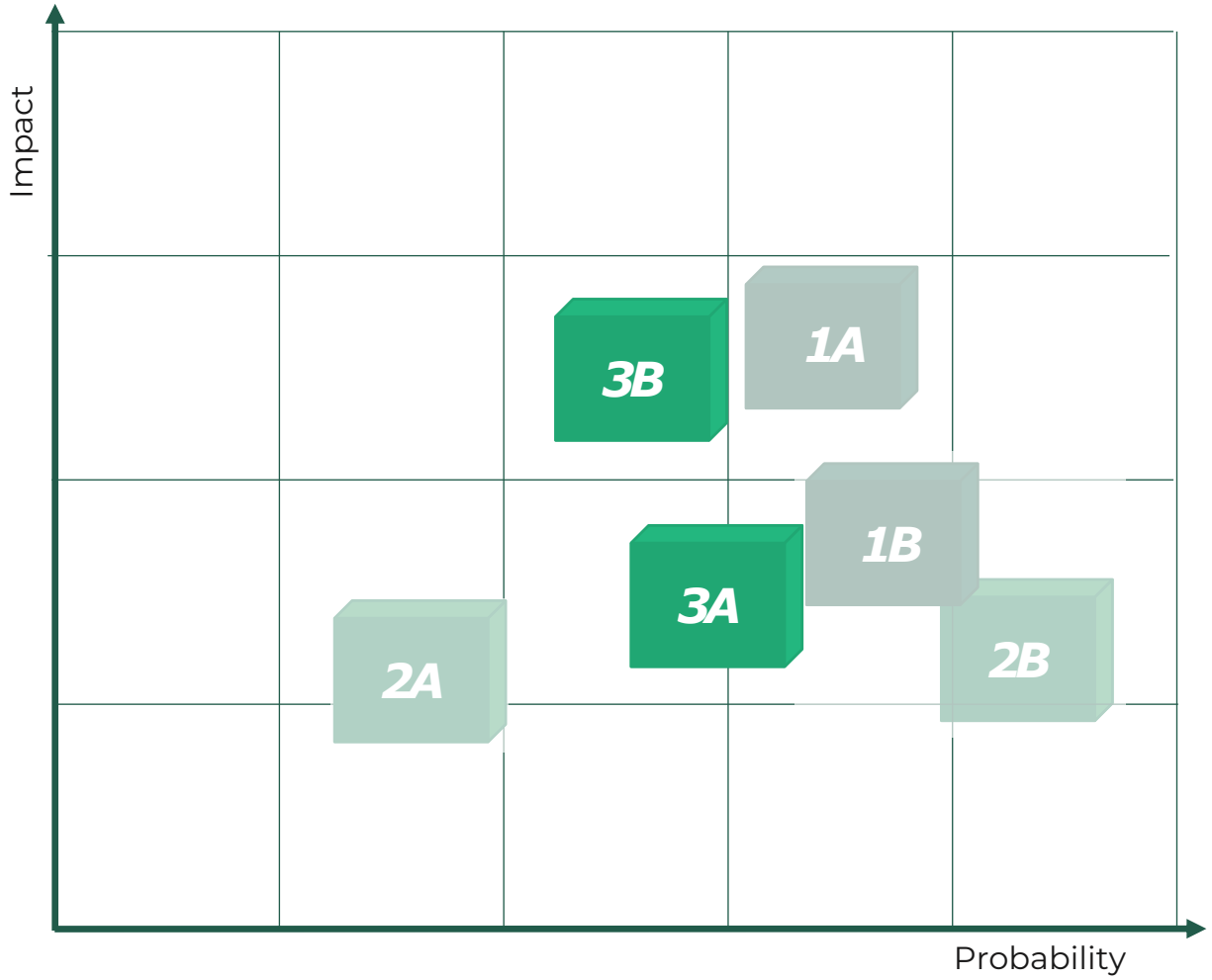
B. Damage to facilities and equipaments

Risk Assesment

The matrix produces an intriguing analysis between the **probability** and the **impact** of a certain risk.

Risk Matrix

Highlight on the potential of an **equipment malfunction**, which **could slow down the production**.



1. Macroeconomic Risks

A. Fluctuation in the Brent Price

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2. Industry Risks

A. Increase in taxes and royalties

B. Energy Transition

3. Specific Risks

A. High debt

B. Damage to facilities and equipaments



Q&A

Presentation Summary

Overview

1. 3R at a Glance
2. Investigating the Operations
3. Tiny Giants

Sectorial Analysis

1. Looking from the Bigger Picture
2. The World Urges for Oil!
3. Oil Stands the Test of Time
4. Brazil as na ESG Reference
5. Brazilian Black Gold
6. Unnattractive Appearance
7. Promising Future

Qualitative Analysis

1. A Heavyweight Team
2. The M&A Powerhouse
3. Taking a Closer Look
4. Expensive, But Not for Long
5. Keeping an Eye on the Debt

Valuation

1. Scenarios
2. Chasing the Production
3. An Elaborately Simple Recipe
4. Macro Impacts on Revenue
5. Good Results Coming Along!
6. Analysing Costs
7. A Closer Look at the Debt
8. Roots to FCFF
9. Cant Escape the Curve
10. 3R Scenario Distribution
11. Cheking One More Time
12. Exit Multiples Optics
13. Risk Assesment

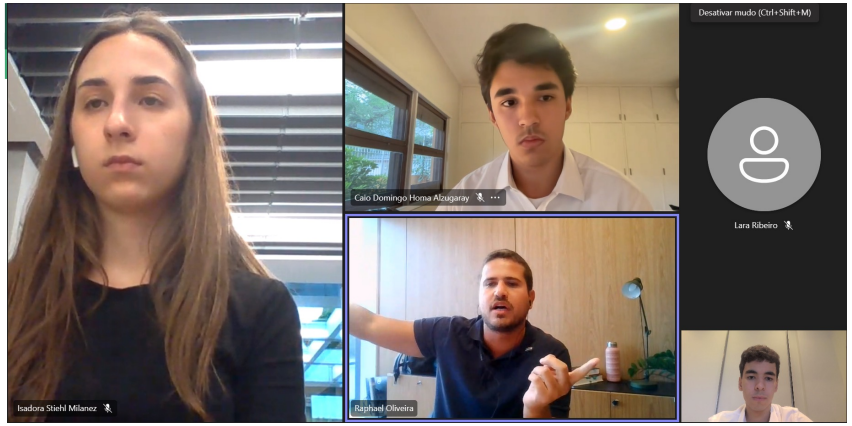
Appendix Summary

1. Field Research
2. Import x Export
3. Petrobras Desinvestment Plan
4. PPI Policy
5. How to Extract the Oil
6. Paramount Resource
7. OGX
8. WACC, Kd & Ke
9. Revenue Build-up
10. Income Statement
11. Balance Sheet
12. Cash Flow
13. FCFF & FCFE
14. Hedge Policy
15. Assumptions
16. Monte Carlo
17. SWOT & Porter 5 Forces
18. Debt
19. Python Code for Upside Graph

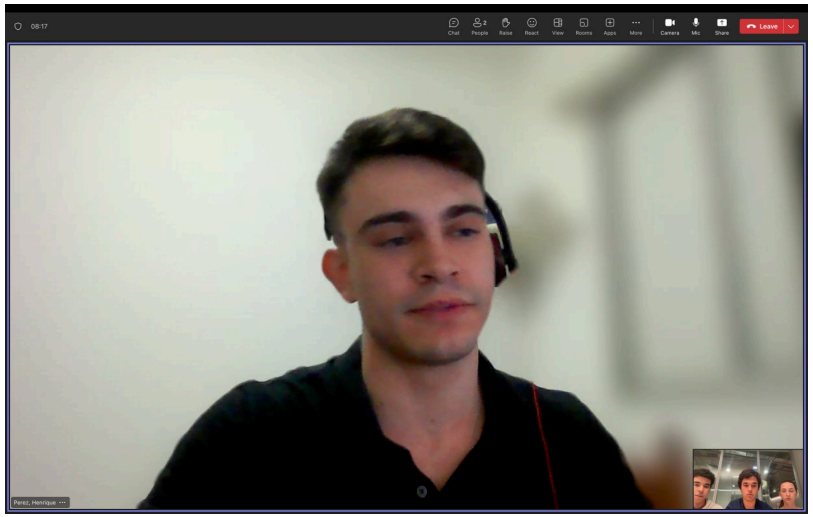


Field Research

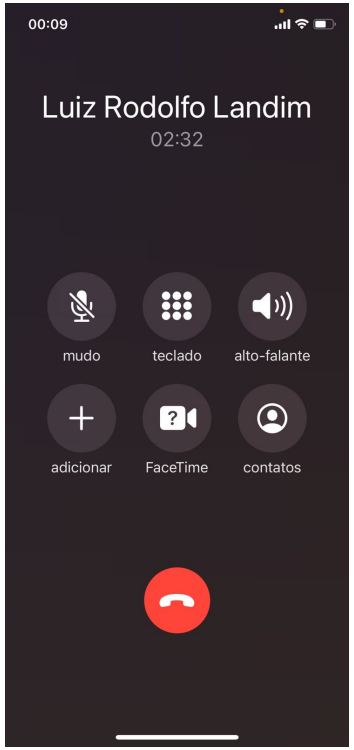
Field Research



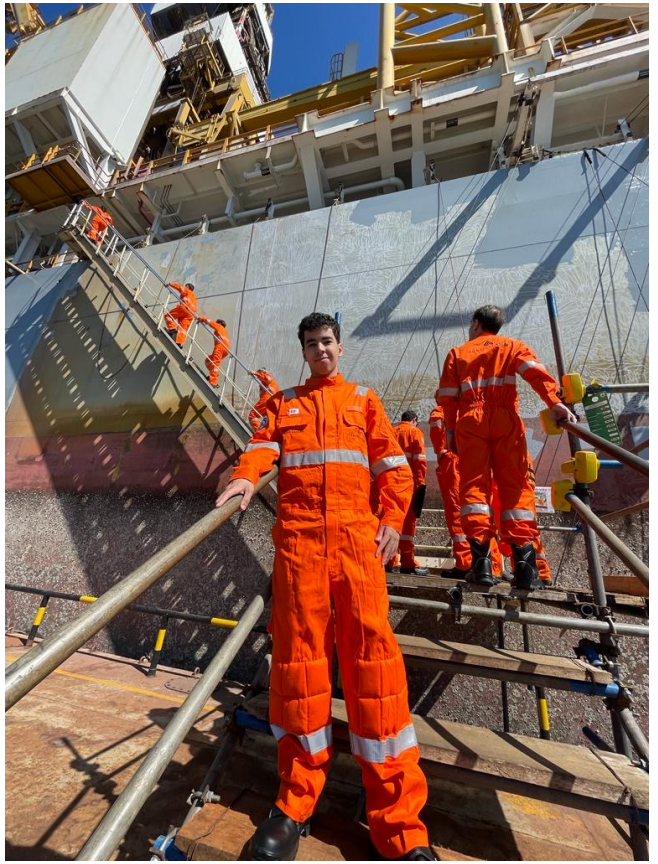
Raphael Oliveira – 3R Petroleum RI



Henrique Perez – Equity Research BTG



Luiz Rodolpho Landim – President of Flamengo & Founder of Ouro Preto Óleo e Gás SA (3R)



We visited the platform

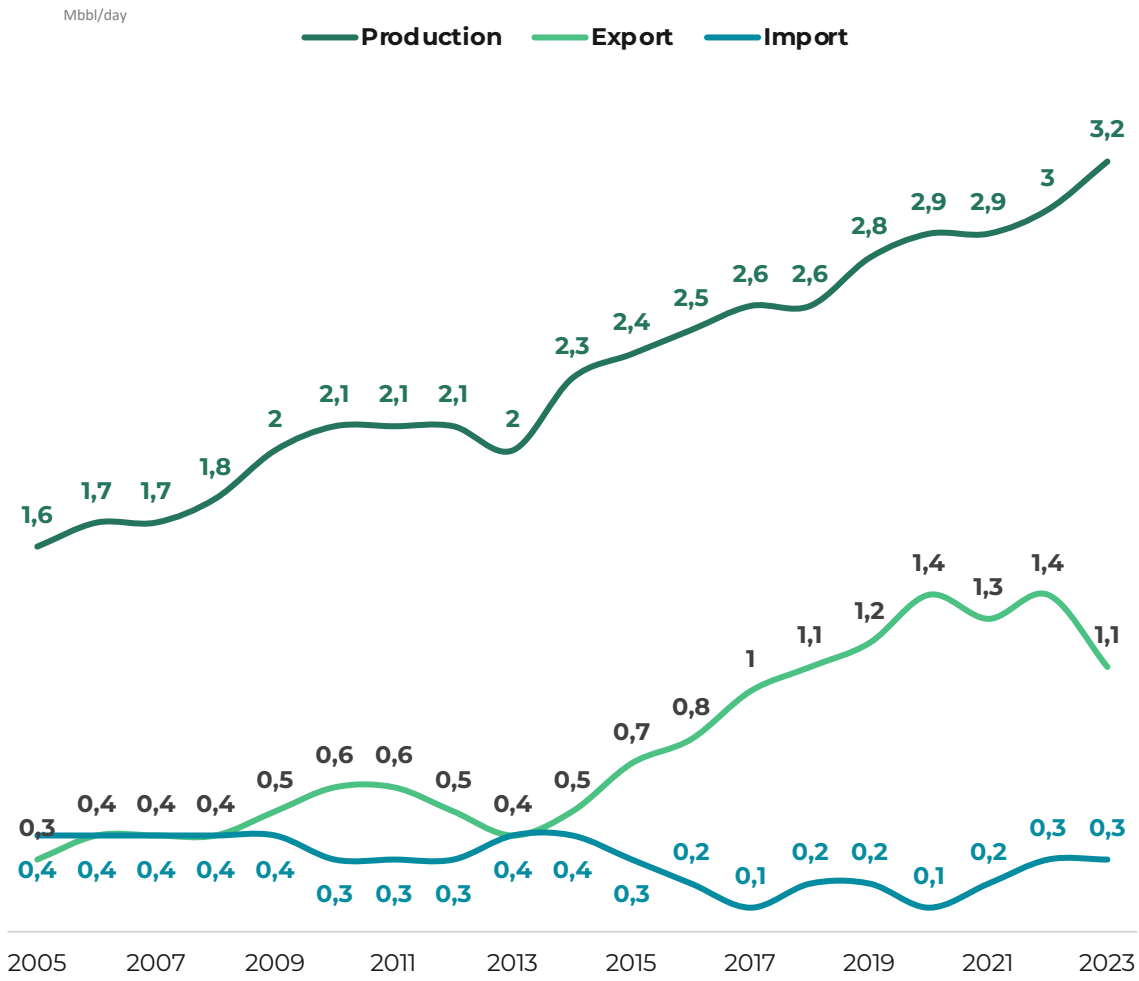
Import x Export

An X-ray of the Oil Extraction

Brazil produces approximately **3.2 million barrels of oil per day** on average and **exports about one-third**, needing to repurchase a portion of light oil

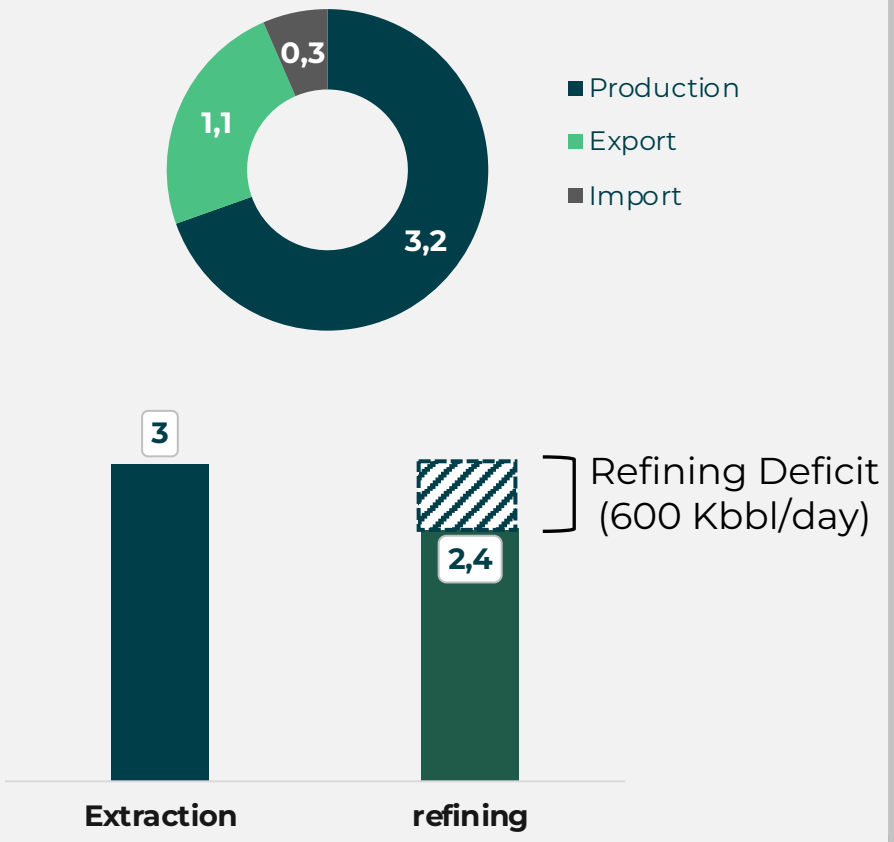
Evolution of Oil production, export and import

Brazilian production has been increasing over the past two decades, with exports following the trend



Average daily production, export, and import in 2022

Brazil still has a refining deficit of approximately 20%, leading the country to repurchase refined products



Sources: IBP

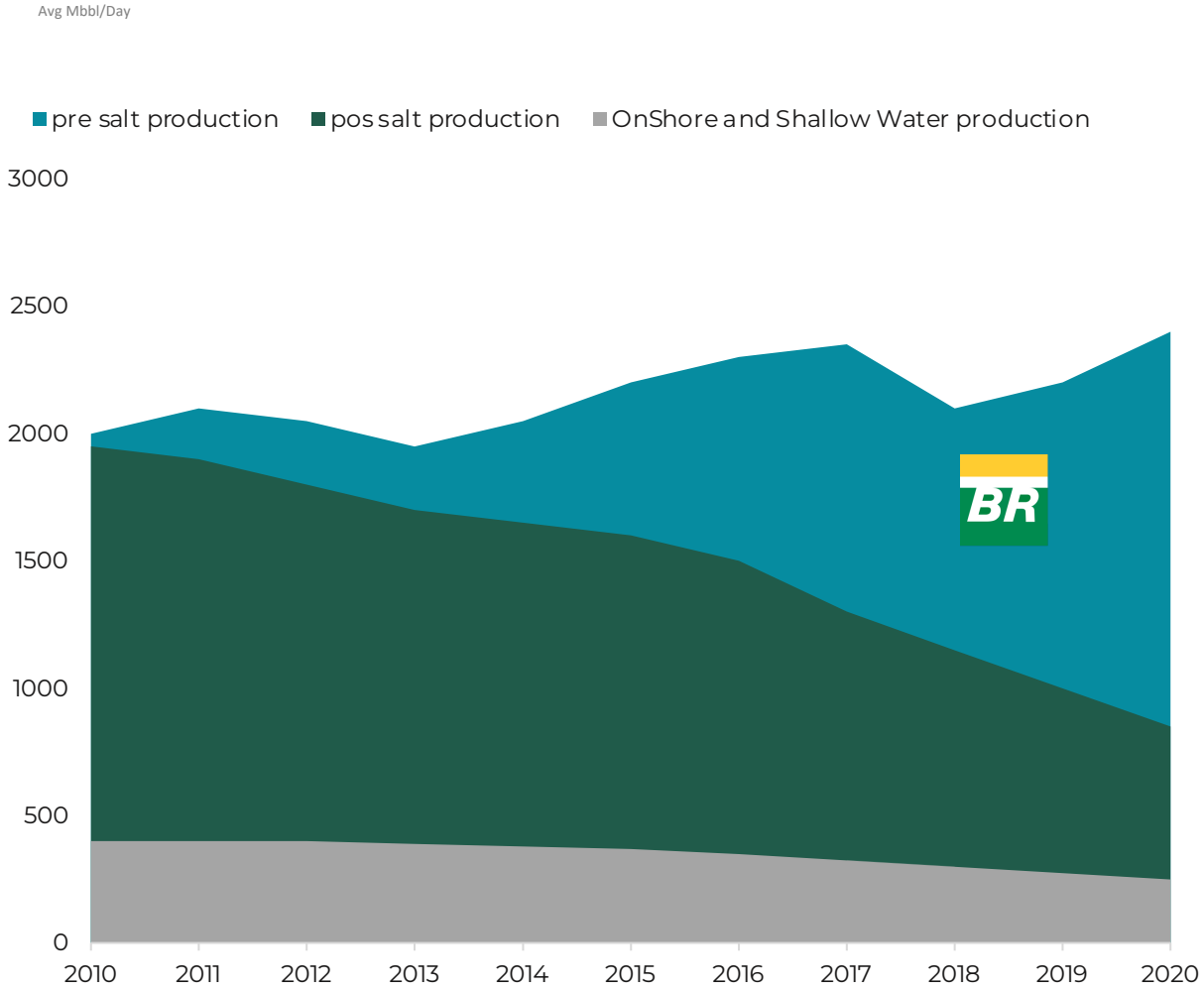
Petrobras Desinvestment Plan

The Holy Grail of Junior Oils

first and second phases consisted of **asset sales projects** and focus on pre-salt, leaving space for Junior Oils to enter the sector and increasing competition. Also **raising a significant amount of capital**

Petrobas focusing on pre salt assets, leaving room to Jrs

Petrobras is focusing on pre-salt and leaving shallower layers for other companies



Petrobras Desinvestment Plan - Phases

The original plan, conceived in 2015, during the Dilma government, is different from the project put into practice in 2019.

Initicial Plan (2015)

Total Expected Revenue (2015-2019):

US$ 58Bn

New Plan (2019)

Total Expected Revenue (2019-2024)

US$ 20-30Bn

Sources: Petrobras

The Holy Grail of Junior Oils

Government presses for the **end of the company's divestment plan**, interrupting the sale of assets and **focusing on modernization**. The new government has a more **state-oriented bias for Petrobras**, which should focus on keeping its assets in government ownership but opens the door to **poor management** at the operational level. Possibly resulting in **less efficiency**

PODER 360

Lula quer frear privatizações na Petrobras

Grupo de Minas e Energia pedirá à estatal que barre a privatizações

agênciaBrasil

Petrobras encerrará processos de desinvestimento em quatro polos

Decisão foi anunciada hoje no Rio de Janeiro

Governamental Pressure:

Lula **is against the sale of assets** by Petrobras. Suggesting **plants to be modernized** instead

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Lula **is against the sale of assets** by Petrobras. Suggesting **plants to be modernized** instead

Sources: Petrobras

PPI Policy

About Petrobras PPI Policy

The **end of the PPI policy** may change the course of the **O&G prices**. The future in **uncertain**, it can be seen by an optimistic ou by a negative point of view. There are no numbers no analyse its impact yet.

g1

Petrobras anuncia fim da paridade de preços de importação do petróleo e nova política de preço para combustíveis

Estatal anuncia que serão feitos sem periodicidade definida, será a depender das necessidades

CNN

Com o fim da paridade de preços da Petrobras, Janela de importação de combustíveis está fechada há 70 dias, diz Abicom

Alta dos preços do petróleo aumentou diferença emtre valores praticados no mercado interno e externo

What is PPI?

Policy adopted by Petrobras from 2016 until 2023, called Import Parity Price (PPI), which linked the price of oil derivatives in refineries to the behavior of the product's price in dollars on the international market.



"Essa é uma promessa de campanha de Lula e significa muito para o povo trabalhador. Os brasileiros ganham em real, mas estavam pagando pelo combustível cotado em dólar"

Senador Humberto Costa (PT-PE)

Sources Carta Capital, CNN, G1

How to Extract the Oil

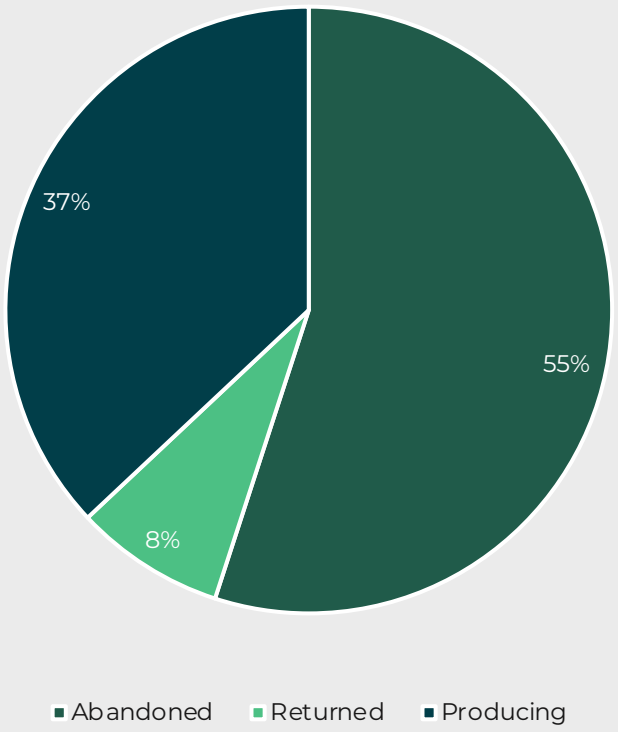
How do they do it?

The company has a **solid** and **efficient operation method**. With **6 steps**, the asset is ready to produce its best, the **black gold**, and in their best conditions.

- 3R Macau S.A.
- 3R Potiguar S.A.
- 3R Pescada S.A.
- 3R Fazenda Belém S.A.
- 3R Areia Branca S.A.
- 3R Candeias S.A.
- 3R Rio Ventura S.A.
- 3R OffShore S.A.
- 3R LUX S.A.

1. **Reactivation** of abandoned wells

Currently, 37% of all of the company's wells are **abandoned** due to **high deterioration**. Through time, 3R pretends to **keep reactivating** them.



Source: RI 3R

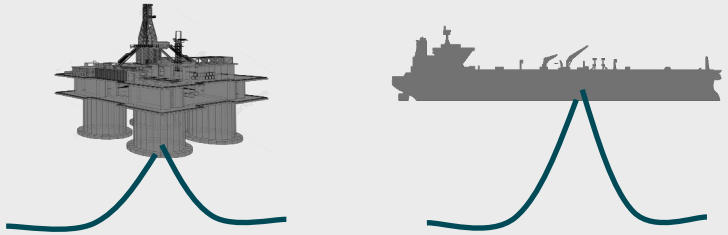
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2. **Pull-in** interventions

Done in **OffShore** platforms, it's the **substitution** of piping and pumps, to connect the field and the platform by a **duct**



3. **Workover** activities

General repairs in all types of wells:

- **Cleansing** and **unclogging** of the well;
- Exchange and repair of the **equipment**;
- **Repair of damage** caused by corrosion or by natural wear;
- Stimulating well production through techniques such as **hydraulic fracturing** (fracking).

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4. Reservoir's **pressurization** by the **injection** of fluids

The fluids ate **injected in the rocks** around the reserves, and, by pressurizing, they **fracture**, **facilitating** the oil extraction

5. **Debottlenecking** of the field

Huge **analysis** of the field to find the best ways to **increase** production, storage, processing and disposal facilities

6. **Densification** of the well network

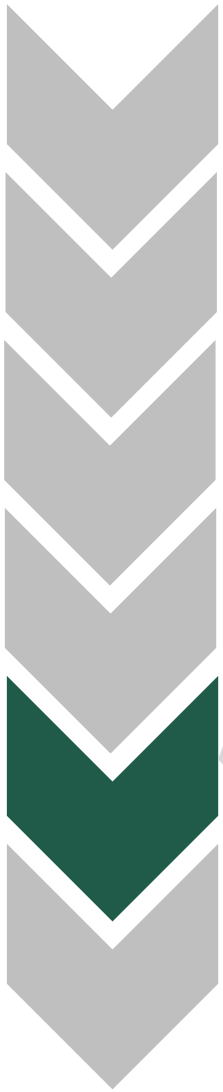
Perforation of **new wells** in 1P or 2P areas.

Source: RI 3R

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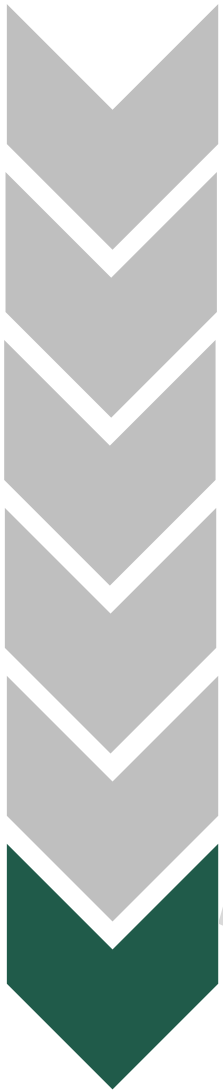
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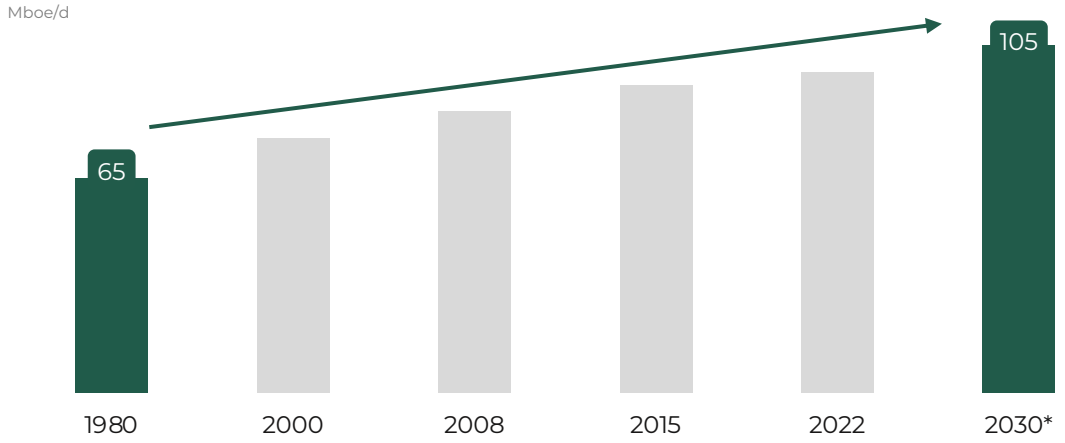
Paramount Resource

Paramount Resource

Oil and Gas are **vital commodities** for all economies. Despite some arguments about transitioning to renewable energies, oil consumption **will continue to grow in the medium term, but not at the same rate.**

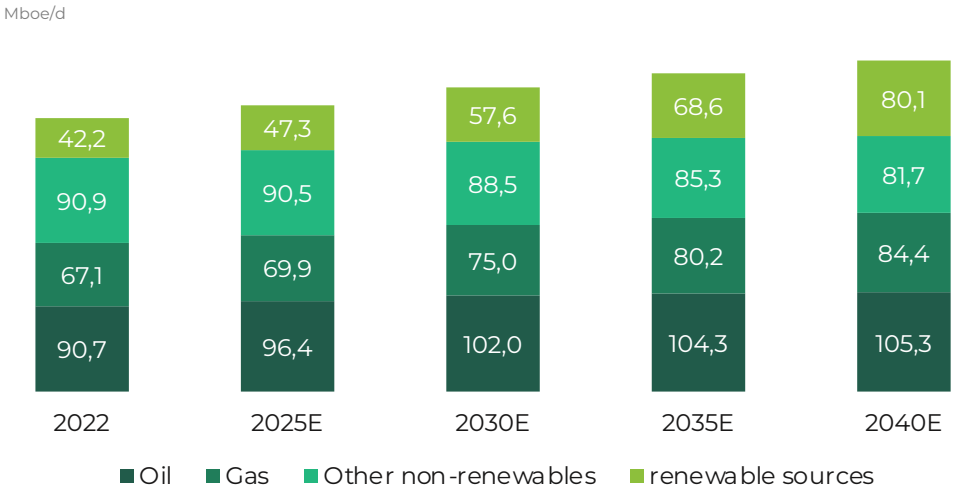
Oil consumption forecast

The consumption of oil will **continue to increase**, but **not at the same rate**



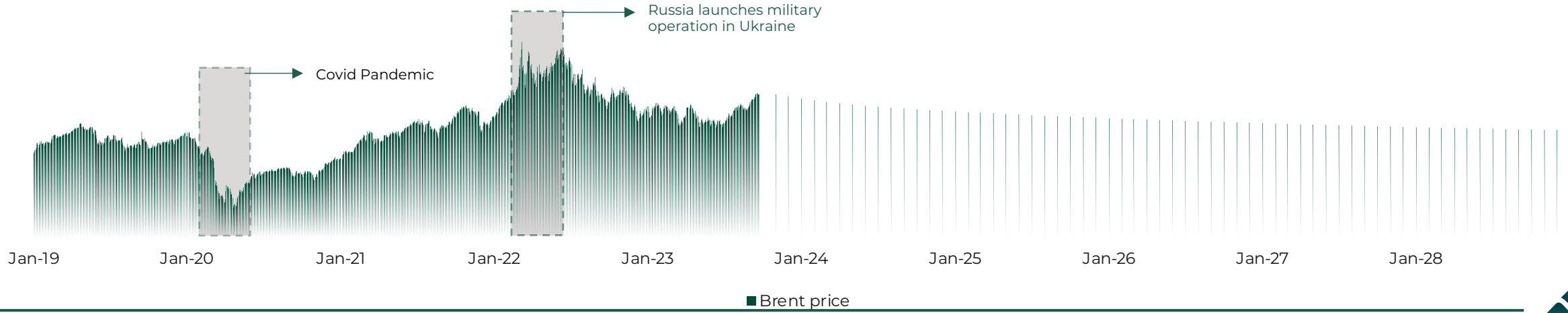
Primary energy demand worldwide

Both renewable energy and oil&gas **will continue to grow** in the coming years.



Brent Price

Oil and gas are fundamental to the world, and their **demand continues to rise despite high prices**.

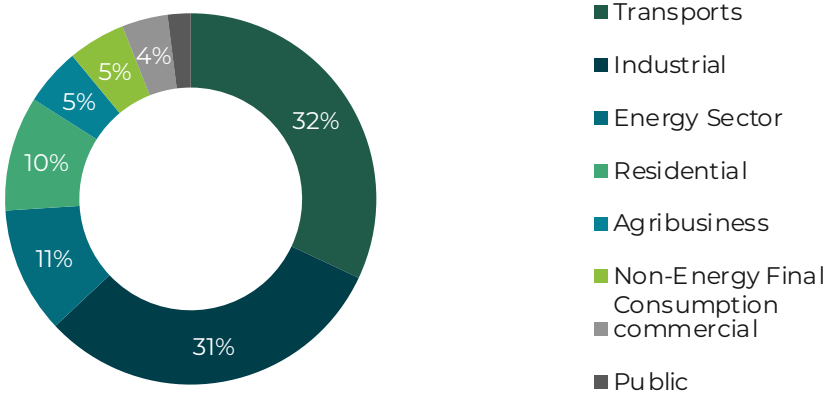


Source: Statista, Bloomberg

Paramount Resource

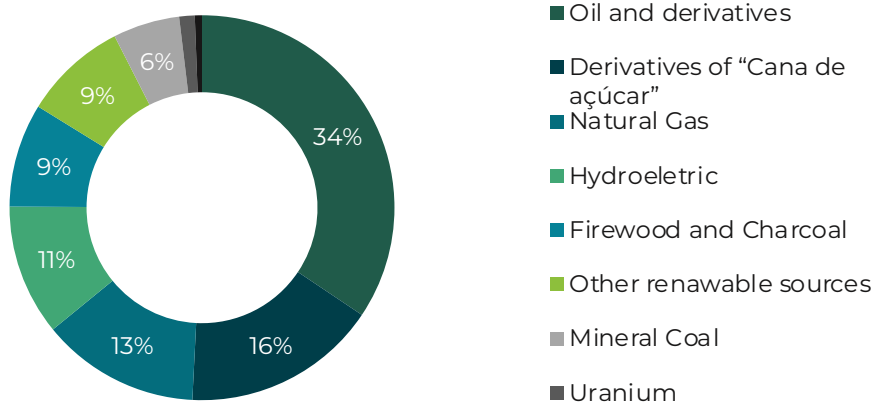
Use of Oil & Gas in Brazil

In Brazil, oil and gas drive crucial industries



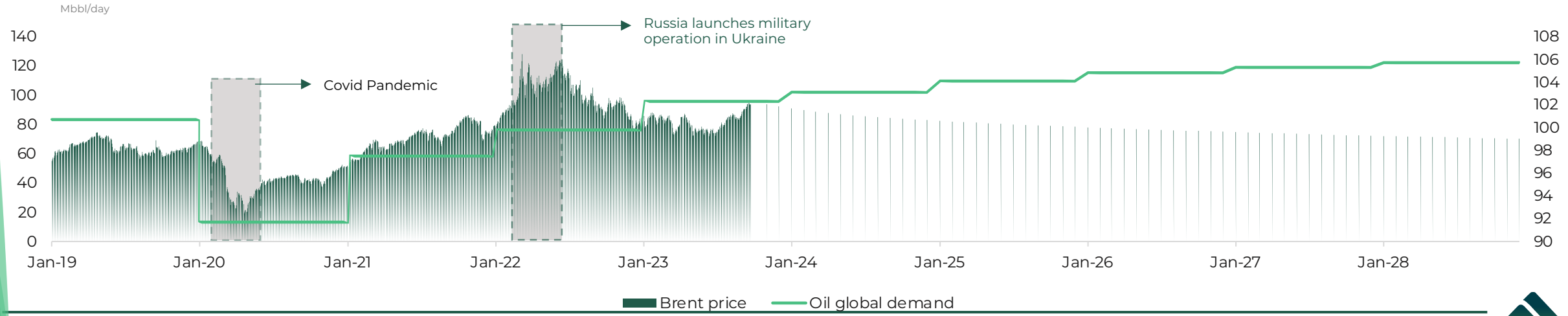
Energy Matrix in Brazil

Oil and gas constitute the majority of the energy matrix, which representes the diverse energy sources that power vehicles, fuel stoves for cooking and produce electricity



Global Oil Demand x Brent Price

As shown in the previous charts, oil and gas are fundamental to the world, and their demand continues to rise despite high prices.



Source: Bloomberg EPE, 3R Ri

OGX

Far from the next OGX

Even with the same narrative of the dream team and a lot of capital raised, the main difference is that **OGX never even found Oil,** while **3R operates directly in the E&P** area, having already overcome this stage.



ESTADÃO

The new company OGX ofuscates Petrobras in a R$2.1bi ANP auction.

OGX spent R$2.1bi in 21 assets, while Petrobras, the largest O&G Brazilian company, spent R$290mi in 27 assets.

uol

OGX estimated 1.5bi barrils for a petroleum field which never produced.

There was no production at Tubarão Azul field, in Campos basin, their only offshore field.





FOLHA DE S.PAULO

OGX completes their second month without producing any petroleum barril

Source: Estadão, Uol, Folha de São Paulo

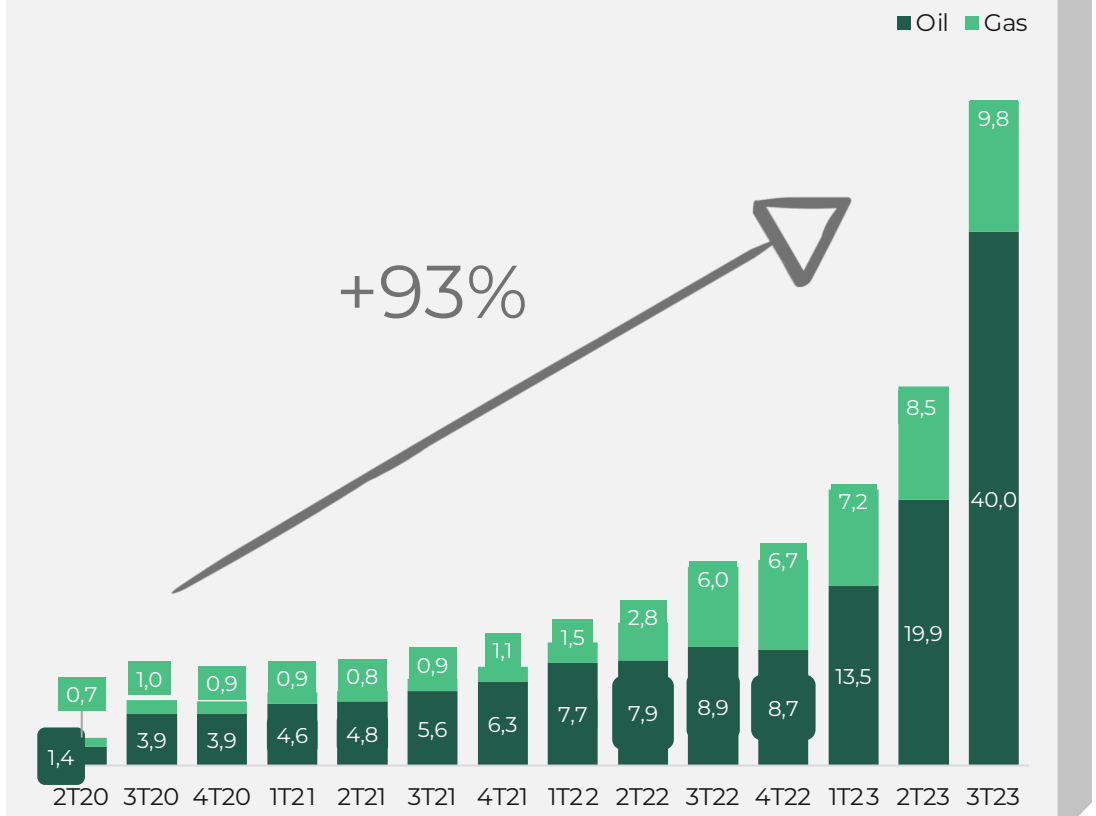
Far from the next OGX

In a meeting with **Rodolfo Landim**, who participated in the **founding of both companies**, he affirmed the different purposes and challenges of each one, **ensuring that 3R is already at a level that OGX never reached**.

3R's Daily Oil&Gas Avarage Production

Since 2020, 3R grew their production in 23 times

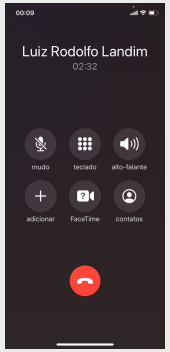
Kbbl/d and Kboe/d





Luiz Rodolfo Landim

Co-Founder of OGX Petróleo e Gás and founder of Ouro Preto Óleo e Gás SA



"OGX and Ouro Preto may **seem similar**, but they are quite **different**. OGX's objective was to **discover new wells**, while Ouro Preto's was to **revitalize** the already used and proven ones. Besides, OGX never got to the E&P part, Ouro Preto, the new 3R, has been doing it since day one."

WACC, Kd & Ke

Kd and Ke

The FCFF is transformed in **present value** through the **WACC** (Weighted Average Cost of Capital)

Cost of Equity

Calculation of Ke

4,44% + 4,57 + 1,21 × 5% = 15,04%

Ke US

Beta
IBOV x 3R

(1+Ke US) * (1+3,91%) / (1+2,5%) -1 = 16,62%

Ke BR

Cost of Debt

Calculation of Kd

8,416% + IPCA = 13,24%

Kd

RRRP13 –> Debenture issued on 15/10/2033

WACC Breakdown

Calculation of WACC

Ke US	15,04%
Ke BR	16,62%
Kd	13,24%
Tax	15,25%
E/(D+E)	72,53%
D/(D+E)	27,47%
WACC	**13,99%**

Source: Anbima

WACC

WACC Calculation

Kd	13,24%
Ke US	15,04%
Ke BR	16,62%
Equity	7.633.920
Debt	2.891.095
IR	15,25%
Risk Free	11,50%
Risk Premium	9,6%
Beta	1,21
WACC	**13,99%**

Ke x [S/(B+S)] + [Kd x (1 – IR)] x [B/(B+S)]

Ke – retorno mínimo esperado pelos acionistas para investir em ações da empresa;

S/(B+S) – porcentagem do financiamento provindo de capital próprio (acionistas);

Kd – custo associado à obtenção de financiamento por meio de empréstimos ou emissão de títulos, líquido do Imposto de Renda;

B/(B+S) – porcentagem do financiamento provindo de capital de terceiros (dívida);

Last Debenture

RRRP13 LEI 12.431

Emissor 3R PETROLEUM OLEO E GAS S.A.

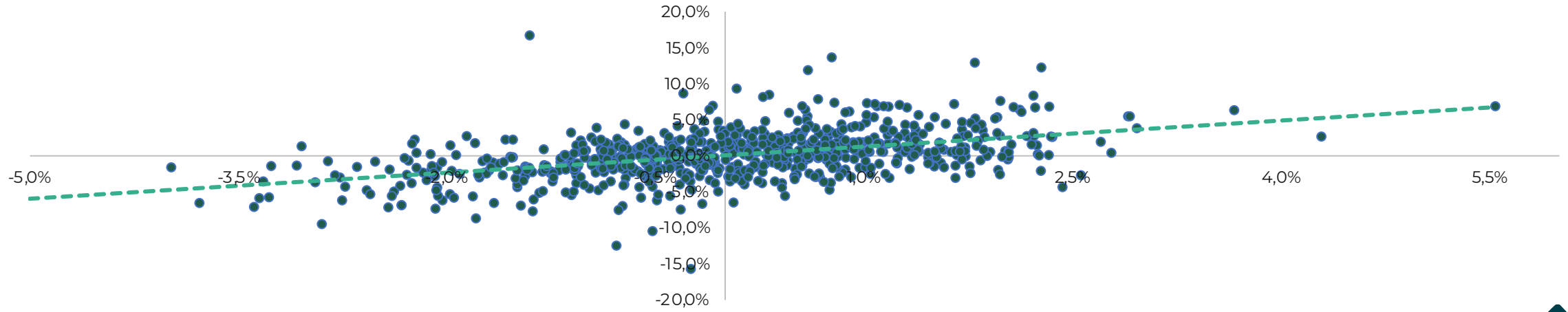
Setor Petróleo e Gás

Data de emissão 15/10/2023

Data de vencimento 15/10/2033

Remuneração IPCA + 8,4166%

Beta – Ibovespa x 3R



Source: Anbima, Yahoo Finance

Revenue Build-up

Revenue Build Up

Conversion factors	
m3 to bbl	6,2893
MMBTU to Km3	37,3
gas linked to oil (%)	13,57%

Scenarios	
Production	Base
Brent Price	Base
Forex	Base

Discounts	
Guidance	10%
Potiguar Depletion	7,88%
Recôncavo Depletion	12,87%
Offshore Depletion	10,04%

	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E
(Kbbl/d)												
Potiguar Reserves 2P - Oil	26,43	35,10	43,08	47,91	50,24	49,66	48,01	44,91	41,17	37,15	33,49	30,28
Recôncavo Reserves 2P - Oil	5,77	9,05	11,20	13,31	13,55	13,16	12,29	10,77	9,08	7,64	6,66	5,96
Offshore Reserves 2P - Oil	11,22	13,30	16,85	20,33	21,38	19,68	18,19	17,39	18,09	16,68	14,93	13,48
Potiguar Oil	21,92	29,10	35,71	39,72	38,37	37,93	36,67	34,30	31,44	28,37	25,58	23,13
Recôncavo Oil	4,52	7,09	8,78	10,44	9,26	8,99	8,39	7,36	6,20	5,22	4,55	4,07
Offshore Oil	9,09	10,77	13,64	16,46	15,57	14,33	13,25	12,67	13,18	12,15	10,88	9,82
Oil Extraction (Kbbl/d)	**35,5**	**47,0**	**58,1**	**66,6**	**63,2**	**61,3**	**58,3**	**54,3**	**50,8**	**45,7**	**41,0**	**37,0**
(Km3/d)												
Potiguar Gas Extraction	472,78	627,86	770,45	856,88	827,72	818,26	791,01	739,97	678,30	612,07	551,88	498,92
Recôncavo Gas Extraction	97,56	153,06	189,50	225,13	199,68	193,95	181,10	158,77	133,80	112,66	98,20	87,84
OffShore Gas Extraction	196,01	232,25	294,34	355,06	335,91	309,18	285,80	273,23	284,28	262,10	234,62	211,76
Gas Extraction (Kboe/d)	**4,8**	**6,4**	**7,9**	**9,0**	**8,6**	**8,3**	**7,9**	**7,4**	**6,9**	**6,2**	**5,6**	**5,0**
Total Production (Kboe/d)	**40,34**	**53,34**	**66,03**	**75,65**	**71,77**	**69,56**	**66,22**	**61,70**	**57,72**	**51,95**	**46,57**	**42,04**

Conversion factors	
m3 to bbl	6,2893
MMBTU to Km3	37,3
gas linked to oil (%)	13,57%

Scenarios	
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Brent Price	Base
Forex	Base

Discounts	
Guidance	10%
Potiguar Depletion	7,88%
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Offshore Depletion	10,04%

	2035E	2036E	2037E	2038E	2039E	2040E	2041E	2042E	2043E	2044E	2045E	2046E	2047E	2048E	2049E	2050E
(Kbbl/d)																
Potiguar Reserves 2P - Oil	27,69	25,57	23,73	22,04	20,44	19,03	17,76	16,62	15,58	14,74	14,03	13,35	12,78	12,22	11,57	10,84
Recôncavo Reserves 2P - Oil	5,38	4,84	4,40	4,05	3,72	3,36	2,95	2,87	2,66	2,45	2,26	2,05	1,96	1,87	1,77	1,69
Offshore Reserves 2P - Oil	12,26	11,14	10,24	9,47	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00
Potiguar Oil	21,15	19,53	18,12	16,83	15,61	14,53	13,56	12,69	11,90	11,26	10,72	10,20	9,76	9,33	8,84	8,28
Recôncavo Oil	3,67	3,31	3,00	2,77	2,55	2,30	2,02	1,96	1,82	1,67	1,54	1,40	1,34	1,28	1,21	1,15
Offshore Oil	8,93	8,12	7,46	6,90	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00
Oil Extraction (Kbbl/d)	**33,8**	**31,0**	**28,6**	**26,5**	**18,2**	**16,8**	**15,6**	**14,7**	**13,7**	**12,9**	**12,3**	**11,6**	**11,1**	**10,6**	**10,0**	**9,4**
(Km3/d)																
Potiguar Gas Extraction	456,26	421,24	390,98	363,09	336,76	313,55	292,60	273,79	256,69	242,81	231,22	220,01	210,60	201,32	190,64	178,64
Recôncavo Gas Extraction	79,24	71,40	64,78	59,74	54,91	49,55	43,47	42,31	39,20	36,12	33,24	30,17	28,86	27,56	26,15	24,90
OffShore Gas Extraction	192,73	175,10	160,84	148,77	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00
Gas Extraction (Kboe/d)	**4,6**	**4,2**	**3,9**	**3,6**	**2,5**	**2,3**	**2,1**	**2,0**	**1,9**	**1,8**	**1,7**	**1,6**	**1,5**	**1,4**	**1,4**	**1,3**
Total Production (Kboe/d)	**38,34**	**35,15**	**32,46**	**30,09**	**20,62**	**19,11**	**17,69**	**16,64**	**15,58**	**14,68**	**13,92**	**13,17**	**12,61**	**12,05**	**11,41**	**10,72**

Revenue Build Up

	2022	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E
Revenue Build-up												
Brent Price	99,0	81,7	86,2	80,2	76,4	73,4	71,1	70,0	70,0	70,0	70,0	70,0
Potiguar Discounted Oil Price	95,1	77,0	82,0	76,2	72,4	69,4	67,1	66,0	66,0	66,0	66,0	66,0
Recôncavo Discounted Oil Price	89,6	76,4	82,7	76,7	72,9	69,9	67,6	66,5	70,0	70,0	70,0	70,0
Offshore Discounted Oil Price	94,0	76,3	81,0	75,2	71,4	68,4	66,1	65,0	70,0	70,0	70,0	70,0
USD/BRL	5,22	5,10	5,25	5,30	5,30	5,30	5,30	5,30	5,30	5,30	5,30	5,30
Gas Prices ($/MMBTU)	9,5	8,55	8,55	8,55	8,55	8,55	8,55	8,55	8,55	8,55	8,55	8,55
Potiguar Revenue - **Oil** (R$ K)		3.140.792	4.570.328	5.268.570	5.566.541	5.153.508	4.923.719	4.682.773	4.380.585	4.015.521	3.623.454	3.267.093
Recôncavo Revenue - **Oil** (R$ K)		642.870	1.124.371	1.304.387	1.472.578	1.252.207	1.175.739	1.080.235	996.849	840.120	707.375	616.589
Offshore Revenue - **Oil** (R$ K)		1.291.135	1.671.472	1.985.271	2.274.742	2.061.311	1.832.717	1.666.315	1.715.566	1.784.922	1.645.653	1.473.131
Potiguar Revenue - **Gas** (R$ K)		552.692	755.574	936.263	1.041.299	1.005.862	994.366	961.253	899.221	824.283	743.802	670.650
Recôncavo Revenue - **Gas** (R$ K)		114.053	184.197	230.289	273.577	242.658	235.689	220.077	192.934	162.600	136.908	119.337
Offshore Revenue - **Gas** (R$ K)		229.143	279.497	357.684	431.480	408.208	375.726	347.314	332.038	345.461	318.507	285.116
Revenue Oil (R$ K)		5.074.796	7.366.171	8.558.227	9.313.861	8.467.026	7.932.175	7.429.323	7.093.000	6.640.563	5.976.483	5.356.813
Revenue Gas (R$ K)		895.888	1.219.268	1.524.236	1.746.356	1.656.728	1.605.781	1.528.644	1.424.194	1.332.345	1.199.217	1.075.104
Total Revenue (R$ K)		**5.970.684**	**8.585.439**	**10.082.463**	**11.060.217**	**10.123.753**	**9.537.955**	**8.957.967**	**8.517.194**	**7.972.908**	**7.175.700**	**6.431.917**

Revenue Build Up

	2034E	2035E	2036E	2037E	2038E	2039E	2040E	2041E	2042E	2043E	2044E	2045E	2046E	2047E	2048E	2049E	2050E
Revenue Build-up																	
Brent Price	70,0	70,0	70,0	70,0	70,0	70,0	70,0	70,0	70,0	70,0	70,0	70,0	70,0	70,0	70,0	70,0	70,0
Potiguar Discounted Oil Price	66,0	66,0	66,0	66,0	66,0	66,0	66,0	66,0	66,0	66,0	66,0	66,0	66,0	66,0	66,0	66,0	66,0
Recôncavo Discounted Oil Price	70,0	70,0	70,0	70,0	70,0	70,0	70,0	70,0	70,0	70,0	70,0	70,0	70,0	70,0	70,0	70,0	70,0
Offshore Discounted Oil Price	70,0	70,0	70,0	70,0	70,0	70,0	70,0	70,0	70,0	70,0	70,0	70,0	70,0	70,0	70,0	70,0	70,0
USD/BRL	5,30	5,30	5,30	5,30	5,30	5,30	5,30	5,30	5,30	5,30	5,30	5,30	5,30	5,30	5,30	5,30	5,30
Gas Prices ($/MMBTU)	8,55	8,55	8,55	8,55	8,55	8,55	8,55	8,55	8,55	8,55	8,55	8,55	8,55	8,55	8,55	8,55	8,55
Potiguar Revenue - **Oil** (R$ K)	2.953.597	2.701.069	2.493.704	2.314.612	2.149.459	1.993.598	1.856.190	1.732.175	1.620.800	1.519.625	1.437.421	1.368.807	1.302.469	1.246.767	1.191.815	1.128.575	1.057.533
Recôncavo Revenue - **Oil** (R$ K)	551.521	497.531	448.280	406.767	375.108	344.737	311.130	272.962	265.685	246.144	226.784	208.721	189.415	181.205	173.025	164.158	156.356
Offshore Revenue - **Oil** (R$ K)	1.329.575	1.210.093	1.099.423	1.009.883	934.078	-	-	-	-	-	-	-	-	-	-	-	-
Potiguar Revenue - **Gas** (R$ K)	606.297	554.460	511.893	475.130	441.229	409.234	381.028	355.571	332.708	311.940	295.066	280.981	267.363	255.929	244.649	231.667	217.084
Recôncavo Revenue - **Gas** (R$ K)	106.744	96.294	86.762	78.728	72.600	66.722	60.218	52.830	51.422	47.640	43.893	40.397	36.660	35.071	33.488	31.772	30.262
Offshore Revenue - **Gas** (R$ K)	257.332	234.207	212.787	195.457	180.785	-	-	-	-	-	-	-	-	-	-	-	-
Revenue Oil (R$ K)	4.834.693	4.408.693	4.041.407	3.731.263	3.458.645	2.338.335	2.167.320	2.005.137	1.886.485	1.765.769	1.664.206	1.577.528	1.491.884	1.427.971	1.364.841	1.292.733	1.213.890
Revenue Gas (R$ K)	970.373	884.961	811.442	749.315	694.614	475.956	441.245	408.401	384.130	359.580	338.958	321.378	304.024	291.000	278.137	263.439	247.346
Total Revenue (R$ K)	**5.805.066**	**5.293.654**	**4.852.849**	**4.480.578**	**4.153.259**	**2.814.292**	**2.608.565**	**2.413.538**	**2.270.615**	**2.125.348**	**2.003.164**	**1.898.905**	**1.795.907**	**1.718.972**	**1.642.978**	**1.556.172**	**1.461.236**

Income Statement

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in thousand of RM	2020	2021	2022	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E	2035E	2036E
Income Statement																	
Net Revenue	**166.924**	**727.799**	**1.720.419**	**5.970.684**	**8.585.439**	**10.082.463**	**11.060.217**	**10.123.753**	**9.537.955**	**8.957.967**	**8.517.194**	**7.972.908**	**7.175.700**	**6.431.917**	**5.805.066**	**5.293.654**	**4.852.849**
Net Revenue of Oil	149.456	695.175	1.405.224	5.074.796	7.366.171	8.558.227	9.313.861	8.467.026	7.932.175	7.429.323	7.093.000	6.640.563	5.976.483	5.356.813	4.834.693	4.408.693	4.041.407
Net Revenue of Gas	17.468	32.624	315.195	895.888	1.219.268	1.524.236	1.746.356	1.656.728	1.605.781	1.528.644	1.424.194	1.332.345	1.199.217	1.075.104	970.373	884.961	811.442
YoY Growth (%)	-	336%	137%	247%	44%	17%	10%	-8%	-6%	-6%	-5%	-6%	-10%	-10%	-10%	-9%	-8%
Cost of Goods Sold (COGS)	**(90.251)**	**(319.530)**	**(835.544)**	**(2.100.534)**	**(3.413.970)**	**(3.995.693)**	**(4.389.589)**	**(4.080.055)**	**(3.882.510)**	**(3.680.922)**	**(3.516.382)**	**(3.302.692)**	**(2.987.704)**	**(2.694.406)**	**(2.447.923)**	**(2.245.975)**	**(2.072.115)**
COGS Margin(%)	*-54%*	*-44%*	*-49%*	*-35%*	*-40%*	*-40%*	*-40%*	*-40%*	*-41%*	*-41%*	*-41%*	*-41%*	*-42%*	*-42%*	*-42%*	*-42%*	*-43%*
Operational Costs	(29.233)	(76.484)	(246.648)	(1.619.603)	(2.854.658)	(3.352.419)	(3.677.522)	(3.366.148)	(3.171.370)	(2.978.524)	(2.831.967)	(2.650.992)	(2.385.920)	(2.138.612)	(1.930.184)	(1.760.140)	(1.613.572)
As % of the revenue	*18%*	*11%*	*14%*	*33%*	*36%*	*36%*	*36%*	*36%*	*36%*	*36%*	*36%*	*36%*	*36%*	*36%*	*36%*	*36%*	*36%*
Rental of Area	(1.080)	(10.830)	(29.164)	(33.392)	(34.694)	(36.082)	(37.525)	(39.026)	(40.587)	(42.211)	(43.899)	(41.094)	(36.985)	(33.151)	(29.920)	(27.285)	(25.013)
Royalties	(16.654)	(78.670)	(144.078)	(130.082)	(144.614)	(162.377)	(174.725)	(173.881)	(163.122)	(152.967)	(142.315)	(132.832)	(119.380)	(107.550)	(98.078)	(89.162)	(81.498)
Depreciation and Amortization	(30.348)	(110.189)	(248.747)	-	(14.500)	(34.000)	(56.000)	(70.500)	(84.500)	(95.750)	(102.250)	(107.250)	(111.930)	(116.143)	(119.919)	(123.327)	(126.434)
Water treatment and electric	(12.063)	(30.237)	(76.781)	(66.533)	(66.533)	(66.533)	(66.533)	(66.533)	(66.533)	(66.533)	(66.533)	(62.281)	(56.054)	(50.244)	(45.347)	(41.352)	(37.909)
Gas Processing and Transportation	-	-	(27.234)	(133.108)	(181.155)	(226.466)	(259.468)	(246.151)	(238.581)	(227.121)	(211.602)	(197.955)	(178.176)	(159.735)	(144.175)	(131.484)	(120.561)
Licensing and Environmental Costs	-	-	(11.514)	(45.995)	(45.995)	(45.995)	(45.995)	(45.995)	(45.995)	(45.995)	(45.995)	(43.056)	(38.751)	(34.734)	(31.349)	(28.587)	(26.207)
Personal Costs	-	(5.887)	(36.404)	(71.821)	(71.821)	(71.821)	(71.821)	(71.821)	(71.821)	(71.821)	(71.821)	(67.231)	(60.509)	(54.237)	(48.951)	(44.639)	(40.922)
Others	(873)	(7.233)	(14.974)	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Gross Income	**76.673**	**408.269**	**886.878**	**3.870.150**	**5.171.468**	**6.086.770**	**6.670.628**	**6.043.698**	**5.655.445**	**5.277.045**	**5.000.812**	**4.670.216**	**4.187.996**	**3.737.510**	**3.357.142**	**3.047.679**	**2.780.734**
Gross Margin (%)	46%	56%	52%	65%	60%	60%	60%	60%	59%	59%	59%	59%	58%	58%	58%	58%	57%
SG&A Expenses	**(239.856)**	**(145.632)**	**(615.821)**	**(1.715.167)**	**(2.319.277)**	**(2.703.567)**	**(2.954.559)**	**(2.714.166)**	**(2.563.790)**	**(2.414.906)**	**(2.301.759)**	**(2.162.03**	**(1.957.393)**	**(1.766.462)**	**(1.605.548)**	**(1.474.267)**	**(1.361.112)**
Operating Cost Margin (%)	-144%	-20%	-36%	-29%	-27%	-27%	-27%	-27%	-27%	-27%	-27%	-27	-27%	-27%	-28%	-28%	-28%
Sales Expenses	**(34.236)**	**(159.583)**	**(339.003)**	**(1.176.505)**	**(1.691.734)**	**(1.986.718)**	**(2.179.381)**	**(1.994.854)**	**(1.879.424)**	**(1.765.140)**	**(1.678.287)**	**(1.571.037)**	**(1.413.949)**	**(1.267.389)**	**(1.143.870)**	**(1.043.098)**	**(956.239)**
As % of the Revenue	*-21%*	*-22%*	*-20%*	*-20%*													
Personnel	(25.499)	(108.059)	(203.307)	(352.787)	(448.897)	(527.170)	(578.293)	(529.329)	(498.700)	(468.375)	(445.328)	(416.870)	(375.187)	(336.298)	(303.523)	(276.783)	(253.735)
Office Rental and Maintenance	(859)	(978)	(978)	(978)	(978)	(978)	(978)	(978)	(978)	(978)	(978)	(978)	(978)	(978)	(978)	(978)	(978)
Third Party Services	(3.307)	(19.911)	(40.023)	(44.003)	(63.273,33)	(74.306,16)	(81.512,06)	(74.610,47)	(70.293,23)	(66.018,81)	(62.770,38)	(58.759,07)	(52.883,78)	(47.402,21)	(42.782,42)	(39.013,40)	(35.764,74)
Provision	2.054	(2.857)	(465)	(346)	(346)	(346)	(346)	(346)	(346)	(346)	(346)	(346)	(346)	(346)	(346)	(346)	(346)
Tax expenses	(1.279)	(1.978)	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
IT	(224)	(4.966)	(20.818)	(33.856)	(33.856)	(33.856)	(33.856)	(33.856)	(33.856)	(33.856)	(33.856)	(33.856)	(33.856)	(33.856)	(33.856)	(33.856)	(33.856)
Other Expenses	(4.224)	(13.661)	(43.835)	(36.163)	(36.163)	(36.163)	(36.163)	(36.163)	(36.163)	(36.163)	(36.163)	(36.163)	(36.163)	(36.163)	(36.163)	(36.163)	(36.163)
Other operating expenses/income	(5.528)	5.940	(153.500)	(44.030)	(44.030)	(44.030)	(44.030)	(44.030)	(44.030)	(44.030)	(44.030)	(44.030)	(44.030)	(44.030)	(44.030)	(44.030)	(44.030)
(Loss) / Reversal in Recoverable Value of Assets	(166.755)	160.421	(123.318)	(26.500)	-	-	-	-	-	-	-	-	-	-	-	-	-

IS

in thousand of RM	2037E	2038E	2039E	2040E	2041E	2042E	2043E	2044E	2045E	2046E	2047E	2048E	2049E	2050E
Income Statement														
Net Revenue	**4.480.578**	**4.153.259**	**2.814.292**	**2.608.565**	**2.413.538**	**2.270.615**	**2.125.348**	**2.003.164**	**1.898.905**	**1.795.907**	**1.718.972**	**1.642.978**	**1.556.172**	**1.461.236**
Net Revenue of Oil	3.731.263	3.458.645	2.338.335	2.167.320	2.005.137	1.886.485	1.765.769	1.664.206	1.577.528	1.491.884	1.427.971	1.364.841	1.292.733	1.213.890
Net Revenue of Gas	749.315	694.614	475.956	441.245	408.401	384.130	359.580	338.958	321.378	304.024	291.000	278.137	263.439	247.346
YoY Growth (%)	-8%	-7%	-32%	-7%	-7%	-6%	-6%	-6%	-5%	-5%	-4%	-4%	-5%	-6%
Cost of Goods Sold (COGS)	**(1.925.479)**	**(1.796.923)**	**(1.263.413)**	**(1.182.612)**	**(1.105.951)**	**(1.049.963)**	**(992.988)**	**(930.711)**	**(870.579)**	**(808.394)**	**(764.084)**	**(720.600)**	**(675.478)**	**(631.796)**
COGS Margin(%)	*-43%*	*-43%*	*-45%*	*-45%*	*-46%*	*-46%*	*-47%*	*-46%*	*-46%*	*-45%*	*-44%*	*-44%*	*-43%*	*-43%*
Operational Costs	(1.489.792)	(1.380.959)	(935.752)	(867.348)	(802.501)	(754.980)	(706.678)	(666.052)	(631.386)	(597.139)	(571.558)	(546.290)	(517.427)	(485.861)
As % of the revenue	*36%*	*36%*	*36%*	*36%*	*36%*	*36%*	*36%*	*36%*	*36%*	*36%*	*36%*	*36%*	*36%*	*36%*
Rental of Area	(23.094)	(21.407)	(14.505)	(13.445)	(12.440)	(11.703)	(10.954)	(10.325)	(9.787)	(9.256)	(8.860)	(8.468)	(8.021)	(7.532)
Royalties	(75.000)	(69.546)	(47.175)	(43.796)	(40.557)	(38.110)	(35.643)	(33.584)	(31.848)	(30.133)	(28.847)	(27.569)	(26.106)	(24.511)
Depreciation and Amortization	(129.283)	(131.913)	(134.352)	(136.004)	(137.535)	(138.952)	(140.285)	(127.033)	(108.709)	(87.823)	(74.378)	(61.387)	(51.101)	(45.515)
Water treatment and electric	(35.001)	(32.444)	(21.984)	(20.377)	(18.854)	(17.737)	(16.602)	(15.648)	(14.834)	(14.029)	(13.428)	(12.834)	(12.156)	(11.415)
Gas Processing and Transportation	(111.331)	(103.203)	(70.716)	(65.559)	(60.679)	(57.073)	(53.425)	(50.361)	(47.749)	(45.171)	(43.236)	(41.325)	(39.141)	(36.750)
Licensing and Environmental Costs	(24.196)	(22.429)	(15.198)	(14.087)	(13.034)	(12.262)	(11.477)	(10.818)	(10.255)	(9.698)	(9.283)	(8.872)	(8.404)	(7.891)
Personal Costs	(37.782)	(35.022)	(23.731)	(21.997)	(20.352)	(19.147)	(17.922)	(16.892)	(16.012)	(15.144)	(14.495)	(13.854)	(13.122)	(12.322)
Others	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Gross Income	**2.555.099**	**2.356.336**	**1.550.878**	**1.425.953**	**1.307.587**	**1.220.652**	**1.132.360**	**1.072.453**	**1.028.326**	**987.514**	**954.887**	**922.378**	**880.694**	**829.440**
Gross Margin (%)	57%	57%	55%	55%	54%	54%	53%	54%	54%	55%	56%	56%	57%	57%
SG&A Expenses	**(1.265.549)**	**(1.181.525)**	**(837.809)**	**(784.998)**	**(734.934)**	**(698.246)**	**(660.955)**	**(629.590)**	**(602.827)**	**(576.387)**	**(556.637)**	**(537.130)**	**(514.846)**	**(490.476)**
Operating Cost Margin (%)	-28%	-28%	-30%	-30%	-30%	-31%	-31%	-31%	-32%	-32%	-32%	-33%	-33%	-34%
Sales Expenses	**(882.884)**	**(818.387)**	**(554.547)**	**(514.010)**	**(475.580)**	**(447.418)**	**(418.793)**	**(394.717)**	**(374.173)**	**(353.878)**	**(338.718)**	**(323.744)**	**(306.639)**	**(287.932)**
As % of the Revenue														
Personnel	(234.271)	(217.157)	(147.148)	(136.391)	(126.194)	(118.721)	(111.126)	(104.737)	(99.286)	(93.900)	(89.878)	(85.904)	(81.366)	(76.402)
Office Rental and Maintenance	(978)	(978)	(978)	(978)	(978)	(978)	(978)	(978)	(978)	(978)	(978)	(978)	(978)	(978)
Third Party Services	(33.021,15)	(30.608,87)	(20.740,89)	(19.224,72)	(17.787,40)	(16.734,08)	(15.663,48)	(14.763,01)	(13.994,63)	(13.235,55)	(12.668,55)	(12.108,49)	(11.468,74)	(10.769,08)
Provision	(346)	(346)	(346)	(346)	(346)	(346)	(346)	(346)	(346)	(346)	(346)	(346)	(346)	(346)
Tax expenses	-	-	-	-	-	-	-	-	-	-	-	-	-	-
IT	(33.856)	(33.856)	(33.856)	(33.856)	(33.856)	(33.856)	(33.856)	(33.856)	(33.856)	(33.856)	(33.856)	(33.856)	(33.856)	(33.856)
Other Expenses	(36.163)	(36.163)	(36.163)	(36.163)	(36.163)	(36.163)	(36.163)	(36.163)	(36.163)	(36.163)	(36.163)	(36.163)	(36.163)	(36.163)
Other operating expenses/income	(44.030)	(44.030)	(44.030)	(44.030)	(44.030)	(44.030)	(44.030)	(44.030)	(44.030)	(44.030)	(44.030)	(44.030)	(44.030)	(44.030)
(Loss) / Reversal in Recoverable Value of Assets	-	-	-	-	-	-	-	-	-	-	-	-	-	-

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in thousand of R$	2020	2021	2022	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E	2035E	2036E
EBIT	**(163.183)**	**262.637**	**271.057**	**2.154.983**	**2.852.191**	**3.383.203**	**3.716.069**	**3.329.532**	**3.091.655**	**2.862.139**	**2.699.054**	**2.508.177**	**2.230.602**	**1.971.048**	**1.751.594**	**1.573.412**	**1.419.622**
EBIT Margin	*-96%*	*36%*	*16%*	*36%*	*33%*	*34%*	*34%*	*33%*	*32%*	*32%*	*32%*	*31%*	*31%*	*31%*	*30%*	*30%*	*29%*
EBITDA	**(132.835)**	**372.826**	**519.804**	**2.170.782**	**2.866.691**	**3.417.203**	**3.772.069**	**3.400.032**	**3.176.155**	**2.957.889**	**2.801.304**	**2.615.427**	**2.342.533**	**2.087.191**	**1.871.513**	**1.696.738**	**1.546.057**
EBITDA Margin (%)	*-80%*	*51%*	*30%*	*36%*	*33%*	*34%*	*34%*	*34%*	*33%*	*33%*	*33%*	*33%*	*33%*	*32%*	*32%*	*32%*	*32%*
Financial Revenue	52.903	197.072	319.817	94.052	716.869	486.286	577.728	751.597	1.008.784	1.307.244	1.624.653						
Financial Expenses	(136.065)	(656.699)	(805.007)	(975.632)	(914.539)	(546.679)	(199.351)	(49.817)	(4.402)	(66)	(8)						
Net Financial Result (Expenses)	(83.162)	(459.627)	(485.190)	(881.580)	(197.671)	(60.393)	378.377	701.780	1.004.382	1.307.179	1.624.645	-	-	-	-	-	-
EBT	**(246.345)**	**(196.990)**	**(214.133)**	**1.273.403**	**2.654.521**	**3.322.811**	**4.094.446**	**4.031.312**	**4.096.037**	**4.169.318**	**4.323.698**	**2.508.177**	**2.230.602**	**1.971.048**	**1.751.594**	**1.573.412**	**1.419.622**
Current income tax and social contribution	(4.783)	(3.850)	(71.675)	(111.616)	(404.814)	(506.729)	(624.403)	(614.775)	(624.646)	(635.821)	(659.364)	(382.497)	(340.167)	(300.585)	(267.118)	(239.945)	(216.492)
Effective Tax Rate				15,25%													
Net Income	**(224.106)**	**(136.416)**	**127.716**	**1.368.308**	**2.249.706**	**2.816.082**	**3.470.043**	**3.416.537**	**3.471.391**	**3.533.497**	**3.664.334**	**2.125.680**	**1.890.435**	**1.670.463**	**1.484.476**	**1.333.466**	**1.203.130**
Dividends	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Payout (%)				0,00%													
Retained Earnings				1.368.308	2.249.706	2.816.082	3.470.043	3.416.537	3.471.391	3.533.497	3.664.334	2.125.680	1.890.435	1.670.463	1.484.476	1.333.466	1.203.130

in thousand of R$	2037E	2038E	2039E	2040E	2041E	2042E	2043E	2044E	2045E	2046E	2047E	2048E	2049E	2050E
EBIT	**1.289.550**	**1.174.811**	**713.070**	**640.955**	**572.652**	**522.406**	**471.405**	**442.863**	**425.499**	**411.127**	**398.250**	**385.248**	**365.847**	**338.964**
EBIT Margin	*29%*	*28%*	*25%*	*25%*	*24%*	*23%*	*22%*	*22%*	*22%*	*23%*	*23%*	*23%*	*24%*	*23%*
EBITDA	**1.418.833**	**1.306.725**	**847.421**	**776.959**	**710.187**	**661.358**	**611.690**	**569.895**	**534.208**	**498.950**	**472.628**	**446.635**	**416.948**	**384.479**
EBITDA Margin (%)	*32%*	*31%*	*30%*	*30%*	*29%*	*29%*	*29%*	*28%*	*28%*	*28%*	*27%*	*27%*	*27%*	*26%*
Financial Revenue														
Financial Expenses														
Net Financial Result (Expenses)	-	-	-	-	-	-	-	-	-	-	-	-	-	-
EBT	**1.289.550**	**1.174.811**	**713.070**	**640.955**	**572.652**	**522.406**	**471.405**	**442.863**	**425.499**	**411.127**	**398.250**	**385.248**	**365.847**	**338.964**
Current income tax and social contribution	(196.656)	(179.159)	(108.743)	(97.746)	(87.329)	(79.667)	(71.889)	(67.537)	(64.889)	(62.697)	(60.733)	(58.750)	(55.792)	(51.692)
Effective Tax Rate														
Net Income	**1.092.894**	**995.652**	**604.326**	**543.209**	**485.323**	**442.739**	**399.516**	**375.326**	**360.610**	**348.430**	**337.517**	**326.498**	**310.056**	**287.272**
Dividends	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Payout (%)														
Retained Earnings	1.092.894	995.652	604.326	543.209	485.323	442.739	399.516	375.326	360.610	348.430	337.517	326.498	310.056	287.272

Balance Sheet

BS

in thousand of R$	2020	2021	2022	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E	2035E
Balance Sheet																
Assets	**2.109.064**	**5.453.481**	**7.971.135**	**15.842.048**	**15.789.382**	**16.455.546**	**18.836.688**	**21.712.847**	**25.049.234**	**28.489.754**	**32.080.465**	**34.112.238**	**35.864.540**	**37.406.298**	**38.782.510**	**40.027.398**
Current Assets	**771.817**	**2.713.458**	**1.570.700**	**9.151.613**	**8.723.447**	**8.983.611**	**11.130.753**	**13.797.412**	**16.993.299**	**20.399.569**	**23.992.530**	**26.037.943**	**27.817.926**	**29.400.310**	**30.828.285**	**32.134.346**
Cash and cash equivalents	287.942	118.725	800.442	7.545.988	6.483.818	6.419.197	8.351.080	11.208.712	14.524.939	18.051.702	21.738.778	23.902.768	25.856.875	27.601.584	29.166.207	30.583.944
Financial investments	417.520	2.389.374	31.353	31.353	31.353	31.353	31.353	31.353	31.353	31.353	31.353	31.353	31.353	31.353	31.353	31.353
Restricted cash	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Receivables from third parties	44.671	114.559	223.252	774.792	1.114.099	1.308.362	1.435.241	1.313.720	1.237.703	1.162.440	1.105.243	1.034.613	931.162	834.645	753.300	686.937
Accounts receivable - Follow on	-	-	193.011	193.011	193.011	193.011	193.011	193.011	193.011	193.011	193.011	193.011	193.011	193.011	193.011	193.011
Inventories	4.751	16.622	187.472	471.299	765.997	896.518	984.897	915.447	871.123	825.893	788.975	741.029	670.355	604.547	549.243	503.932
Prepaid expenses	2.479	11.347	77.378	77.378	77.378	77.378	77.378	77.378	77.378	77.378	77.378	77.378	77.378	77.378	77.378	77.378
Income tax, social contribution and other recoverable	11.550	54.803	41.021	41.021	41.021	41.021	41.021	41.021	41.021	41.021	41.021	41.021	41.021	41.021	41.021	41.021
Other assets	2.904	8.028	16.771	16.771	16.771	16.771	16.771	16.771	16.771	16.771	16.771	16.771	16.771	16.771	16.771	16.771
Non-Current Assets	**1.337.247**	**2.740.023**	**6.400.435**	**6.690.435**	**7.065.935**	**7.471.935**	**7.705.935**	**7.915.435**	**8.055.935**	**8.090.185**	**8.087.935**	**8.074.295**	**8.046.614**	**8.005.988**	**7.954.226**	**7.893.052**
Financial investments	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Deferred income tax and social contribution	16.489	96.718	500.225	500.225	500.225	500.225	500.225	500.225	500.225	500.225	500.225	500.225	500.225	500.225	500.225	500.225
Advances for assignment of blocks	121.266	158.885	593.549	593.549	593.549	593.549	593.549	593.549	593.549	593.549	593.549	593.549	593.549	593.549	593.549	593.549
Fixed asset	351.827	948.013	2.228.071	2.518.071	2.893.571	3.299.571	3.533.571	3.743.071	3.883.571	3.917.821	3.915.571	3.901.931	3.874.250	3.833.624	3.781.862	3.720.688
Intangible asset	835.658	1.495.112	2.997.516	2.997.516	2.997.516	2.997.516	2.997.516	2.997.516	2.997.516	2.997.516	2.997.516	2.997.516	2.997.516	2.997.516	2.997.516	2.997.516
Right of use	1.239	25.490	48.875	48.875	48.875	48.875	48.875	48.875	48.875	48.875	48.875	48.875	48.875	48.875	48.875	48.875
Other Assets	10.768	15.805	32.199	32.199	32.199	32.199	32.199	32.199	32.199	32.199	32.199	32.199	32.199	32.199	32.199	32.199
Liabilties	**1.031.964**	**1.278.036**	**3.629.860**	**10.132.466**	**7.830.094**	**5.680.175**	**4.591.273**	**4.050.897**	**3.915.892**	**3.822.915**	**3.749.291**	**3.655.384**	**3.517.251**	**3.388.545**	**3.280.282**	**3.191.703**
Current Liabilities	**114.511**	**566.863**	**891.485**	**4.191.003**	**4.295.058**	**3.405.694**	**2.712.759**	**2.215.896**	**2.081.513**	**1.988.624**	**1.915.001**	**1.821.094**	**1.682.960**	**1.554.255**	**1.445.992**	**1.357.413**
Suppliers	22.445	53.609	246.668	620.116	1.007.867	1.179.602	1.295.888	1.204.508	1.146.189	1.086.676	1.038.101	975.015	882.025	795.438	722.672	663.053
Loans and Debentures	9.566	673	147.457	2.832.093	2.406.426	1.260.554	395.967	43.514	621	89	-	-	-	-	-	-
Labor obligations	3.251	16.428	55.946	155.819	210.701	245.613	268.415	246.576	232.915	219.389	209.110	196.416	177.825	160.479	145.861	133.934
Amounts payable for acquisitions	896	1.429	299.089	299.089	299.089	299.089	299.089	299.089	299.089	299.089	299.089	299.089	299.089	299.089	299.089	299.089
Income tax, social contribution and other taxes payable	25.146	52.462	57.301	198.862	285.950	335.811	368.376	337.186	317.675	298.358	283.677	265.543	238.997	214.224	193.346	176.313
Derivatives	34.349	115.970	32.734	32.734	32.734	32.734	32.734	32.734	32.734	32.734	32.734	32.734	32.734	32.734	32.734	32.734
Other obligations	18.858	326.292	52.290	52.290	52.290	52.290	52.290	52.290	52.290	52.290	52.290	52.290	52.290	52.290	52.290	52.290

BS

in thousand of R$	2036E	2037E	2038E	2039E	2040E	2041E	2042E	2043E	2044E	2045E	2046E	2047E	2048E	2049E	2050E
• Balance Sheet															
Assets	**41.154.240**	**42.182.763**	**43.121.928**	**43.492.931**	**44.000.636**	**44.452.284**	**44.870.401**	**45.244.871**	**45.594.893**	**45.931.847**	**46.256.086**	**46.576.165**	**46.885.522**	**47.177.342**	**47.446.342**
Current Assets	**33.330.645**	**34.435.846**	**35.458.161**	**35.930.473**	**36.543.555**	**37.104.400**	**37.634.810**	**38.124.611**	**38.578.146**	**39.001.514**	**39.392.491**	**39.766.766**	**40.118.219**	**40.442.869**	**40.740.227**
Cash and cash equivalents	31.876.453	33.062.863	34.156.497	34.922.266	35.580.174	36.183.527	36.745.046	37.266.481	37.749.845	38.200.234	38.618.529	39.012.729	39.383.800	39.729.838	40.049.318
Financial investments	31.353	31.353	31.353	31.353	31.353	31.353	31.353	31.353	31.353	31.353	31.353	31.353	31.353	31.353	31.353
Restricted cash	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Receivables from third parties	629.735	581.427	538.952	365.200	338.503	313.195	294.649	275.798	259.943	246.413	233.048	223.064	213.203	201.938	189.619
Accounts receivable - Follow on	193.011	193.011	193.011	193.011	193.011	193.011	193.011	193.011	193.011	193.011	193.011	193.011	193.011	193.011	193.011
Inventories	464.923	432.022	403.178	283.473	265.344	248.144	235.582	222.798	208.825	195.333	181.380	171.439	161.682	151.558	141.757
Prepaid expenses	77.378	77.378	77.378	77.378	77.378	77.378	77.378	77.378	77.378	77.378	77.378	77.378	77.378	77.378	77.378
Income tax, social contribution and other recoverable	41.021	41.021	41.021	41.021	41.021	41.021	41.021	41.021	41.021	41.021	41.021	41.021	41.021	41.021	41.021
Other assets	16.771	16.771	16.771	16.771	16.771	16.771	16.771	16.771	16.771	16.771	16.771	16.771	16.771	16.771	16.771
Non-Current Assets	**7.823.594**	**7.746.918**	**7.663.767**	**7.562.458**	**7.457.082**	**7.347.884**	**7.235.591**	**7.120.260**	**7.016.746**	**6.930.333**	**6.863.595**	**6.809.400**	**6.767.303**	**6.734.473**	**6.706.115**
Financial investments	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Deferred income tax and social contribution	500.225	500.225	500.225	500.225	500.225	500.225	500.225	500.225	500.225	500.225	500.225	500.225	500.225	500.225	500.225
Advances for assignment of blocks	593.549	593.549	593.549	593.549	593.549	593.549	593.549	593.549	593.549	593.549	593.549	593.549	593.549	593.549	593.549
Fixed asset	3.651.230	3.574.554	3.491.403	3.390.094	3.284.718	3.175.520	3.063.227	2.947.896	2.844.382	2.757.969	2.691.231	2.637.036	2.594.939	2.562.109	2.533.751
Intangible asset	2.997.516	2.997.516	2.997.516	2.997.516	2.997.516	2.997.516	2.997.516	2.997.516	2.997.516	2.997.516	2.997.516	2.997.516	2.997.516	2.997.516	2.997.516
Right of use	48.875	48.875	48.875	48.875	48.875	48.875	48.875	48.875	48.875	48.875	48.875	48.875	48.875	48.875	48.875
Other Assets	32.199	32.199	32.199	32.199	32.199	32.199	32.199	32.199	32.199	32.199	32.199	32.199	32.199	32.199	32.199
Liabilties	**3.115.415**	**3.051.045**	**2.994.557**	**2.761.233**	**2.725.730**	**2.692.054**	**2.667.432**	**2.642.386**	**2.617.082**	**2.593.426**	**2.569.235**	**2.551.797**	**2.534.657**	**2.516.420**	**2.498.148**
Current Liabilities	**1.281.125**	**1.216.754**	**1.160.267**	**926.943**	**891.439**	**857.764**	**833.142**	**808.095**	**782.791**	**759.135**	**734.944**	**717.507**	**700.366**	**682.130**	**663.858**
Suppliers	611.727	568.437	530.485	372.983	349.129	326.497	309.969	293.148	274.763	257.011	238.653	225.572	212.734	199.414	186.518
Loans and Debentures	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Labor obligations	123.654	114.972	107.339	76.113	71.315	66.767	63.434	60.046	57.197	54.765	52.363	50.569	48.797	46.773	44.559
Amounts payable for acquisitions	299.089	299.089	299.089	299.089	299.089	299.089	299.089	299.089	299.089	299.089	299.089	299.089	299.089	299.089	299.089
Income tax, social contribution and other taxes payable	161.631	149.232	138.330	93.734	86.882	80.386	75.626	70.788	66.718	63.246	59.815	57.253	54.722	51.831	48.669
Derivatives	32.734	32.734	32.734	32.734	32.734	32.734	32.734	32.734	32.734	32.734	32.734	32.734	32.734	32.734	32.734
Other obligations	52.290	52.290	52.290	52.290	52.290	52.290	52.290	52.290	52.290	52.290	52.290	52.290	52.290	52.290	52.290

BS

in thousand of R$	2020	2021	2022	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E	2035E
Non-Current Liabilities	**917.453**	**711.173**	**2.738.375**	**5.941.462**	**3.535.036**	**2.274.481**	**1.878.514**	**1.835.001**	**1.834.379**	**1.834.290**	**1.834.290**	**1.834.290**	**1.834.290**	**1.834.290**	**1.834.290**	**1.834.290**
Loans and Debentures	623.219	47.539	904.085	4.107.172	1.700.745	440.191	44.224	710	89	-	-	-	-	-	-	-
Derivatives	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Amounts payable for acquisitions	-	-	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Abandonment provision	282.841	413.964	1.112.985	1.112.985	1.112.985	1.112.985	1.112.985	1.112.985	1.112.985	1.112.985	1.112.985	1.112.985	1.112.985	1.112.985	1.112.985	1.112.985
Other obligations	11.393	249.670	721.305	721.305	721.305	721.305	721.305	721.305	721.305	721.305	721.305	721.305	721.305	721.305	721.305	721.305
Equity	**1.077.100**	**4.175.445**	**4.341.275**	**5.709.582**	**7.959.289**	**10.775.371**	**14.245.414**	**17.661.951**	**21.133.342**	**24.666.839**	**28.331.173**	**30.456.854**	**32.347.289**	**34.017.752**	**35.502.228**	**36.835.695**
Capital stock	1.228.618	4.146.616	4.154.406	4.154.406	4.154.406	4.154.406	4.154.406	4.154.406	4.154.406	4.154.406	4.154.406	4.154.406	4.154.406	4.154.406	4.154.406	4.154.406
Capital reserve	-	114.976	37.136	37.136	37.136	37.136	37.136	37.136	37.136	37.136	37.136	37.136	37.136	37.136	37.136	37.136
Asset and liability valuation adjustments	108.980	115.852	106.383	106.383	106.383	106.383	106.383	106.383	106.383	106.383	106.383	106.383	106.383	106.383	106.383	106.383
Loss carryover	- 260.498	- 269.554	- 15.486	1.352.822	3.602.528	6.418.610	9.888.653	13.305.190	16.776.581	20.310.078	23.974.413	26.100.093	27.990.529	29.660.992	31.145.468	32.478.934
Minoritary shareholder participation	-	67.555	58.836	58.836	58.836	58.836	58.836	58.836	58.836	58.836	58.836	58.836	58.836	58.836	58.836	58.836
Equity + Liabilities	**2.109.064**	**5.453.481**	**7.971.135**	**15.842.048**	**15.789.382**	**16.455.546**	**18.836.688**	**21.712.847**	**25.049.234**	**28.489.754**	**32.080.465**	**34.112.238**	**35.864.540**	**37.406.298**	**38.782.510**	**40.027.398**
check	##########	##########	##########	VERDADEIRO	VERDADEIRO	VERDADEIRO	VERDADEIRO	VERDADEIRO	VERDADEIRO	VERDADEIRO	VERDADEIRO	VERDADEIRO	VERDADEIRO	VERDADEIRO	VERDADEIRO	VERDADEIRO
	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

in thousand of R$	2036E	2037E	2038E	2039E	2040E	2041E	2042E	2043E	2044E	2045E	2046E	2047E	2048E	2049E	2050E
Non-Current Liabilities	**1.834.290**	**1.834.290**	**1.834.290**	**1.834.290**	**1.834.290**	**1.834.290**	**1.834.290**	**1.834.290**	**1.834.290**	**1.834.290**	**1.834.290**	**1.834.290**	**1.834.290**	**1.834.290**	**1.834.290**
Loans and Debentures	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Derivatives	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Amounts payable for acquisitions	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Abandonment provision	1.112.985	1.112.985	1.112.985	1.112.985	1.112.985	1.112.985	1.112.985	1.112.985	1.112.985	1.112.985	1.112.985	1.112.985	1.112.985	1.112.985	1.112.985
Other obligations	721.305	721.305	721.305	721.305	721.305	721.305	721.305	721.305	721.305	721.305	721.305	721.305	721.305	721.305	721.305
Equity	**38.038.825**	**39.131.719**	**40.127.371**	**40.731.698**	**41.274.907**	**41.760.230**	**42.202.969**	**42.602.485**	**42.977.811**	**43.338.421**	**43.686.851**	**44.024.368**	**44.350.866**	**44.660.921**	**44.948.193**
Capital stock	4.154.406	4.154.406	4.154.406	4.154.406	4.154.406	4.154.406	4.154.406	4.154.406	4.154.406	4.154.406	4.154.406	4.154.406	4.154.406	4.154.406	4.154.406
Capital reserve	37.136	37.136	37.136	37.136	37.136	37.136	37.136	37.136	37.136	37.136	37.136	37.136	37.136	37.136	37.136
Asset and liability valuation adjustments	106.383	106.383	106.383	106.383	106.383	106.383	106.383	106.383	106.383	106.383	106.383	106.383	106.383	106.383	106.383
Loss carryover	33.682.064	34.774.958	35.770.611	36.374.937	36.918.146	37.403.469	37.846.208	38.245.724	38.621.050	38.981.661	39.330.091	39.667.607	39.994.105	40.304.161	40.591.433
Minoritary shareholder participation	58.836	58.836	58.836	58.836	58.836	58.836	58.836	58.836	58.836	58.836	58.836	58.836	58.836	58.836	58.836
Equity + Liabilities	**41.154.240**	**42.182.763**	**43.121.928**	**43.492.931**	**44.000.636**	**44.452.284**	**44.870.401**	**45.244.871**	**45.594.893**	**45.931.847**	**46.256.086**	**46.576.165**	**46.885.522**	**47.177.342**	**47.446.342**
check	VERDADEIRO	VERDADEIRO	VERDADEIRO	VERDADEIRO	VERDADEIRO	VERDADEIRO	VERDADEIRO	VERDADEIRO	FALSO	VERDADEIRO	VERDADEIRO	VERDADEIRO	VERDADEIRO	FALSO	VERDADEIRO
	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

Cash Flow

CF

in thousand of R$	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E
Cash Flow Statement												
Net Income	1.368.308	2.249.706	2.816.082	3.470.043	3.416.537	3.471.391	3.533.497	3.664.334	2.125.680	1.890.435	1.670.463	1.484.476
(+) D&A	-	14.500	34.000	56.000	70.500	84.500	95.750	102.250	107.250	111.930	116.143	119.919
(-)ΔWK	(220.485)	(104.283)	(68.277)	(43.605)	46.562	28.849	28.138	20.580	24.669	35.991	33.620	28.384
Cash Flow from Operating Activities	1.147.823	2.159.923	2.781.805	3.482.438	3.533.599	3.584.740	3.657.385	3.787.165	2.257.599	2.038.357	1.820.226	1.632.779
(-) CAPEX	(290.000)	(390.000)	(440.000)	(290.000)	(280.000)	(225.000)	(130.000)	(100.000)	(93.610)	(84.250)	(75.517)	(68.157)
Cash Flow from Investing Activities	(290.000)	(390.000)	(440.000)	(290.000)	(280.000)	(225.000)	(130.000)	(100.000)	(93.610)	(84.250)	(75.517)	(68.157)
(+/-)Δ Debt	5.887.723	(2.832.093)	(2.406.426)	(1.260.554)	(395.967)	(43.514)	(621)	(89)	–	–	–	–
(-) Dividends	-	-	-	-	-	-	-	-	-	-	-	-
Cash Flow from Financing Activities	5.887.723	(2.832.093)	(2.406.426)	(1.260.554)	(395.967)	(43.514)	(621)	(89)	–	–	–	–
Cash BOP	800.442	7.545.988	6.483.818	6.419.197	8.351.080	11.208.712	14.524.939	18.051.702	21.738.778	23.902.768	25.856.875	27.601.584
Delta Cash	6.745.546	(1.062.169)	(64.622)	1.931.883	2.857.632	3.316.227	3.526.763	3.687.076	2.163.990	1.954.107	1.744.709	1.564.622
Cash EOP	**7.545.988**	**6.483.818**	**6.419.197**	**8.351.080**	**11.208.712**	**14.524.939**	**18.051.702**	**21.738.778**	**23.902.768**	**25.856.875**	**27.601.584**	**29.166.207**

in thousand of R$	2035E	2036E	2037E	2038E	2039E	2040E	2041E	2042E	2043E	2044E	2045E	2046E	2047E	2048E	2049E	2050E
Cash Flow Statement																
Net Income	1.333.466	1.203.130	1.092.894	995.652	604.326	543.209	485.323	442.739	399.516	375.326	360.610	348.430	337.517	326.498	310.056	287.272
(+) D&A	123.327	126.434	129.283	131.913	134.352	136.004	137.535	138.952	140.285	127.033	108.709	87.823	74.378	61.387	51.101	45.515
(-)ΔWK	23.097	19.922	16.839	14.832	60.133	9.322	8.833	6.487	6.588	4.524	3.365	3.127	2.488	2.477	3.152	3.849
Cash Flow from Operating Activities	1.479.890	1.349.487	1.239.016	1.142.398	798.811	688.535	631.691	588.178	546.389	506.883	472.684	439.381	414.382	390.362	364.309	336.636
(-) CAPEX	(62.153)	(56.977)	(52.606)	(48.763)	(33.042)	(30.627)	(28.337)	(26.659)	(24.954)	(23.519)	(22.295)	(21.086)	(20.182)	(19.290)	(18.271)	(17.156)
Cash Flow from Investing Activities	(62.153)	(56.977)	(52.606)	(48.763)	(33.042)	(30.627)	(28.337)	(26.659)	(24.954)	(23.519)	(22.295)	(21.086)	(20.182)	(19.290)	(18.271)	(17.156)
(+/-)Δ Debt	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
(-) Dividends	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Cash Flow from Financing Activities	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Cash BOP	29.166.207	30.583.944	31.876.453	33.062.863	34.156.497	34.922.266	35.580.174	36.183.527	36.745.046	37.266.481	37.749.845	38.200.234	38.618.529	39.012.729	39.383.800	39.729.838
Delta Cash	1.417.737	1.292.510	1.186.409	1.093.635	765.768	657.908	603.354	561.519	521.435	483.364	450.389	418.295	394.200	371.072	346.038	319.479
Cash EOP	**30.583.944**	**31.876.453**	**33.062.863**	**34.156.497**	**34.922.266**	**35.580.174**	**36.183.527**	**36.745.046**	**37.266.481**	**37.749.845**	**38.200.234**	**38.618.529**	**39.012.729**	**39.383.800**	**39.729.838**	**40.049.318**

FCFF & FCFE

FCFF

FCFF 2023E-2033E

FCFF	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E
(=) EBIT	2.154.983	2.837.691	3.349.203	3.660.069	3.259.032	3.007.155	2.766.389	2.596.804	2.400.927	2.123.355	1.863.803
(-) Tax	15,25%	15,25%	15,25%	15,25%	15,25%	15,25%	15,75%	16,25%	16,75%	17,25%	17,75%
(=) NOPAT	1.826.348	2.404.943	2.838.450	3.101.909	2.762.029	2.548.564	2.330.683	2.174.823	1.998.772	1.757.076	1.532.978
(+) Depreciation	–	14.500	34.000	56.000	70.500	84.500	95.750	102.250	107.250	111.928	116.138
(-) Capex	(290.000)	(390.000)	(440.000)	(290.000)	(280.000)	(225.000)	(130.000)	(100.000)	(93.551)	(84.203)	(75.489)
(+/-) Delta WC	(220.485)	(103.256)	(66.896)	(42.047)	47.589	29.841	28.935	21.041	25.023	35.991	33.620
FCFF	**1.315.863**	**1.926.188**	**2.365.554**	**2.825.862**	**2.600.119**	**2.437.905**	**2.325.368**	**2.198.114**	**2.037.494**	**1.820.791**	**1.607.247**

FCFF 2034E – 2050E

2034E	2035E	2036E	2037E	2038E	2039E	2040E	2041E	2042E	2043E	2044E	2045E	2046E	2047E	2048E	2049E	2050E
1.644.351	1.466.169	1.312.381	1.182.309	1.067.569	605.827	533.694	465.374	415.111	364.094	321.036	284.159	247.773	220.383	193.369	162.706	129.312
18,25%	18,75%	19,25%	19,75%	20,25%	20,75%	21,25%	21,75%	22,25%	22,75%	23,25%	23,75%	24,25%	24,75%	25,25%	25,75%	26,25%
1.344.257	1.191.263	1.059.748	948.803	851.387	480.118	420.284	364.155	322.748	281.262	246.396	216.671	187.688	165.838	144.543	120.809	95.367
119.912	123.319	126.426	129.275	131.905	134.344	136.015	137.564	138.998	140.346	141.609	142.799	143.927	144.994	146.016	146.992	147.917
(68.135)	(62.138)	(56.976)	(52.613)	(48.772)	(33.419)	(30.982)	(28.676)	(26.972)	(25.248)	(23.800)	(22.566)	(21.347)	(20.433)	(19.529)	(18.497)	(17.367)
28.384	23.096	19.922	16.839	14.832	60.133	9.323	8.834	6.488	6.589	5.553	4.748	4.687	3.516	3.470	3.950	4.310
1.424.418	**1.275.540**	**1.149.120**	**1.042.303**	**949.351**	**641.175**	**534.640**	**481.877**	**441.262**	**402.950**	**369.757**	**341.652**	**314.955**	**293.916**	**274.500**	**253.254**	**230.227**

FCFE

in thousand of R$	2020	2021	2022	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E
Debt Schedule											
Debt BoP				1.051.542	6.939.265	4.107.172	1.700.745	440.191	44.224	710	89
(-) Debt Paid				(1.824.446)	(2.832.093)	(2.406.426)	(1.260.554)	(395.967)	(43.514)	(621)	(89)
(+) Debt Issued				7.712.169							
Debt EoP			1.051.542	6.939.265	4.107.172	1.700.745	440.191	44.224	710	89	-
Interest				(975.632)	(914.539)	(546.679)	(199.351)	(49.817)	(4.402)	(66)	(8)

FCFF	**1.315.863**	**1.926.188**	**2.365.554**	**2.825.862**	**2.600.119**	**2.437.905**	**2.325.368**	**2.198.114**	**2.037.494**	**1.820.791**	**1.607.247**
Change in net borrowing	7.712.169	–	–	–	–	–	–	–	–	–	–
Debt Amortization	(1.824.446)	(2.832.093)	(2.406.426)	(1.260.554)	(395.967)	(43.514)	(621)	(89)	–	–	–
Cost of net financial debt	(826.848)	(775.072)	(463.310)	(168.950)	(42.220)	(3.731)	(56)	(7)	–	–	–
FCFE	**6.376.738** -	**1.680.977** -	**504.183**	**1.396.358**	**2.161.932**	**2.390.660**	**2.324.691**	**2.198.018**	**2.037.494**	**1.820.791**	**1.607.247**

Source: ITR

Hedge Policy

Hedge

Hedge	Quantidade *(Mil Barris)*	Preço Médio	Vencimento
NDF			
	415	$ 85,6	**4T23**
	429	$ 83,4	**1T24**
	412	$ 81,3	**2T24**
	383	$ 79,8	**3T24**
	358	$ 78,3	**4T24**
	165	$ 76,6	**1T25**
	22	$ 74,5	**2T25**
	-	-	**3T25**
Total	**2.184**	**$ 81,3**	**-**

Hedge	Quantidade *(Mil Barris)*	Preço Médio		Vencimento
Collar		Put	Call	
	863	$ 59,0	$ 98,5	**4T23**
	843	$ 59,1	$ 98,5	**1T24**
	848	$ 55,3	$ 95,6	**2T24**
	670	$ 51,7	$ 98,3	**3T24**
	565	$ 50,5	$ 98,6	**4T24**
	721	$ 50,4	$ 98,2	**1T25**
	1097	$ 53,2	$ 92,4	**2T25**
	315	$ 66,3	$ 87,1	**3T25**
Total	**5.921**	**$ 55,1**	**$ 96,3**	**-**

Fonte:

Assumptions

Premises

The main premises that support the model.

Scenario Control Center	
Production Scenario (Discount %)	1
Base (1)	10%
Bear (2)	15%
Bull (3)	5%
Brent Scenario (Discount %)	1
Base (1)	
Bear (2)	10%
Bull (3)	-10%
Forex Scenario	1
Base (1)	
Bear (2)	
Bull (3)	

Date-Year	2022	2023	2024	2025	2026	2027	2028	2029	2030
Brent Price	99,0	81,7	86,2	80,2	76,4	73,4	71,1	70,0	70,0
Brent Price Scenario	99,0	81,7	86,2	80,2	76,4	73,4	71,1	70,0	70,0

FOREX - USD/BRL	2022	2023	2024	2025	2026	2027	2028	2029	2030
Scenario	5,2	5,1	5,3	5,3	5,3	5,3	5,3	5,3	5,3

Pessimist

Indicador	2022	2023	2024	2025	2026	2027	2028	2029	2030
GDP growth (%)	2,9%	2,9%	1,8%	1,8%	1,8%	1,8%	1,8%	1,8%	1,8%
GDP per capita (US$)	9441,8	10434,0	10872,9	11517,7	11517,7	11517,7	11517,7	11517,7	11517,7
Unemployment (%)	8,5%	7,8%	8,4%	8,2%	8,2%	8,2%	8,2%	8,2%	8,2%
IPCA (%)	6%	4,7%	3,8%	3,5%	3,5%	3,5%	3,5%	3,5%	3,5%
IGPM (%)	5%	-3,9%	3,4%	3,5%	3,5%	3,5%	3,5%	3,5%	3,5%
Selic (%)	13,75%	11,5%	9,0%	9,0%	9,0%	9,0%	9,0%	9,0%	9,0%
Exchange (US$/Real)	5,2	5,1	4,9	5,0	5,0	5,0	5,0	5,0	5,0
Exportation (US$ bi)	334,1	335,0	346,8	362,5	362,5	362,5	362,5	362,5	362,5
Importation (US$ bi)	272,6	243,5	247,0	252,6	252,6	252,6	252,6	252,6	252,6

Base

Indicador	2022	2023	2024	2025	2026	2027	2028	2029	2030
GDP growth (%)	2,9%	2,8%	1,5%	1,8%	1,8%	1,8%	1,8%	1,8%	1,8%
GDP per capita (US$)	9441,8	9937,1	10355,1	10969,2	10969,2	10969,2	10969,2	10969,2	10969,2
Unemployment (%)	8,5%	8,0%	8,0%	8,4%	8,2%	8,2%	8,2%	8,2%	8,2%
IPCA (%)	5,8%	4,8%	3,9%	4,0%	4,0%	4,0%	4,0%	4,0%	4,0%
IGPM (%)	5,5%	-3,4%	3,0%	3,0%	4,0%	4,0%	4,0%	4,0%	4,0%
CDI(%)	0,1	0,1	0,1	0,1	0,1	0,1	0,1	0,1	0,1
Selic (%)	13,75%	11,8%	9,5%	7,5%	9,0%	9,0%	9,0%	9,0%	9,0%
Exchange (US$/Real)	5,2	5,1	5,3	5,3	5,3	5,3	5,3	5,3	5,3
Exportation (US$ bi)	334	332	328	326	352	352	352	352	352
Importation (US$ bi)	272,6	251,5	266,8	280,5	280,5	280,5	280,5	280,5	280,5

Optimisti

Indicador	2022	2023	2024	2025	2026	2027	2028	2029	2030
GDP growth (%)	2,9%	2,5%	1,0%	1,5%	1,5%	1,5%	1,5%	1,5%	1,5%
GDP per capita (US$)	9441,8	9440,3	9837,4	10420,8	10420,8	10420,8	10420,8	10420,8	10420,8
Unemployment (%)	8,5%	8,6%	9,5%	9,3%	9,0%	9,0%	9,0%	9,0%	9,0%
IPCA (%)	5,8%	4,9%	4,1%	4,5%	4,0%	4,0%	4,0%	4,0%	4,0%
IGPM (%)	5,5%	-3,1%	3,2%	3,3%	4,4%	4,4%	4,4%	4,4%	4,4%
Selic (%)	13,75%	11,8%	10,0%	9,0%	9,0%	9,0%	9,0%	9,0%	9,0%
Exchange (US$/Real)	5,2	5,1	5,3	5,4	5,5	5,5	5,5	5,5	5,5
Exportation (US$ bi)	334,1	319,5	307,0	311,6	311,6	311,6	311,6	311,6	311,6
Importation (US$ bi)	272,6	252,0	267,6	262,3	273,8	273,8	273,8	273,8	273,8

Monte Carlo

Brent Prices Monte Carlo Simulation Code

```
Sub SimulacaoMonteCarlo()
    Dim wsPremises As Worksheet
    Dim wsDCF As Worksheet
    Dim wsMonteCarlo As Worksheet
    Dim i As Integer
    Dim j As Integer
    Dim iteracoes As Integer
    Dim precoBrent(1 To 9) As Double
    Dim precoBrentOriginal(1 To 9) As Double

    Set wsPremises = ThisWorkbook.Sheets("Premises")
    Set wsDCF = ThisWorkbook.Sheets("DCF")
    Set wsMonteCarlo = ThisWorkbook.Sheets("Monte Carlo")

    ' Preços originais do Brent de 2022 a 2030
    precoBrentOriginal(1) = 99
    precoBrentOriginal(2) = 81.7
    precoBrentOriginal(3) = 86.2
    precoBrentOriginal(4) = 80.2
    precoBrentOriginal(5) = 76.4
    precoBrentOriginal(6) = 73.4
    precoBrentOriginal(7) = 71.1
    precoBrentOriginal(8) = 70
    precoBrentOriginal(9) = 70

    ' Define o número de iterações da simulação
    iteracoes = 20000

    ' Inicia a simulação de Monte Carlo
    For i = 1 To iteracoes
        ' Gera preços aleatórios do Brent com variação de -10% a +10%
        For j = 1 To 9
            precoBrent(j) = precoBrentOriginal(j) * (1 + Rnd() * 0.2 - 0.1)
            wsPremises.Cells(3, 4 + j).Value = precoBrent(j)
        Next j

        ' Recalcula o modelo DCF
        wsDCF.Calculate

        ' Armazena o resultado do upside na planilha Monte Carlo
        wsMonteCarlo.Cells(i, 1).Value = wsDCF.Cells(49, 4).Value

        ' Restaura os preços originais do Brent para a próxima iteração
        For j = 1 To 9
            wsPremises.Cells(3, 4 + j).Value = precoBrentOriginal(j)
        Next j
    Next i

    MsgBox "Simulação concluída"
End Sub
```

SWOT & Portes 5 Forces

SWOT Analysis

3R Petroleum doesn't have as established a track record as its competitors, but it is **already delivering a production increase and has a high growth potential**.

SWOT Analysis

The SWOT Analysis provide a framework to **evaluate the firm's advantages and vulnerabilities**

Strength: The company has excellent exploratory wells, with a large production capacity. A new highly skilled and experienced management team. High growth potencial.



Weakness: The market questions the execution capability of 3R, as it has not delivered all the expected results and does not have a strong track record.



Opportunity: The company has already acquired its assets, allowing the new management team to concentrate on organic growth, increasing the production.



Threats: The Brent price is at a high level. If it drops significantly, it could affect the company's results. Damage to facilities and equipaments.

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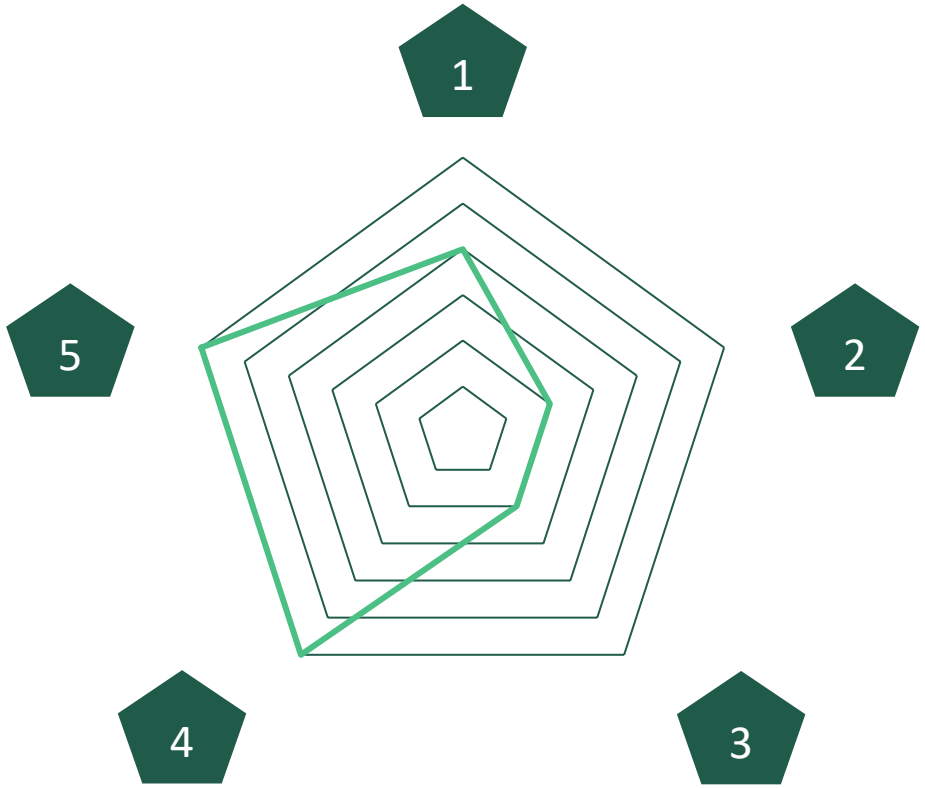


Opportunity: The company has already acquired its assets, allowing the new management team to concentrate on organic growth, increasing the production.

Threats: The Brent price is at a high level. If it drops significantly, it could affect the company's results. Damage to facilities and equipaments.



Porter 5 forces



1

Rivarly among existing competitors: 2

Rivalry has little influence in the sector, it only matter during the acquisition of exploratory wells through auctions.

2

Potencial of new entrants into the industry: 1

Entering the industry was already challenging due to the high investment required, and with the end of Petrobras divestment policy, it will become harder.

3

Bargaining power of suppliers: 1

Suppliers provide services to the company, so they do not have significant bargaining power

4

Bargaining power of buyers: 3

Most of the O&G sold by junior companies are sold in large volumes to major corporations, allowing them to demand discounts or special conditions

5

Threat of substitute services: 3

The energy transition will not significantly impact the sector in the medium term.

Debt

Keeping

To finance the signific
incur a **lot of debt** fro

on, the company had to

Among other goals, the company purchased mature fields from Petrobras, its specialty.

690m

IPO Priced at R$ 21 per share

Avg. Daily Productio

Even if the CapEx is high, this

K/Bbl & Million of Reais

Total Production (Kbbl/day

Plans

The company invested in the purchase of assets and in CapEx.

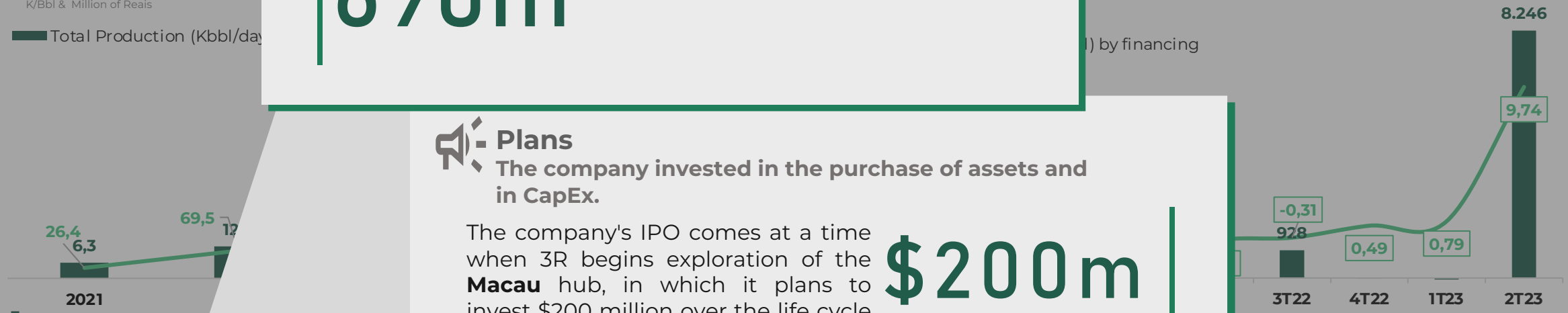
The company's IPO comes at a time when 3R begins exploration of the **Macau** hub, in which it plans to invest $200 million over the life cycle of the wells.

$200m

ebt & Net Debt/Ebitda

ached maximum peak shot in 2Q23 pushing the multiple.

l) by financing



Financing Histo

The company's main

4T20 → 2T21 → 4T21 → 3T22 → 2T23

Source: Morgan Stanley, Bloomberg, RI 3R

Keeping

To finance the signific
incur a **lot of debt** fro

on, the company had to

Initially resources going into its cash

822m

Follow on

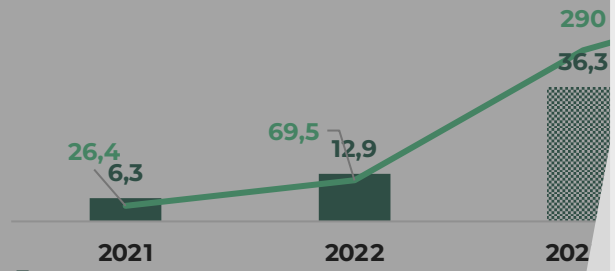
Priced at R$ 36 per share

Avg. Daily Productio

Even if the CapEx is high, this

K/Bbl & Million of Reais

Total Production (Kbbl/da



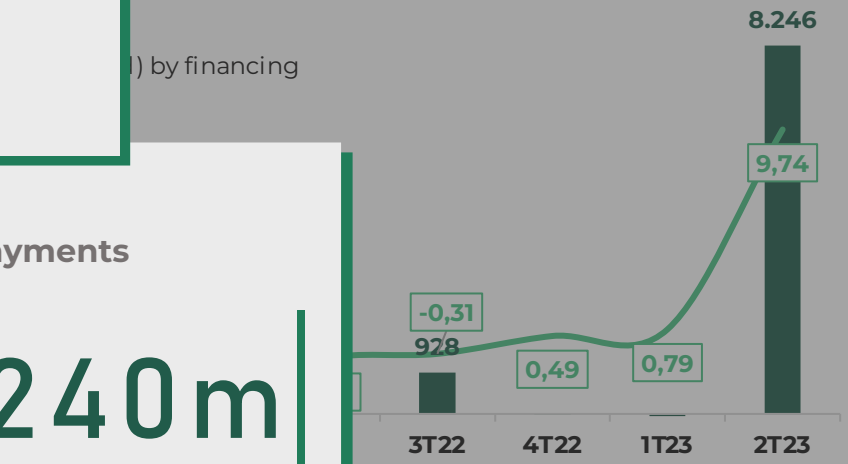
Financing History

The company's main financings since its IPO

ebt & Net Debt/Ebitda

ached maximum peak shot in 2Q23 pushing the multiple.

l) by financing

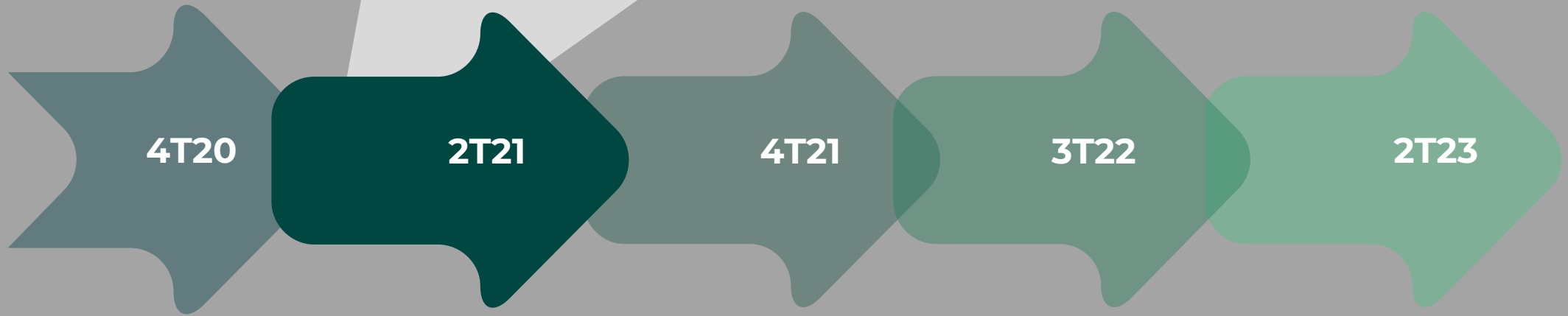


Plans

Company prepares for finalizing the asset payments

The amount should be used for the payment of the closing consideration of US$ 240 million to Petrobras at the time of completion of the purchase of **Recôncavo Cluster**.

U$240m



Source: Morgan Stanley, Bloomberg, RI 3R

Keeping

To finance the signific
incur a **lot of debt** fro

on, the company had to

Company raises capital for Potiguar purchase and provides partial exit to Starboard

2,1Bn

Follow on

Priced at R$ 33 per share

Avg. Daily Productio

Even if the CapEx is high, this

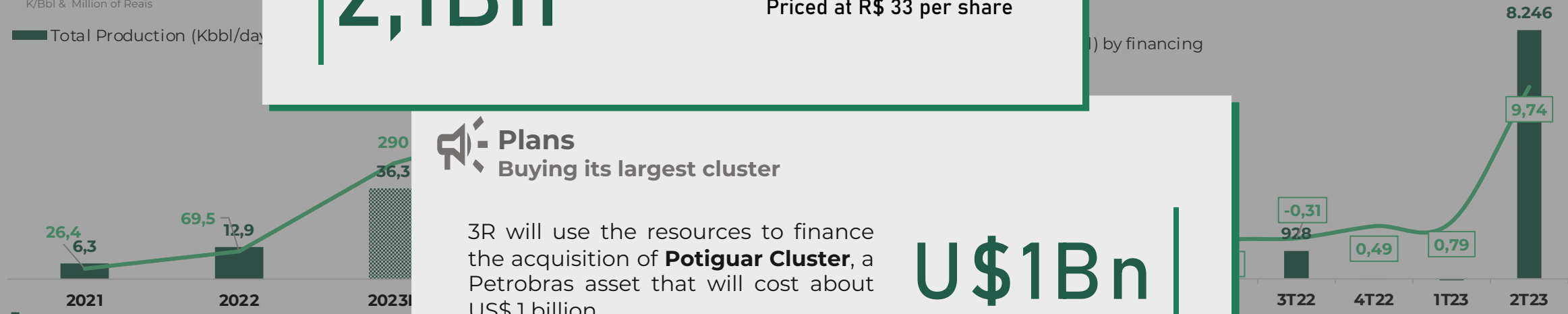
K/Bbl & Million of Reais

Total Production (Kbbl/day

ebt & Net Debt/Ebitda

ached maximum peak shot in 2Q23 pushing the multiple.

l) by financing



Financing History

The company's main financings since its IPO, wi

Plans

Buying its largest cluster

3R will use the resources to finance the acquisition of **Potiguar Cluster**, a Petrobras asset that will cost about US$ 1 billion.

U$1Bn



Source: Morgan Stanley, Bloomberg, RI 3R

Keeping

To finance the signific
incur a **lot of debt** fro

on, the company had to

Priority remains the payment of the
Potiguar Cluster and strengthening of cash

900m

Debenture

Maturity Sept 2025

DI + 3% a.a

Avg. Daily Productio

Even if the CapEx is high, this

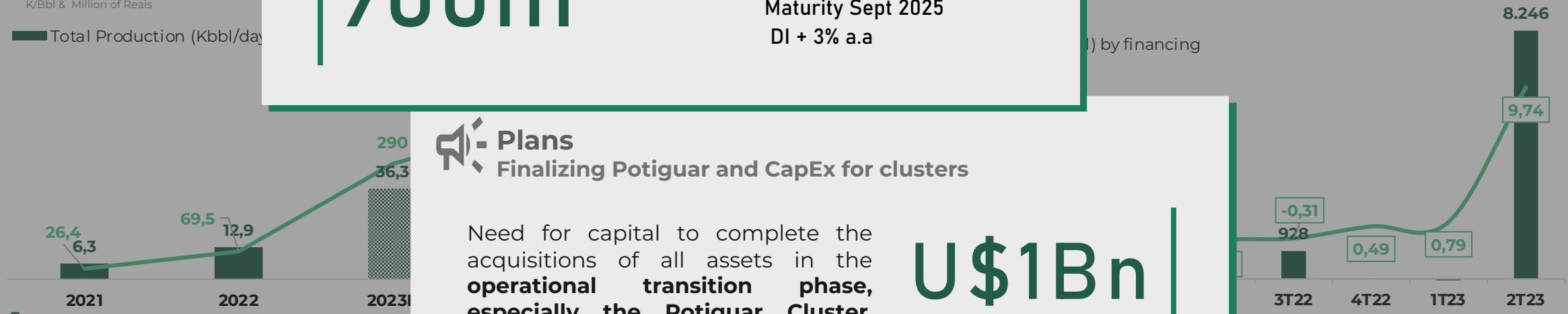
K/Bbl & Million of Reais

Total Production (Kbbl/day

ebt & Net Debt/Ebitda

ached maximum peak shot in 2Q23 pushing the multiple.

l) by financing



Financing History

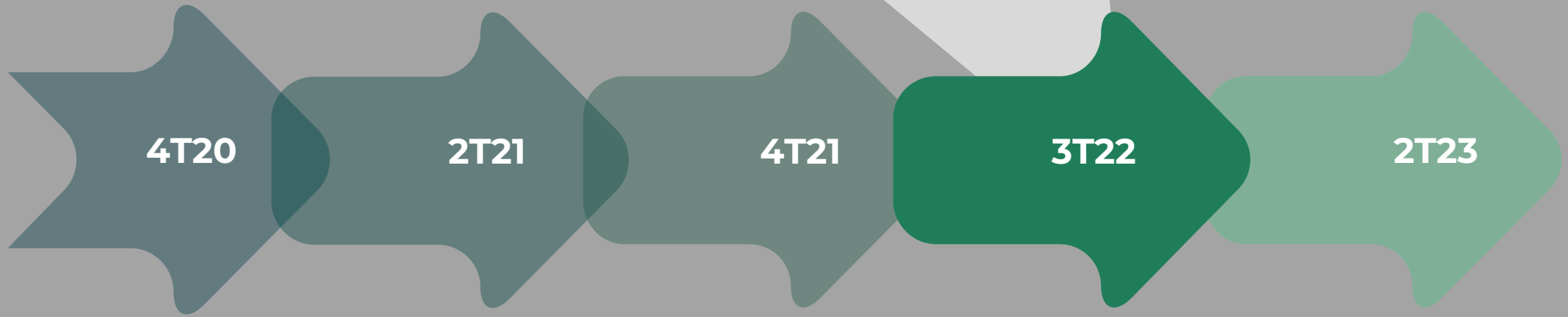
The company's main financings since its IPO, wi

Plans

Finalizing Potiguar and CapEx for clusters

Need for capital to complete the acquisitions of all assets in the **operational transition phase, especially the Potiguar Cluster**, whose closing is estimated for 1Q23.

U$1Bn



Source: Morgan Stanley, Bloomberg, RI 3R

Keeping

To finance the signific
incur a **lot of debt** fro

The debentures are expected to pay about 9.49% to 10.16% per year, which can be considered a quite positive premium.

8,5Bn

Debenture 5,1Bn
Follow On 900M
Loan 2,5Bn

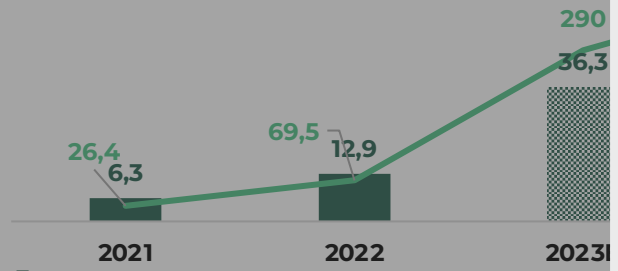
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Avg. Daily Productio

Even if the CapEx is high, this

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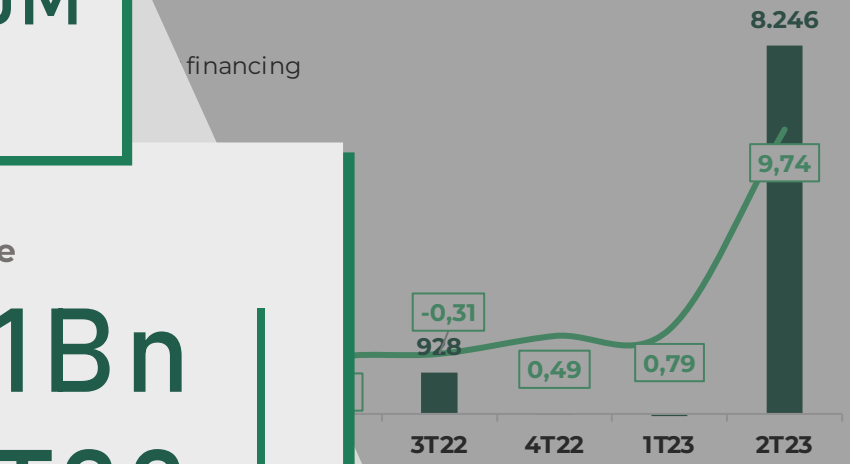
Financing History

The company's main financings since its IPO, wi

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financing

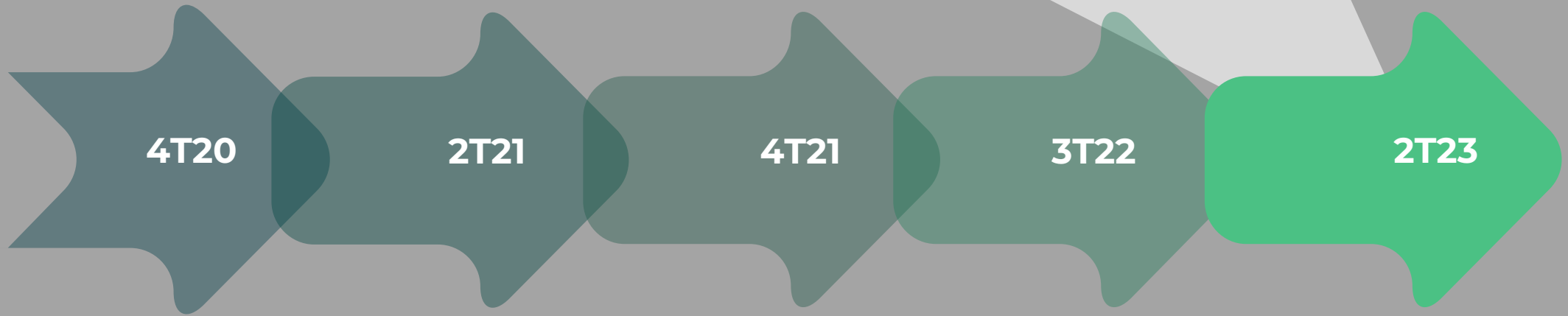


Plans

Strengthening of cash, reduction of Leverage (Follow On) and Cluster Payments.

Increase the **cash position**, optimize the capital structure, and **reduce leverage** and cope with expenses and investments. As well as the **final payment of all its clusters.**

+1Bn
3T23



Source: Morgan Stanley, Bloomberg, RI 3R

Python

Python Code for Upside Graph

```
historical_price_color = (0.063, 0.639, 0.498)
upside_projection_color = (0.078, 0.475, 0.376)

file_path = '/Users/caioalzugaray/Downloads/RRRP3.SA.csv'
df_prices = pd.read_csv(file_path, sep=';', parse_dates=['Date'], dayfirst=True)

upsides = [1.11, 1.07, 0.96, 0.93, 0.91, 0.87, 0.83, 0.79, 0.73, 0.71, 0.69, 0.65, 0.61, 0.57, 0.50, 0.44, 0.41, 0.39, 0.35,

last_price = df_prices['Close'].iloc[-1]
last_price_date = df_prices['Date'].iloc[-1]

additional_days = (last_price_date - df_prices['Date'].min()).days * 0.3
extended_end_date = last_price_date + timedelta(days=additional_days)

fig, ax = plt.subplots(figsize=(12, 8))

ax.plot(df_prices['Date'], df_prices['Close'], label='Closing Price', linestyle='-', linewidth=2, color=historical_price_colo

ax.xaxis.set_major_locator(mdates.MonthLocator(interval=3))
ax.xaxis.set_major_formatter(mdates.DateFormatter('%m/%y'))
ax.set_xlim([df_prices['Date'].min(), extended_end_date])

ax.grid(False)

# Adjusting the chart scale
minimum_price = df_prices['Close'].min()
maximum_price = df_prices['Close'].max()
margin = (maximum_price - minimum_price) * 0.5
ax.set_ylim([minimum_price - margin, maximum_price + margin])

for upside in upsides:
    future_price = last_price * (1 + upside)
    future_date = last_price_date + timedelta(days=additional_days)
    ax.plot([last_price_date, future_date], [last_price, future_price], '--', color=upside_projection_color)

    if upside == max(upsides) or upside == min(upsides):
        ax.text(future_date, future_price, f'R$ {future_price:.2f}', fontsize=8, color=upside_projection_color)

ax.set_xlabel('Data')
ax.set_ylabel('Preço (R$)')
ax.set_title('Preço da Ação RRRP3 com Projeções de Upside')
ax.legend()

png_file_path = '/Users/caioalzugaray/Desktop/rrrp3_stock_price_projection.png'
plt.savefig(png_file_path, format='png', transparent=True, dpi=300)

plt.show()
```